

Taxation and Reallocation in the Secondary Market for Capital Goods

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This paper studies how the tax system shapes capital reallocation. I expand the neoclassical model to feature vintage capital, a market for used capital, a realistic tax system, and a closure for entry. I show that investment subsidies like bonus depreciation yield ambiguous results on reallocation and entry. Because subsidies increase investment, the supply of used assets may increase through a capital stock expansion channel, leading to greater reallocation and entry. However, subsidies also increase the effective capital gains tax rate on sale, which may reduce reallocation and entry through a lock-in effect. The lock-in channel is more likely to dominate when new and old vintages are sufficiently complementary in production, while the capital stock expansion channel is more likely to dominate when they are highly substitutable—the baseline assumption in national accounting. I use asset-level data from financed equipment transactions in Texas, firm-level data from Compustat, and industry-level data from the Business Dynamics Statistics to test the theory. I find that equipment from bonus vintages subsequently transacts less, firms more exposed to investment subsidies invest more but have lower capital sale proceeds, and more-exposed industries experience persistently lower entry. These patterns are consistent with the lock-in effect dominating.

JEL-Classification: E22, G31, H25, L26

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1 Introduction

Capital misallocation is widespread (Hsieh and Klenow 2009). One way to reduce the resulting burden on productivity, output, and welfare is to reduce frictions in secondary markets for capital goods. This paper studies how a concrete policy wedge—the tax treatment of used capital purchases and sales—can create or reduce additional frictions in the reallocative process. I show theoretically that tax policies which stimulate new capital formation can impede its subsequent reallocation and provide evidence that this tension is empirically important.

Investment subsidies, which stimulate new capital accumulation (Zwick and Mahon 2017), shape capital reallocation through two opposing forces. First, by stimulating new investment, a subsidy may expand the capital stock, leaving a larger quantity of capital goods to be sold in the future, which would increase reallocation (Lanteri and Rampini 2023). I refer to this as the capital stock expansion channel. However, when a firm sells a depreciable asset, it is taxed on the difference between the sale price and the portion of the original purchase price that has not yet been deducted—the tax basis. Because this tax is realized only when the asset is sold, it can discourage reallocation and lock assets into their current owners (Feldstein, Slemrod, and Yitzhaki 1980; Hassett and Viard 2007; Hines 2016). Policies like bonus depreciation strengthen this force by reducing the tax basis carried by the subsidized asset, raising the effective capital gains tax rate on sale. I refer to this as the lock-in channel.¹ Whether an investment subsidy ultimately increases reallocation therefore cannot be inferred from its effect on new investment alone.

To study these two channels, I develop a new theoretical framework for secondary capital markets in a two-vintage model incorporating the tax treatment of capital purchases and sales.² In the model, capital lives for two periods. Incumbent firms are taxed at the corporate rate and produce using “new” and “old” capital, where old capital is the result of investment in the previous period. Each physical unit of old capital is less efficient than a unit of new capital, so the firm may prefer to sell it at the exogenous used price q and replace it with new capital. New investment receives a cohort-specific depreciation schedule that can be accelerated through bonus depreciation, while the tax due upon sale depends on the asset’s remaining basis. The resale channel contrasts with the canonical Hall and Jorgenson (1967), in which firms invest in a unit of capital and hold it until it geometrically decays to zero. Nevertheless, I show that the model reduces to a stylized version of that benchmark when resale is shut down, yielding the standard result that increases in bonus reduce the user cost of capital and increase investment.

1. The purchaser of a used asset also receives a new tax basis equal to the price it pays, so depreciation allowances for used purchases affect demand for used capital.

2. The two-vintage structure is related to Edgerton (2011), Lanteri (2018), and Lanteri and Rampini (2023). The key distinction is that I attach a depreciation schedule to each purchase cohort and jointly model the corporate tax rate, the taxation of resale, and depreciation allowances for new and used purchases.

The model’s central prediction is that investment subsidies for new capital have ambiguous effects on reallocation. Holding the price of used capital fixed, the firm’s two optimality conditions drive the two competing forces. On the one hand, the firm’s optimality condition relates the marginal product of new capital to a version of the Jorgensonian cost of capital less the option value of selling it in the next period. If bonus goes up, then the cost of capital declines and the firm expands the stock of new capital. With a larger capital stock, more capital is eligible to be sold in the following period, which is the capital stock expansion channel. On the other hand, the firm’s optimality condition for how much capital to sell equates the marginal product of retaining the capital to the foregone benefits from sale, which include the tax benefit of recognizing its remaining basis earlier. By reducing that remaining basis, bonus reduces the tax benefit of selling and strengthens the incentive to retain the asset. This is the lock-in channel. The net effect on reallocation depends on which channel dominates. I show that when the new and old vintages are complementary, additional new investment raises the marginal product of old capital, reinforcing lock-in. The opposite holds under the typical benchmark of high substitutability, in which the firm shifts to relying more on new capital to produce output. Therefore, bonus depreciation can create more capital while causing less of it to be subsequently reallocated.

I test the predictions of the partial equilibrium model using exogenous variation in bonus depreciation, which was repeatedly varied by policymakers from 2001–2016.³ I rely on two datasets. First, I study firm-level capital expenditures and capital sale proceeds using Compustat. This allows for a direct comparison between the capital stock expansion and lock-in channels. However, Compustat reports dollar expenditures and sale proceeds and therefore cannot distinguish whether these outcomes changed because prices or quantities adjusted. To address this, I study the quantity of physical transactions using a second dataset: Uniform Commercial Code filings from Texas from 2001–2017. These filings contain the universe of financed equipment transactions in the state and have previously been used in other settings to study used asset transactions (Edgerton 2011; Ma, Murfin, and Pratt 2022).

In Compustat, I follow Zwick and Mahon (2017) in exploiting variation across firms in their predetermined exposure to bonus depreciation to estimate the effect of the subsidies on investment and capital sale proceeds. The identifying assumption is that, absent changes in bonus depreciation, investment and capital sale proceeds would not have changed differentially across industries according to this predetermined exposure. I find that a one-percentage-point increase in exposure to bonus depreciation raises investment by approximately 3.5 percent but decreases capital sale proceeds by approximately 11 percent. Thus, the same policy that encourages firms to

3. The 2017 Tax Cuts and Jobs Act provides a complementary policy experiment because it changed the corporate tax rate, used investment subsidies, and new investment subsidies at once. That also makes the resulting evidence less clean to interpret. I provide a complete analysis of the policy episode in Appendix G.

purchase new capital also reduces their capital sale proceeds, consistent with the lock-in channel dominating the capital stock expansion channel for the dollar value of sales.

Next, I find that vintages of capital originally purchased during bonus years are less likely to transact than vintages purchased during non-bonus years. Treating model year as a proxy for the original purchase vintage, I compare subsequent transactions involving equipment from bonus and nonbonus vintages within manufacturers at the same asset age. I find that equipment from bonus vintages appears in approximately 13 percent fewer subsequent financed transactions. A local comparison around the temporary lapse in bonus depreciation from 2005 through 2007 produces a similar negative transaction gap. I also test the transaction gap by age. Since all equipment in my sample has its ordinary tax basis expire after either five or seven years, the transaction gap should not persist after age seven. I find that this holds in the data, with bonus vintages being substantially less likely to transact between ages two and seven and null effects thereafter. Together, the Compustat and UCC results show that bonus depreciation raises investment while reducing subsequent reallocation, consistent with the lock-in channel dominating the capital stock expansion channel.

The Compustat and UCC evidence corresponds to the region of the partial equilibrium model in which the lock-in channel dominates the capital stock expansion channel. Partial equilibrium, however, cannot determine what this decline in reallocation means for buyers of used capital or how adjustment in the used price feeds back onto incumbent sales. My closure is motivated by evidence from Ma, Murfin, and Pratt (2022), which shows that the typical used asset transaction involves an incumbent firm selling to a young and financially constrained firm. I therefore close the model with a fringe of potential entrants that purchase only used assets and allow the price of used capital to adjust to clear the market. Because greater purchases per entrant raise both the entry cutoff and the mass of active entrants, market clearing links reallocation directly to entry. I show that equilibrium price adjustment attenuates but does not reverse the partial equilibrium response of reallocation. Consequently, when the lock-in channel dominates, bonus depreciation can raise investment while reducing both reallocation and entry.

Finally, I test the model's downstream prediction that the tax system influences entry through reallocation. Using industry-level data from the Census Bureau's Business Dynamics Statistics and the same predetermined industry exposure to bonus depreciation from Zwick and Mahon (2017), I estimate the industry-level response of the entry rate to a one-percentage-point increase in the present value of depreciation deductions. In a panel local projections framework, the entry response is initially small but becomes statistically significant after one year and is largest three to four years after the reform, consistent with subsidized machines reaching the secondary market only after they have aged. More importantly, I show that this plausibly runs through reallocation. I interact the bonus exposure with a measure of how liquid the industry's secondary market for

capital is from Kermani and Ma (2022) and find that the decline in entry is larger in industries whose equipment retains more value in secondary markets. That mediating factor suggests that the mechanism for the entry decline is secondary markets. Together, the results trace the model’s full chain from investment subsidies to subsequent reallocation and, ultimately, firm entry.

In sum, I find that subsidies for new investment can be a source of misallocation rather than a remedy for it. This result is quantitatively and economically significant. Aggregated upward, back-of-the-envelope calculations suggest that lower-bonus or no-bonus counterfactuals would have increased reallocation by roughly \$13–\$20 billion per year from 2012–2017. These findings stand in contrast to the standard argument that these policies expand the capital stock and, consequently, the supply of used assets to young and financially constrained firms (Smith 2009; Lanteri, Medina, and Tan 2023).

Related Literature. This paper relates to three main strands of literature.

First, my paper builds on the literature on secondary markets for capital and capital reallocation (Eisfeldt and Rampini 2006). In my model, a population of entrants purchases used capital from incumbent firms. This structure is motivated by Eisfeldt and Rampini (2007), who show that financially constrained firms rely disproportionately on used capital, and Ma, Murfin, and Pratt (2022), who show that startups purchase older machines and that local used capital availability shapes startup hiring and subsequent investment. Darmouni and Sutherland (2024) similarly show that shortages of new equipment cause intermediate-age firms to turn toward used capital, increasing the competition faced by younger firms. The willingness of these firms to absorb used capital depends on its price. Following Lanteri (2018), I endogenize the price of used assets to clear the market, and show that this equilibrium feedback propagates into investment, used capital purchases, and entry. The closest theoretical paper is Lanteri and Rampini (2023). They show that subsidizing new investment expands the future supply of old capital, which generates the capital stock expansion channel. While I retain this force, I add an opposing lock-in channel generated by how investment subsidies affect the tax treatment of future resale.⁴ Accelerated depreciation encourages new investment, but it also leaves the subsidized capital with less tax basis when it subsequently reaches the secondary market, strengthening its owner’s incentive to retain it. Whether reallocation rises therefore depends on which force dominates. Investment subsidies can consequently create more capital while causing less of it to be reallocated.

Second, my paper contributes to the literature on how taxation affects reallocation of durable assets, including housing and financial capital. For example, on the seller side, realization-based

4. Baley, Blanco, and Oviedo (2025) study how corporate taxes interact with fixed adjustment costs and an exogenous purchase–resale price wedge. They allow capital losses upon resale to be deducted, but do not track purchase-cohort tax basis, depreciation recapture, a buyer-side basis reset, or an endogenous secondary market price.

taxation can discourage sales because tax on any gain becomes due only when an asset is sold. This lock-in effect has been documented for corporate stock (Feldstein, Slemrod, and Yitzhaki 1980; Ivković, Poterba, and Weisbenner 2005), corporate dispositions of property, plant, and equipment (Desai and Gentry 2004), and depreciable business assets more generally (Hines 2016).⁵ In my model, this seller-side wedge is determined jointly by the corporate tax rate and the asset's remaining basis, so an investment subsidy can create a subsequent incentive to retain the subsidized asset. On the buyer side, a taxable transfer establishes a new depreciable basis, and the resulting deductions can encourage trade. Prior work shows that this force can generate tax-motivated transfers of real estate and structures (Hendershott and Ling 1984; Gordon, Hines, and Summers 1987; Hassett and Viard 2007). I incorporate both sides of the transaction: the seller carries the basis created by the asset's original depreciation schedule, while the buyer receives a new basis equal to the purchase price, whose deductions may themselves be accelerated. A smaller empirical literature shows that tax policy toward new capital can affect contemporaneous used equipment prices and sales (Smith 2009; Edgerton 2011; Basu, Kim, and Singh 2024). I ask what happens one step later, when the tax-favored machine itself becomes old and can return to the market. Because it carries the depreciation schedule from its purchase vintage, the subsidy that encouraged its initial purchase may discourage its subsequent sale. I place these seller- and buyer-side forces in an equilibrium market in which the used price, new investment, and entry are endogenous and test the model's vintage-specific predictions using individual equipment transactions.

Finally, this paper contributes to an empirical literature on the effects of business tax policy on investment and entry. Following Hall and Jorgenson (1967), this literature typically studies the effects of tax policy through its effect on the user cost of capital. For example, House and Shapiro (2008) show that temporary bonus depreciation creates strong incentives to move investment into the subsidy window, while Kitchen and Knittel (2011), Zwick and Mahon (2017), and Ohrn (2019) provide empirical evidence that accelerated depreciation raises investment. I recover this investment response, but treat it as the beginning rather than the end of the analysis. Because accelerated depreciation also reduces the tax basis of the capital being purchased, it changes the incentive to sell that capital later. I therefore follow the same reforms from new investment into capital sales, physical transactions in used capital, and firm entry. Whereas the investment literature asks how strongly tax policy stimulates capital formation, I show that this response does not reveal whether the capital it creates is subsequently reallocated. Moreover, Sapollnik and Swonder (2025) and Fairlie et al. (2026) show, at a different level of aggregation than I use,

5. Related work on housing finds that transaction taxes distort prices and reduce trade (Kopczuk and Munroe 2015; Best and Kleven 2018). Because those taxes apply to the sale price rather than the realized gain, they provide evidence on taxes as barriers to trade rather than realization lock-in itself.

that corporate tax cuts increase business entry. I study a different policy margin and show that, at the industry level, investment subsidies reduce entry through the secondary market for capital goods.

Roadmap. The remainder of the paper proceeds as follows. Section 2 describes the tax treatment of new and used assets, and Section 3 develops the partial equilibrium model. Sections 4–6 introduce the data and present the evidence on investment and capital reallocation. Section 7 closes the used capital market and develops the model’s implications for entry, which I test in Section 8. Section 9 gauges the aggregate magnitude of the reallocation response, and Section 10 concludes.

2 The Tax Treatment of New and Used Assets

Tax policy analysis tends to start and end at the point when a firm purchases a machine. However, the tax treatment of the machine does not: the rules that apply when it is purchased interact with those that apply if it is later sold. This section develops that connection in two steps, beginning with the rules for buying new and used capital and then turning to the rules governing resale.

Rules for Buying New Capital. Historically, the government has subsidized investment through a mixture of tax credits and accelerated depreciation policies. The latter has been dominant since the Tax Reform Act of 1986. Under the Modified Accelerated Cost Recovery System (MACRS), capital is depreciated over a prescribed recovery period that typically understates its economic useful life. Since firms discount the future at a positive rate (Gormsen and Huber 2022), one dollar of tax depreciation spread over several years has less than one dollar of value in present value terms. To see that concretely, suppose a non-negative fraction d_t of a dollar of investment can be deducted in each period t , with $\sum_{t=0}^T d_t = 1$ and some deductions occurring after the purchase period. Then, with a discount rate $r > 0$, the present value of those deductions is

$$z = \sum_{t=0}^T \frac{d_t}{(1+r)^t} < 1.$$

If the asset is held through its scheduled recovery period, the associated tax savings have present value τz , where τ is the business income tax rate. With an investment tax credit c , the tax-adjusted price of a dollar of investment is therefore $1 - c - \tau z$. The value of the subsidy depends on both the timing of the deductions, summarized by z , and the tax rate at which they are claimed.

Bonus depreciation, which has been studied extensively with respect to new capital (House and Shapiro 2008; Kitchen and Knittel 2011; Zwick and Mahon 2017), allows firms to deduct

a percentage θ of investment immediately, with the remainder following the standard MACRS schedule. The modified present value of depreciation allowances is

$$\tilde{z} = \theta + (1 - \theta)z.$$

The associated tax savings are $\tau\tilde{z}$, so a higher bonus rate lowers the tax-adjusted purchase price. Historically, bonus depreciation has consequently raised new investment. Bonus was first introduced at 30% in 2002, raised to 50% in 2003, allowed to lapse in 2005, reinstated at 50% in 2008, temporarily raised to 100% in 2011, and raised again to 100% for several years following the 2017 Tax Cuts and Jobs Act. Appendix Table A.1 illustrates how 50-percent bonus changes the timing and present value of deductions for a five-year asset. Before 2018, bonus depreciation was limited by the original-use requirement and therefore generally applied only to new capital. Purchasers of used equipment could still deduct their purchase cost under the ordinary MACRS schedule, but they could not claim bonus depreciation. The TCJA extended bonus depreciation to qualifying used property acquired after September 27, 2017, making used equipment eligible for bonus depreciation for the first time.

As these deductions are claimed, the portion of the asset's purchase price that has not yet been deducted declines. This remaining amount is the asset's tax basis. The depreciation rules that apply when a machine is purchased therefore determine the basis it carries into a subsequent sale. The next subsection explains how that basis is treated when the machine is sold.

Rules for Selling Used Capital. The statutory rules governing sales of used capital have been relatively stable since the Tax Reform Act of 1986 (TRA86), but the tax due when an asset is sold has varied with depreciation policy and the business income tax rate. Since TRA86, the basic principle has been that firms pay tax on the difference between the sale price and the asset's remaining tax basis.

Suppose that an asset purchased in period s is sold at the beginning of period t , before the period- t deduction is claimed. If q_s is its original purchase price and d_i is the fraction of that price deducted in period i , its basis immediately before sale is

$$B_t = q_s \left(1 - \sum_{i=s}^{t-1} d_i \right).$$

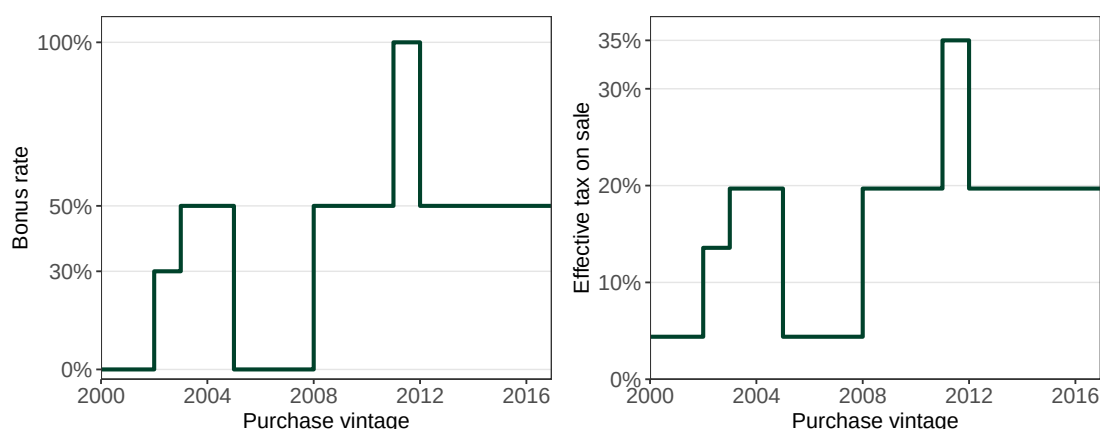
If the firm sells the asset for q_t , its after-tax proceeds are

$$q_t - \tau(q_t - B_t) = (1 - \tau)q_t + \tau B_t.$$

The tax on the gain $q_t - B_t$ is known as depreciation recapture.⁶ This expression describes the tax due conditional on sale. Holding the sale price fixed, the liability is larger when prior deductions have reduced B_t or when the business income tax rate is higher. Bonus depreciation therefore increases the tax due upon a subsequent sale by reducing the asset's basis more quickly. Under full expensing, $B_t = 0$, so the entire sale price is included in taxable income.

Whether the firm chooses to sell, however, depends on the alternative of retaining the asset. If the asset is retained, its remaining basis is recovered through the remaining depreciation deductions. If it is sold, that basis offsets taxable sale proceeds immediately. The relevant comparison therefore depends on the deductions whose recovery sale brings forward and on the tax rate that determines their value, rather than simply on total basis at the sale date.⁷

Figure 1: Bonus Depreciation and the Effective Tax on Capital Sales



Notes: The left panel plots the statutory bonus depreciation rate from 2000–2016. The right panel reports the effective tax on gross sale proceeds for a \$100 asset in the seven-year MACRS class sold for $q = \$50$ after three depreciation deductions. The calculation holds the corporate tax rate fixed at $\tau = 0.35$ and defines the effective tax rate as $\tau(q - B)/q$, where B is the asset's remaining adjusted basis at sale. Bonus depreciation applies the ordinary MACRS schedule to the basis remaining after the bonus deduction.

Synthesizing the Effects of Investment Subsidies on Capital Investment and Sales. The structure of the federal tax system systematically links subsidies on new capital with taxes on their sale proceeds. Figure 1 makes this apparent. The left panel plots the statutory bonus rate θ

6. Strictly, §1245 recapture applies only to the portion of the gain attributable to prior depreciation deductions, i.e., $\min(q_t - B_t, q_s - B_t)$. Any gain above the original purchase price, $q_t - q_s$, is a §1231 capital gain. For used equipment, the sale price almost always falls below the original purchase price, so the entire gain is recapture. For corporations, the distinction is immaterial in either case because both are taxed at rate τ .

7. Until 2018, sellers could also defer recapture by exchanging equipment for like-kind property under Section 1031. TCJA changed several of these rules simultaneously: it increased bonus depreciation for new capital, extended bonus depreciation to used property, reduced the corporate tax rate from 35 to 21 percent, and repealed like-kind treatment for equipment.

from 2000–2016, while the right panel calculates an illustrative tax rate on sale for a \$100 seven-year asset that is sold for \$50 after claiming depreciation for three years. With no bonus, the capital gains tax on sale is small, but at 100% bonus, the tax rate rises to 35%. Consequently, bonus may stimulate new investment, thereby increasing the stock of old assets, but it also may generate lock-in by raising the effective tax rate on sale. The following section formalizes when each of those separate forces may dominate.

3 A Model of Capital Investment and Sales in Partial Equilibrium

This section studies a firm’s choices over investment in new capital and sales of used capital in partial equilibrium and develops predictions for the empirical effects of investment subsidies on each variable. I use the stylized two-period model to separate two competing margins. On the one hand, a literature on the effects of investment subsidies would suggest that investment subsidies would increase reallocation by increasing the stock of new assets (Lanteri and Rampini 2023). However, theoretical and empirical studies of the effects of higher capital gains tax rates on housing (Kopczuk and Munroe 2015) and financial assets (Ivković, Poterba, and Weisbenner 2005) would suggest that the higher tax rates following bonus would generate lock-in, leading to lower reallocation. The model I develop below delineates the conditions under which each margin may dominate, and shows that it is possible for investment subsidies to simultaneously lead to a larger stock of new capital *and* lower reallocation. Later, in Section 7, I extend the model to general equilibrium and show how it affects entry.

3.1 Environment

Technology. I study a representative firm in discrete time $t = 0, 1, \dots$, which discounts future cash flows at rate $\beta \in (0, 1)$. The output good is the numeraire. The firm produces output by combining “new” and “old” machines. At the beginning of period t , it purchases

$$I_t = K_t^n$$

units of new capital, which are immediately available for production and are elastically supplied at a price of one. In period $t + 1$, this capital becomes old capital:

$$K_{t+1}^o = K_t^n = I_t.$$

Each physical unit of old capital provides $\phi \in (0, 1)$ efficiency units of capital. After producing as old capital in period $t + 1$, the cohort is retired in period $t + 2$ and has no productive or resale value. Thus, the lifecycle of capital purchased in period t is

$$\underbrace{I_t = K_t^n}_{\text{new capital produces at } t} \implies \underbrace{K_{t+1}^o = K_t^n}_{\phi K_{t+1}^o \text{ efficiency units at } t+1} \implies \underbrace{0}_{\text{retired at } t+2}.$$

At the beginning of period t , the firm may sell some of its inherited old capital to another firm at an exogenous price q_t per physical machine. Because $K_t^o = K_{t-1}^n$, if the firm sells S_t units, it retains

$$\tilde{K}_t^o = K_t^o - S_t = K_{t-1}^n - S_t \geq 0. \quad (1)$$

In words, the total stock of old capital K_t^o , the amount of that sold S_t , and the remainder $\tilde{K}_t^o$ are measured in physical units, while $\phi \tilde{K}_t^o$ measures the efficiency units supplied by retained old capital. Putting that together, output is

$$Y_t = F(K_t^n, \phi \tilde{K}_t^o) = F(K_t^n, \phi (K_{t-1}^n - S_t)).$$

I impose standard regularity conditions on the production technology.

Assumption 1 (Technology). *The production function $F : \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$ is twice continuously differentiable on $\mathbb{R}_{++}^2$, strictly increasing, and concave.*

I impose no sign restriction on F_{12} . New capital and the efficiency services of retained old capital may therefore be complements, $F_{12} > 0$, or substitutes, $F_{12} < 0$.

Tax Policy. The firm's tax treatment has three relevant components. First, operating income is taxed at rate $\tau \in [0, 1)$.

Second, the firm faces a tax depreciation schedule. Its ordinary schedule, which is the stylized model counterpart to MACRS, is

$$\bar{d}_0, \quad \bar{d}_1, \quad \bar{d}_2 \geq 0, \quad \bar{d}_0 + \bar{d}_1 + \bar{d}_2 = 1.$$

If the firm purchases a unit of capital in period t , a fraction $\bar{d}_0$ would be deducted immediately, $\bar{d}_1$ in the following period, and the remainder is deducted upon retirement. I am interested in how bonus depreciation, which brings those deductions forward in time, affects the firm's decisions about investment and sales of capital. I model bonus depreciation as

$$d_0(\theta) = \theta + (1 - \theta)\bar{d}_0, \quad d_j(\theta) = (1 - \theta)\bar{d}_j, \quad j = 1, 2. \quad (2)$$

When $\theta = 0$, the firm faces the ordinary depreciation schedule and when $\theta = 1$, all new investment is immediately tax-deductible. Intermediate values pull a proportional amount of depreciation forward to the present.

Third, I assume that the firm can use all deductions at rate τ and that gains and losses from asset sales are taxed symmetrically. Suppose a machine purchased in the previous period is sold in period t . Then its remaining basis immediately before sale is $d_1(\theta) + d_2(\theta)$. Mirroring the actual treatment of recapture in the tax code, the seller's after-tax proceeds from selling the capital are

$$(1 - \tau)q_t + \tau [d_1(\theta) + d_2(\theta)].$$

Conversely, if the old capital is retained rather than sold, the firm deducts $d_1(\theta)$ in period t , leaving $d_2(\theta)$ to be deducted when the firm retires the capital in period $t + 1$.

Given this setup, I proceed in two steps. I first study the firm's investment choice when capital cannot be resold, which provides the neoclassical benchmark. I then introduce the option to sell old capital at the exogenous price q_t and study the firm's joint choices over investment, retention, and sales.

3.2 Benchmark: No Secondary Market

I start with the classic problem in which firms choose a quantity of investment in a capital good, and the capital good is held until it is retired. I show that this is an abbreviated, two-period variant of the Hall and Jorgenson (1967) model of investment, in which a unit of capital purchased in period t is held until it asymptotically has no value under geometric depreciation. In the present setting's stylized framework, that is equivalent to retaining every machine through its old period. Given inherited capital K_{t-1}^n and tax policy, the firm solves

$$V^N(K_{t-1}^n) = \max_{K_t^n \geq 0} \left\{ (1 - \tau)F(K_t^n, \phi K_{t-1}^n) - (1 - \tau d_0(\theta))K_t^n + \tau d_1(\theta)K_{t-1}^n + \beta [\tau d_2(\theta)K_{t-1}^n + V^N(K_t^n)] \right\}. \quad (3)$$

At an interior stationary allocation, $K_t^n = K_{t-1}^n = K^n$, the firm's optimal investment satisfies

$$F_1(K^n, \phi K^n) + \beta \phi F_2(K^n, \phi K^n) = \frac{1 - \tau(d_0(\theta) + \beta d_1(\theta) + \beta^2 d_2(\theta))}{1 - \tau}. \quad (4)$$

This Euler equation is similar to a consumption-savings problem in a two-period model. The left-hand side sums the discounted net present value of marginal products of a unit of new capital over its lifetime, while the right-hand side is the tax-adjusted user cost of capital over the two periods it lives. Let $u(\theta)$ denote the user cost of capital, which is the right-hand side of (4). As

long as $\tau > 0$, $\beta < 1$, and some depreciation is delayed, the user cost is decreasing in θ :

$$u'(\theta) = -\frac{\tau}{1-\tau} [(1-\beta)\bar{d}_1 + (1-\beta^2)\bar{d}_2] < 0.$$

If the lifetime marginal product on the left-hand side of (4) is decreasing in K^n , the reduction in user cost must be met by an increase in the capital stock. A perfect substitutes benchmark makes this channel transparent. Suppose

$$F(K^n, \phi K^n) = (K^n + \phi K^n)^\nu = ((1 + \phi) K^n)^\nu, \quad 0 < \nu < 1.$$

This formulation resembles a stylized version of the neoclassical benchmark with ν taking the role of the capital share. With perfect substitutes across vintages, the capital stock is well-defined as $K = K^n + \phi K^n$ (Fisher 1965). At a stationary allocation, the optimal quantity of capital is

$$K(\theta) = \left[\frac{\nu(1 + \beta\phi)}{u(\theta)} \right]^{\frac{1}{1-\nu}}.$$

In this example,

$$\frac{d \log K(\theta)}{d\theta} = -\frac{1}{1-\nu} \frac{u'(\theta)}{u(\theta)} > 0.$$

Here, just like in the Hall and Jorgenson (1967) model, the semi-elasticity of the capital stock with respect to bonus is given by the output elasticity of capital and the semi-elasticity of the user cost with respect to bonus, with the ultimate prediction that bonus raises the capital stock. This conventional investment channel is what motivates the idea that increases in bonus depreciation increase reallocation by increasing the capital stock. The next section adds only the option to sell old capital.

3.3 A Secondary Market for Capital

Now, suppose the firm has the option to sell some of its old capital. It may want to do that to finance purchasing more efficient new capital. I impose an additional mild regularity condition on the firm's production technology.

Assumption 2 (Interior technology). *The Hessian of F is negative definite, and F satisfies the Inada conditions in each input.*

I assume a negative-definite Hessian so that, together with the Inada conditions, there is a unique and interior choice of the two capital inputs, which will later make comparative statics well-defined. In this partial-equilibrium exercise, I hold $q_t = q$ fixed over time. The firm's Bellman

becomes

$$\begin{aligned}
V(K_{t-1}^n) = \max_{\substack{K_t^n \geq 0 \\ 0 \leq S_t \leq K_{t-1}^n}} & \left\{ (1 - \tau)F(K_t^n, \phi \tilde{K}_t^o) - (1 - \tau d_0(\theta))K_t^n \right. \\
& + \left[(1 - \tau)q + \tau(d_1(\theta) + d_2(\theta)) \right] S_t + \tau d_1(\theta) \tilde{K}_t^o \\
& \left. + \beta \left[\tau d_2(\theta) \tilde{K}_t^o + V(K_t^n) \right] \right\},
\end{aligned} \tag{5}$$

subject to (1). At an interior stationary allocation, write $K^o = \tilde{K}_t^o$ for the quantity of old capital retained by the firm. The optimality conditions are

$$F_1(K^n, \phi K^o) = 1 + \frac{\tau(1 - \beta)}{1 - \tau} (d_1(\theta) + d_2(\theta)) - \beta q, \tag{6}$$

$$\phi F_2(K^n, \phi K^o) = q + \frac{\tau(1 - \beta)}{1 - \tau} d_2(\theta). \tag{7}$$

Equation (6) governs investment in new capital, while (7) governs the decision to sell or retain old capital. The optimality condition for new investment states that its marginal product is equal to its marginal cost. While the marginal cost involves two standard terms—the unit price of new investment and the cost of delayed tax deductions—it also involves a term βq . This captures the option value of being able to sell capital in the following period, which encourages additional investment today. The second optimality condition relates the marginal product of retaining a unit of old capital in the firm to its marginal cost, which is the foregone proceeds from selling the capital. That includes both the sale price q as well as the tax benefit of recognizing d_2 one period earlier by selling.⁸ Studying the optimality conditions individually reveals a tension. When bonus goes up, $d_1(\theta)$ and $d_2(\theta)$ decline. Holding old capital fixed, the first change reduces the marginal cost of new investment, which stimulates more capital creation. But holding new capital fixed, the reduction in $d_2(\theta)$ reduces the benefit of selling before retirement by effectively increasing depreciation recapture on sale. Because capital sales satisfy

$$S = K^n - K^o,$$

the effect on sales depends on which response is larger. To characterize these responses, I first assume a regularity condition which guarantees that bonus increases investment in new capital. This assumption follows a large literature which illustrates this empirically (House and Shapiro 2008; Kitchen and Knittel 2011; Zwick and Mahon 2017) and also maintains continuity with the canonical model outlined in the previous section. The appendix shows that, in order for investment to rise with bonus, the following condition is sufficient.

8. The deduction d_1 is received in the current period whether the machine is sold or retained and therefore cancels.

Assumption 3 (Investment response). *At the stationary allocations considered below,*

$$F_{12} \geq \phi F_{22}.$$

Intuitively, bonus lowers the marginal cost of new investment and retained old capital. When the two inputs are complements, the resulting increase in retention reinforces the investment response; when they are substitutes, the retention response reduces the marginal product of new capital and partially offsets the direct investment incentive. Assumption 3 rules out substitution strong enough to reverse that response.

With that, we can characterize the response of capital sales to bonus. Differentiating (6) and (7) with respect to θ , holding q fixed, gives the following result.

Proposition 1 (Bonus depreciation in partial equilibrium). *Suppose $\tau > 0$, $\beta < 1$, $\theta < 1$, and $d_1(\theta) > 0$. At an interior stationary allocation, under Assumptions 2 and 3, a marginal increase in bonus depreciation raises investment. It increases reallocation through higher capital sales if and only if*

$$d_2(\theta)F_{11} - \phi d_1(\theta)F_{12} - \phi^2[d_1(\theta) + d_2(\theta)]F_{22} > 0, \quad (8)$$

where the derivatives of F are evaluated at the stationary allocation.

Proof. Appendix B.3.

The intuition follows from evaluating the optimality conditions. Fixing q , bonus raises K^n and expands the stock of machines that become available for sale, which is close to the intuition guiding the stock-expansion channel. However, accounting for the lock-in effect generated by a lower $d_2(\theta)$, which raises the effective capital gains tax on sale, it also raises desired retention and reduces reallocation. If new and old capital are close substitutes, then the increase in new investment weakens the retention response and shifts the composition of capital toward new, thereby increasing reallocation. If the two vintages are complementary, then additional new capital raises the value of retaining old capital and reinforces the lock-in effect. Reallocation falls when the retention response exceeds the increase in capital creation.

Studying a CES production function, as in Lanteri (2018), makes this concrete. Suppose

$$F(K^n, \phi K^o) = \left[\omega (K^n)^{\frac{\sigma-1}{\sigma}} + (1-\omega)(\phi K^o)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\nu\sigma}{\sigma-1}}, \quad 0 < \omega < 1, \quad 0 < \nu < 1, \quad \sigma > 0,$$

where the Cobb–Douglas case is understood as the limit when $\sigma = 1$. The marginal rate of substitution is

$$\frac{F_1(K^n, \phi K^o)}{F_2(K^n, \phi K^o)} = \frac{\omega}{1-\omega} \left(\frac{K^n}{\phi K^o} \right)^{-1/\sigma}.$$

Here, σ governs how strongly the firm changes its mix of new and retained capital in response

to their relative marginal costs, while ν governs the scale response. When the condition

$$F_{12} > 0 \iff \sigma < \frac{1}{1 - \nu}$$

holds, new investment raises the marginal value of retained old capital and reinforces retention. Approaching instead the neoclassical limit in which $\sigma \rightarrow \infty$, new and old capital approach perfect substitutes, and reallocation likely rises when bonus increases.

Thus, it is ultimately an ambiguous question as to whether bonus increases reallocation or not. The next empirical sections test this.

4 Data

I test the partial equilibrium model's predictions using two primary datasets. First, I use Compustat to examine the joint behavior of capital sales and new capital investments. However, the firm-level data allow me to only analyze the dollar volume of capital sales and investment rather than prices or quantities separately. As a result, I use a second dataset: Uniform Commercial Code (UCC) filings in Texas. They provide asset-level records of equipment entering secured lending relationships, allowing me to identify the vintage, type, and age of individual machines that are sold on secondary markets.

4.1 Compustat

First, I use firm-level data from Compustat to measure capital expenditures and cash proceeds from sales of property, plant, and equipment. In terms of the model, capital expenditures (item capx) are the closest empirical counterpart to new investment I , while the proceeds from used capital sales (item sppe) correspond to qS , the used price multiplied by the quantity of capital sold. Compustat therefore allows me to examine the joint behavior of capital formation and the value of capital sales. These data have been used extensively to study reallocation (Eisfeldt and Rampini 2006; Eisfeldt and Shi 2018), but they have a major shortcoming because I cannot separately recover used capital prices and quantities. In the model, that is not important because there is a common price q of used assets, but in practice, used capital prices vary considerably across firms and industries.

I study the responses of investment and used capital sales to bonus depreciation using U.S.-domiciled nonfinancial firms with positive sales, assets, and net PPE and nonnegative capital expenditures. I match each firm to its four-digit industry exposure to bonus depreciation as constructed by Zwick and Mahon (2017). It spans 1995–2016 and contains 102,466 firm-years from

11,674 firms. Appendix C describes the outcome construction, sample restrictions, and treatment of missing sale proceeds in detail for this sample.

4.2 Used Capital Transactions in Texas

Compustat’s main shortcoming is that it observes qS , which can change because either the price or quantity changes. Moreover, the model does not make an unconditional prediction about their product. However, my second dataset allows me to observe the quantity of used asset transactions S alone. Many used asset transactions involve financing, and when a business uses equipment as collateral, the lender typically files a financing statement with the state government to document its claim. These Uniform Commercial Code (UCC) filings are a standard feature of commercial lending in the United States and have been used in recent work on equipment markets, capital reallocation, and small business lending (Edgerton 2012; Ma, Murfin, and Pratt 2022; Basu, Kim, and Singh 2024). The filings are a near-complete universe of financed used asset transactions because lenders have a strong incentive to file; it gives them priority in bankruptcy proceedings. The records consequently provide broad coverage of equipment entering secured lending relationships, although they exclude unfinanced purchases. I treat the appearance of a positive-age machine in an itemized filing as a financed used asset transaction. This linkage is not exact because a UCC filing records a security interest rather than a bill of sale, and refinancing may generate a filing without a contemporaneous ownership transfer. I use the universe of UCC filings in Texas, which I obtained from Jensen (2024).

In each filing, I observe the borrower, the secured party, and the collateral in free-form text. Some descriptions only contain a blanket claim on the assets of the borrower, but many filings itemize specific pieces of equipment and report their type, manufacturer, model year, and serial number. These descriptions are sufficient to identify individual machines associated with each financed transaction and recover their characteristics. When a filing lists multiple machines, I count each one as a separate observation. For each observation, I can construct transaction quantities by equipment type and vintage.⁹ I date the age of each machine using the filing event on which it appears and comparing that to the model year recovered from the description. The description also allows me to figure out the equipment type. Many filings contain a serial number or vehicle identification number; when the latter is valid, it can be used to recover the model year. Age is then filing year less the model year. Several types of records contain blanket liens, legal and financial boilerplate, and records that cannot be reconstructed unambiguously; I exclude all such filings. Appendix D.1 describes the construction in detail.

9. UCC filings do not report transaction prices, loan amounts, or loan terms. I therefore observe the quantity and characteristics of equipment involved in financed transactions but not the prices at which they sell.

I am able to link each transaction to the tax system using each observation's equipment type, model year, and asset age. The equipment type allows a linkage to the baseline ordinary MACRS depreciation schedule. For example, construction equipment and trucks are generally five-year property, while agricultural equipment is generally seven-year property, resulting in some variation across assets. Next, since I treat model year as a proxy for the original purchase vintage, that allows me to identify if a particular machine was exposed to bonus when it was originally sold to the first owner. In that case, its effective tax rate on sale would differ from an otherwise equivalent asset initially bought in a year without bonus. Third, the asset age determines how much the asset has progressed through the applicable tax depreciation schedule since its initial purchase. Together, these fields allow me to construct the predicted statutory basis remaining when the equipment appears in the UCC data, which is the empirical counterpart to the model's remaining-basis margin. As an example, a three-year-old tractor belonging to the 2007 non-bonus vintage would have followed an ordinary MACRS depreciation schedule, while a three-year-old machine appearing in 2011 would belong to the 2008 bonus vintage. The machines may be the same age but they would have different predicted remaining basis because only the latter was exposed to bonus depreciation.

I retain equipment assigned to five- or seven-year property with an identified equipment type, model year, and valid age. The resulting sample contains 233,845 equipment items from 2001 through 2019. Using age zero to identify new equipment and positive age to identify used equipment, the sample contains 58,837 new and 175,008 used equipment items. Appendix Tables [D.1](#), [D.2](#), and [D.3](#) report the sample construction, summary statistics, and largest equipment categories.

5 Capital Investment and Sales in Compustat

This section studies the behavior of firm-level investment and capital sales across bonus depreciation episodes.

I begin by studying bonus depreciation for new capital, which was repeatedly varied by policymakers from 2001–2016. In the model, under Assumption [3](#), Proposition [1](#) states that bonus raises capital investment through the capital stock expansion channel, but it is ex ante ambiguous whether capital sales rise. The capital stock expansion channel increases the number of machines that eventually become available for sale, while the lock-in channel increases the incentive to retain them. Compustat provides a useful setting to test these competing forces because it reports capital expenditures jointly with cash proceeds from sales of property, plant, and equipment.

To identify the effect of bonus depreciation on each outcome, I follow the same strategy as Zwick and Mahon ([2017](#)). Firms vary in their production technologies; some firms rely on long-

lived equipment, while others rely to a greater extent on shorter-lived equipment. That creates predetermined variation in exposure to bonus depreciation because the net present value of a dollar of ordinary depreciation deductions is lower for firms with longer-lived equipment, making bonus more valuable to those firms (House and Shapiro 2008). Zwick and Mahon (2017) provide a measure of this exposure at the four-digit NAICS industry level using pre-bonus investment expenditures to form a composite net present value of depreciation deductions z_N . Applying bonus rate θ_t , that becomes

$$\tilde{z}_{N,t} = \theta_t + (1 - \theta_t)z_N.$$

Consequently, I identify the effect of bonus by comparing changes within firms from more- and less-exposed industries while absorbing permanent firm differences and aggregate year effects. The identifying assumption is that, absent changes in bonus depreciation, capital expenditures and capital sales would not have changed differentially across industries according to this predetermined exposure.

For the nonnegative outcomes, I estimate

$$E[Y_{i,t} \mid \tilde{z}_{N(i),t}, \alpha_i, \gamma_t] = \exp(\beta \tilde{z}_{N(i),t} + \alpha_i + \gamma_t), \quad (9)$$

where α_i and γ_t denote firm and year fixed effects. I estimate (9) using PPML and estimate the corresponding extensive-margin specification using a linear probability model. Standard errors are clustered by four-digit NAICS industry, the level at which exposure varies.

Table 1 reports the results for four different outcomes. Columns 1, 3, and 4 decompose capital sale proceeds into the unconditional amount, an indicator for reporting positive proceeds, and proceeds conditional on positive sales. The fifth column estimates the corresponding investment response, which Zwick and Mahon (2017) already show is significantly positive. The results indicate that capital sales and capital investment move in opposite directions. Whereas a one-percentage-point increase in exposure to bonus $\tilde{z}$ raises investment by approximately 3.5 percent, it reduces unconditional capital sale proceeds by approximately 11 percent. Moreover, this difference seems driven by the extensive margin because the probability that a firm reports positive proceeds falls by 0.64 percentage points, whereas it is not possible to distinguish the intensive-margin used capital sale response from zero, although the sign is consistent.¹⁰

10. I do not report an extensive-margin result because practically all Compustat firms report positive investment.

Table 1: Bonus Depreciation, Capital Sale Proceeds, and Investment

	Capital Sale Proceeds (sppe)				Investment (capx)	
	Baseline	Quadratic Trends	Any Proceeds	Positive Proceeds	Baseline	Quadratic Trends
	(1)	(2)	(3)	(4)	(5)	(6)
$\tilde{z}$	-11.858*** (2.286)	-4.809** (2.214)	-0.641*** (0.242)	-2.470 (1.580)	3.402** (1.684)	1.082** (0.477)
Estimator	PPML	PPML	LPM	PPML	PPML	PPML
Sample	All	All	All	Sellers	All	All
Firm FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X
Industry trends		X				X
Observations	69,445	69,445	102,466	30,917	102,210	102,210

Notes: Capital sale proceeds are cash proceeds from sales of property, plant, and equipment (sppe); investment is capital expenditures (capx). $\tilde{z}$ is the present value of depreciation deductions from Zwick and Mahon (2017). Columns (2) and (6) include two-digit NAICS-specific quadratic time trends. The sample consists of U.S. nonfinancial Compustat firms from 1995–2016. Parenthesized values are standard errors clustered by four-digit NAICS industry. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Toward supporting a parallel trends interpretation of the results, I also include specifications for both capital sales (column 2) and investment (column 6) that include quadratic trends for two-digit NAICS industries. This guards against the possibility that industries with different exposure to bonus were concentrated in broad sectors experiencing different underlying trends. The estimates attenuate but retain their signs and statistical significance. I find that firms more exposed to bonus reduced their capital sales by 4.8 percent more and increased investment by approximately 1.1 percent. Thus, the opposing responses cannot be explained solely by smooth differences in trends across broad industries.

The positive investment response confirms the capital stock expansion channel, while the decline in capital sale proceeds is consistent with the lock-in channel dominating for the dollar value of sales. However, Compustat has two limitations in this context. First, it does not distinguish between sales of equipment and structures. Second, it measures qS rather than S or q separately. In the partial equilibrium analysis, q is fixed, and in the data a common change in q

would be absorbed by the year fixed effects. If q responds differently across firms or industries, however, the estimated effect combines changes in prices and quantities. The UCC analysis in the next section addresses both limitations by focusing on itemized equipment and measuring financed used-equipment activity with counts rather than sale proceeds.

6 Used Equipment Reallocation in UCC Filings

I compare transactions involving equipment whose model year assigns it to a bonus purchase vintage—which I call *bonus vintages*—with equipment assigned to nonbonus purchase vintages. The model highlights that the comparison between the two is *ex ante* ambiguous. On the one hand, bonus encourages firms to create more capital, expanding the cohort that may eventually reach the secondary market. On the other hand, the resulting machines face higher effective tax rates on sale, which strengthens the retention incentive. The analysis here asks which channel dominates in the subsequent circulation of bonus-treated vintages.

To study which channel dominates, I aggregate the data into cells organized by equipment type, MACRS recovery class, basis schedule, filing year, and asset age. I group these cells by manufacturer because manufacturers may differ in the durability of the goods they produce, the extensiveness of their secondary-market networks, and other characteristics. For example, John Deere, which is one of the largest manufacturers of tractors in the data, has a wide network of secondary market dealers. My preferred specification therefore compares vintages within manufacturers, holding equipment type, MACRS recovery class, and basis schedule fixed. Let $N_{m,t,a}$ denote the number of transactions involving equipment from manufacturer m , filed in year t , at age a . I use model year as a proxy for the machine’s original purchase vintage, $v = t - a$, and estimate

$$\mathbb{E} [N_{m,t,a} \mid \alpha_{m,a}, \gamma_t] = \exp [\alpha_{m,a} + \gamma_t + \beta \text{BonusVintage}_{t-a}], \quad (10)$$

where $\alpha_{m,a}$ allows each manufacturer to have its own transaction profile over the asset lifecycle and γ_t absorbs shocks common to all transactions in a filing year. Thus, the preferred specification compares machines from the same manufacturer at the same age that appear in different filing years and therefore belong to different purchase vintages. I estimate (10) by PPML over filing years 2001–2017 and cluster standard errors by purchase vintage, which is the level at which bonus eligibility changes.

Identification comes from changes in the statutory bonus schedule across vintages. The identifying assumption is that, absent their initial differential tax treatment, bonus and nonbonus vintages within the same manufacturer and equipment type would have otherwise transacted at similar rates at the same age. Table 2 reports the main results.

Table 2: Bonus Depreciation and Used-Equipment Mentions in UCC Filings

	Full sample	Manufacturer observed	Manufacturer	Manufacturer × age
	(1)	(2)	(3)	(4)
Bonus vintage	-0.257*** (0.068)	-0.240*** (0.075)	-0.115** (0.055)	-0.141** (0.071)
Type × class × basis-method FE	X	X		
Manufacturer FE			X	
Manufacturer × age FE				X
Filing-year FE	X	X	X	X
Common asset-age FE	X	X	X	Absorbed
Equipment mentions	147,172	110,453	110,453	110,453
Cells	17,808	14,576	46,465	46,465
Manufacturers	–	2,945	2,945	2,945
Vintage clusters	66	66	66	66

Notes: The outcome is the number of used-equipment mentions in each observed UCC cell. Bonus vintage equals one when the asset's model-year purchase vintage received any positive federal bonus depreciation and zero otherwise. All columns use asset ages greater than zero and filing years 2001–2017. Columns (1) and (2) aggregate mentions to equipment-type × MACRS-class × basis-method × filing-year × asset-age × model-year cells. Columns (2)–(4) use the same manufacturer-observed mentions. Columns (3) and (4) instead form cells within manufacturer × equipment-type × MACRS-class × basis method; column (4) absorbs manufacturer-specific age profiles. Estimates are PPML coefficients. Standard errors clustered by model-year vintage are in parentheses. Reported significance levels use a t distribution with the number of vintage clusters minus one degrees of freedom. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Column (1) reports the broad comparison using equipment-type-by-MACRS-class-by-basis-schedule cells. Bonus-vintage equipment appears approximately 23 percent less often in subsequent used asset transactions than nonbonus-vintage equipment. Column (2) restricts the sample to observations for which I identify a manufacturer and yields a similar estimate. Column (3) then compares transactions within manufacturers, which attenuates the implied difference to approxi-

mately 11 percent. Finally, column (4) allows each manufacturer to have its own asset-age profile. This is my preferred specification and implies that bonus-vintage equipment appears in approximately 13 percent fewer subsequent transactions. While the results attenuate when moving from broad categories to the within-manufacturer design, the differences remain economically and statistically significant, with the sign predicted by the lock-in effect.

The appendix addresses three additional concerns. First, the baseline indicator pools vintages receiving different dosages of bonus depreciation. When restricting to a comparison between only 50-percent and zero-bonus vintages, I find a substantially similar result in Appendix Table D.5. Second, some filings may be repeated and that may cause UCC counts to overstate distinct transactions, but I find that retaining only initial filings or the first observed asset-borrower pair does not change the estimate. Third, Appendix Table D.4 tests whether the results are an artifact of the parser or of sample selection. The estimate remains similar when I exclude age one, omit model-year 2001, exclude farm equipment or trailers, and restrict the sample to equipment receiving direct high-confidence MACRS assignments. Restricting the sample to transactions explicitly described as used produces a smaller and less precise estimate, although this text-based restriction retains only 5,260 equipment items. The result also remains similar when I count trackable machines, unique asset-event appearances, first observed asset-borrower pairs, or first-borrower pairs identified using validated VINs.

At first glance, the results may appear to differ from those in Basu, Kim, and Singh (2024), who find that bonus depreciation increases used asset transactions and lowers their prices. However, my analysis studies a different margin of reallocation than theirs. Their estimates, which also rely on UCC data, capture a replacement margin in which machines purchased prior to bonus are sold as firms replace them with subsidized new capital. That margin does not distinguish between bonus and nonbonus vintages.¹¹ In contrast, my analysis tests whether lock-in subsequently reduces the circulation of the newly created cohort. Bonus may therefore increase contemporaneous sales of inherited machines while reducing the later circulation of the bonus-treated machines that replace them. Moreover, their increase in quantities is not necessarily inconsistent with the Compustat decline in sale proceeds because they also find that used capital prices fall.

However, there is an additional concern with the design. Since there is only variation across purchase vintages over time, I cannot absorb vintage fixed effects. Consequently, the estimates may partly reflect differences in the initial size of purchase cohorts or other economic conditions associated with their purchase years. These concerns point in opposite directions. Bonus is typically used as a stimulative policy and is therefore endogenous to the business cycle, while it also

11. A closely related replacement mechanism appears in automobile markets: Berger et al. (2026) estimate that pandemic transfers raised both new and used car sales and use a quantitative model in which new car purchases generate trade-ins that expand used car supply.

would naturally generate larger cohorts.

Table 3: Bonus Depreciation and Used-Equipment Transactions Around the 2005–07 Lapse

	Baseline	Supported sample	Cohort control	Completed cells
	(1)	(2)	(3)	(4)
Bonus relative to 2005–07 lapse	-0.173*	-0.174*	-0.094	-0.145*
	(0.081)	(0.083)	(0.056)	(0.062)
Exact age-zero cohort support		X	X	X
Log age-zero cohort-size control			X	X
Supported zero cells completed				X
Manufacturer $\times$ age FE	X	X	X	X
Filing-year FE	X	X	X	X
Equipment mentions	29,768	25,496	25,496	25,061
Cells	10,076	7,107	7,107	11,780
Manufacturers	784	550	550	425
Vintage clusters	8	8	8	8

Notes: Each column reports a PPML regression of UCC equipment counts for purchase vintages 2003–10 and asset ages 1–7. The coefficient compares the surrounding 2003–04 and 2008–10 bonus vintages with the 2005–07 lapse. Columns (2)–(4) require the manufacturer purchase vintage to appear at age zero; columns (3)–(4) control flexibly for its log age-zero count, and column (4) fills supported filing-year-by-age cells with zero transactions. Standard errors clustered by purchase vintage are in parentheses.

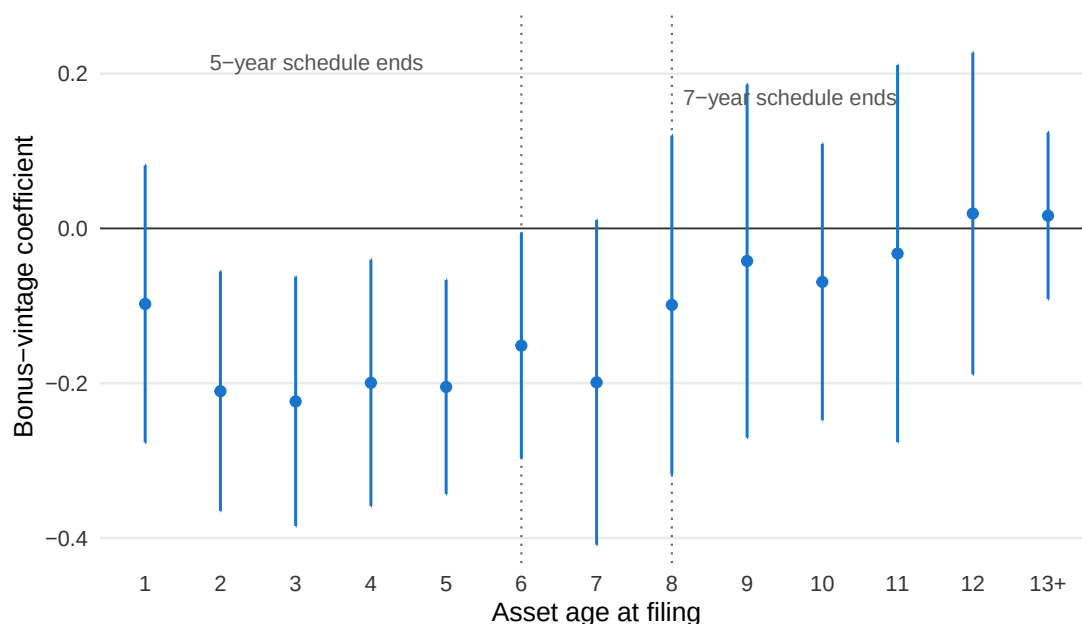
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ using t_{G-1} .

Table 3 addresses the main concern by exploiting the temporary lapse in bonus depreciation from 2005 through 2007. Column (1) compares the 2005–07 lapse vintages at ages one through seven with the surrounding 2003–04 and 2008–10 bonus vintages, and I find that bonus vintages transact approximately 16 percent less than lapse vintages. Requiring exact age-zero cohort support leaves the estimate essentially unchanged in column (2). Flexibly controlling for the age-zero UCC count attenuates the coefficient from -0.174 to -0.094 in column (3), while completing supported zero-transaction cells yields a coefficient of -0.145 in column (4). Across columns, the estimates remain negative, although precision is limited with only eight purchase-vintage clusters. Appendix Table D.6 shows that controlling for quadratic or cubic purchase-vintage cur-

vature does not attenuate the estimate, while also showing that the 2005–07 lapse provides its principal identifying variation.

The results so far show that a pooled regression over ages identifies a large difference in transaction rates between bonus and nonbonus vintages, but the model predicts that we should observe a difference by age. Bonus and nonbonus vintages differ in remaining tax basis only until the ordinary depreciation schedule has been exhausted. Once both schedules are exhausted, the difference in their effective tax treatment on sale disappears, so the tax-related component of the transaction gap should disappear. I test that subsequently in Figure 2.

Figure 2: The Bonus-Vintage Transaction Gap by Asset Age



Notes: The figure plots PPML coefficients comparing bonus and nonbonus vintages at each asset age. Negative coefficients indicate fewer transactions involving bonus vintages. The specification includes manufacturer-by-asset-age and filing-year fixed effects. Standard errors are clustered by model-year purchase vintage, and the bars report 95% confidence intervals. The dotted lines mark ordinary basis exhaustion for five- and seven-year MACRS property.

Figure 2 estimates the bonus-nonbonus transaction gap by asset age and shows an age profile consistent with theory. Since the sample is restricted to five- and seven-year assets, the gap should go away after age seven in the data. I observe exactly that profile: the coefficients from ages two through seven are negative and statistically indistinguishable from one another, averaging -0.198 . From age eight onward, they average only -0.034 and are not jointly distinguishable from zero. Appendix Table D.7 shows that the difference between the early- and late-age averages is -0.164 and statistically significant at $p = 0.008$. Consequently, the age profile supports the prediction that the tax-related transaction gap disappears once bonus and nonbonus equipment

no longer differ meaningfully in remaining basis.

Jointly, the firm-level evidence in Compustat and the asset-level evidence from UCC filings in Texas are supportive of the lock-in channel dominating the capital stock expansion channel, so investment subsidies do not necessarily promote reallocation. The next two sections show why this matters theoretically by linking investment subsidies to entry.

7 Reallocation and Entry in General Equilibrium

Empirically, bonus reduces measures of reallocation across two datasets. The downstream effects of that depend on who buys used capital. Ma, Murfin, and Pratt (2022) show that the typical transaction involves an incumbent firm selling to a young and financially constrained firm. For example, we may observe J.B. Hunt selling an older truck to a newly established owner-operator who may not have the financial capacity to purchase a brand-new truck. This establishes a direct link between reallocation and entry. In the partial equilibrium model, the price of used capital q was held fixed. This section closes the used capital market by introducing a fringe of potential entrants and allowing q to adjust. I show that this price adjustment attenuates but does not reverse the partial-equilibrium response of reallocation, thereby linking the effect of investment subsidies on capital sales to their effect on entry.

7.1 Closing the Model with Entrants

There are two types of firms: a unit mass of incumbents and a unit mass of potential entrants. The incumbents are exactly as in Section 3. They continue to take q as given, but q will now be determined by market clearing.

To close the used capital market, I assume that each period the potential entrants decide whether to operate. An active entrant purchases U physical units of used capital from incumbents and produces for one period according to $h(\phi U)$, after which the machines retire. Thus, each physical unit supplies the same ϕ efficiency units whether it is retained by the incumbent or sold to an entrant. This reflects evidence from Ma, Murfin, and Pratt (2022) that the buyers of used capital are young, financially constrained firms who struggle to acquire new capital, though of course it is a simplification. The assumption of homogeneous technology is likewise a simplification, though it may be reasonable in the relevant settings. In the UCC data I use, the most active industries are in agriculture, trucking, and related fields, while Kermani and Ma (2022) show that industries like agriculture have the most liquid secondary markets for used capital. Those industries have relatively homogeneous entrants.

Entrants do not accumulate productive capital or become incumbents. Although their deci-

sion is repeated each period, the entrant problem is static, so I suppress time subscripts throughout this section. Each potential entrant draws a deductible fixed operating cost c from a continuous distribution G on $\mathbb{R}_+$; that cost governs entry. To ensure that entrants are well-behaved, I make assumptions about both h and G :

Assumption 4 (Entrant Regularity). *h and G are governed by the following regularity conditions:*

- (i) *The function $h : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is continuous on $\mathbb{R}_+$, twice continuously differentiable and strictly increasing on $\mathbb{R}_{++}$, and satisfies $h''(x) < 0$ for every $x > 0$, $h(0) = 0$, and*

$$\lim_{x \downarrow 0} h'(x) = \infty, \quad \lim_{x \rightarrow \infty} h'(x) = 0.$$

- (ii) *G is continuously differentiable with positive density over its support, with $G(0) = 0$ and $0 < G(c) < 1$ for every finite $c > 0$.*

Entrants are taxed at the same corporate tax rate as incumbents. When an entrant purchases an asset, it receives a new tax basis equal to its purchase price q . The entrant may deduct a share $\theta^U \in [0, 1]$ of q in the current period and the remainder when the capital is retired in the following period. I hold θ^U fixed throughout the main analytical results but discuss comparative statics for it in the appendix. Conditional on operating, the entrant solves

$$\pi(c) = \max_{U \geq 0} \left\{ (1 - \tau) [h(\phi U) - c] - q(1 - \tau \theta^U)U + \beta \tau (1 - \theta^U)qU \right\}. \quad (11)$$

An active entrant's interior choice satisfies

$$\phi h'(\phi U) = q \left[1 + \frac{\tau(1 - \beta)}{1 - \tau} (1 - \theta^U) \right]. \quad (12)$$

The right-hand side is the tax-adjusted cost of used capital.

The purchase price is q , while the second term reflects the cost of postponing the deduction for the share $1 - \theta^U$. This term is fixed in the bonus experiment below. Thus, bonus depreciation for new capital does not shift entrant demand directly; it affects entrants through the equilibrium used price q .

Because c is a fixed cost, it does not affect the optimal choice of U , and all active entrants purchase the same quantity. Using (12), the marginal entrant's zero-profit condition is

$$\bar{c} = h(\phi U) - \phi h'(\phi U)U. \quad (13)$$

An entrant operates if and only if $c \leq \bar{c}$. Entrants below the cutoff earn positive profits, while only the marginal entrant earns zero. The active mass, and hence the entry rate, is therefore $\lambda = G(\bar{c})$,

and aggregate demand for used capital is λU .

7.2 Stationary Competitive Equilibrium

To obtain transparent analytical comparative statics, I consider constant policies and evaluate the incumbent's dynamic problem along a stationary path. Prices and allocations are then constant, while the incumbent continues to account for the future resale value of new investment.

Definition 1 (Stationary competitive equilibrium). Given constant policy parameters (τ, θ, θ^U) , a stationary competitive equilibrium consists of a set of allocations $\{K^n, K^o, S, U, \bar{c}, \lambda\}$ and a price q such that, taking q as given, the constant choices $K_t^n = K^n$ and $S_t = S$ solve the incumbent's problem from the stationary inherited stock $K_t^o = K^n$, with retained old capital $K^o = K^n - S$.

Conditional on operating, each entrant optimally chooses U , the entry cutoff is given by (13), and $\lambda = G(\bar{c})$. Finally, the used capital market clears:

$$S = \lambda U. \quad (14)$$

Given policies, individual optimization maps each used price q into incumbent sales $S(q)$ and aggregate entrant demand

$$D(q) \equiv \lambda(q)U(q).$$

The equilibrium price equates these two quantities. To ensure that the resulting supply and demand schedules have a unique interior crossing, I impose the following condition.

Assumption 5 (Stationary used capital market). *The set of used prices at which the stationary incumbent and entrant problems admit finite interior solutions is an interval. At every incumbent allocation induced by a price in this interval,*

$$F_{11} + \phi(1 + \beta)F_{12} + \beta\phi^2F_{22} < 0, \quad (15)$$

where the derivatives of F are evaluated at $(K^n, \phi K^o)$. Moreover, aggregate used capital demand exceeds incumbent sales at some price in this interval, while incumbent sales exceed aggregate demand at a higher price.

Together with the entrant regularity conditions, the assumption guarantees that the supply of used assets slopes upward and the demand for them slopes downward. Along a stationary path, q is both the current sale price of old capital and the anticipated resale price of new investment. Since q enters the incumbent's two optimality conditions, investment and retention therefore adjust together whenever the price changes. Condition (15) governs their net effect on incumbent sales by ensuring that incumbent supply slopes upward, which holds automatically

when new and retained old capital are substitutes and limits the strength of their complementarity otherwise. By contrast, the regularity conditions for entrants ensure that a higher q raises the tax-adjusted cost of used capital, reducing purchases per entrant, the entry cutoff, and participation, so that aggregate demand slopes downward. The final part of the assumption ensures that the two schedules cross in the interior, which yields the following result.

Lemma 1 (Stationary equilibrium). *Under Assumptions 2, 4, and 5, there exists a unique interior stationary competitive equilibrium. In particular,*

$$K^n > 0, \quad K^o > 0, \quad 0 < S < K^n, \quad U > 0, \quad 0 < \lambda < 1.$$

The proof is in Appendix B.4, which also shows that the equilibrium remains interior under sufficiently small changes in the policy parameters. This allows the stationary equilibrium conditions to be differentiated in the comparative statics below.

7.3 Reallocation and Entry in General Equilibrium

In partial equilibrium, I established that investment subsidies have ambiguous effects on reallocation. Closing the capital market links that ambiguity to entry. I start by showing that a perturbation to the economy moves reallocation and entry in the same direction.

Proposition 2 (Reallocation and entry). *Under a local change in the model's policy parameters around an interior stationary equilibrium, S , U , $\bar{c}$, and λ move in the same direction.*

Proof. See Appendix B.5.

Intuition for the result comes from recognizing that the entrant side of the market can be summarized by purchases per entrant U . Since entrants are homogeneous except over fixed costs, all active participants choose the same quantity of used capital U . The cutoff condition (13) then maps U into the fixed cost of the marginal entrant, which earns no profits, and concavity implies

$$\frac{d\bar{c}}{dU} = -\phi^2 U h''(\phi U) > 0.$$

A higher U raises the cutoff $\bar{c}$ and therefore the mass of active entrants $\lambda = G(\bar{c})$, so aggregate entrant demand λU is monotonically increasing in U . By market clearing, capital sales, purchases per entrant, the entry cutoff, and the entry rate must all move in the same direction. Consequently, once the general equilibrium response of reallocation is known, the response of entry follows from Proposition 2.

Given the linkage between reallocation and entry, I now show how the partial equilibrium Proposition 1, which links accelerated depreciation policies to reallocation, translates into general

equilibrium. In general equilibrium, the proposition survives, but it is attenuated. To see that, let

$$S_{\theta}^{PE} \equiv \left. \frac{\partial S(q, \theta)}{\partial \theta} \right|_{q \text{ fixed}}$$

denote the partial equilibrium response of reallocation in Proposition 1. Because market clearing requires $S(q, \theta) = D(q)$, differentiating gives

$$\frac{dq}{d\theta} = \frac{S_{\theta}^{PE}}{D_q - S_q}, \quad \frac{dS}{d\theta} = \frac{D_q}{D_q - S_q} S_{\theta}^{PE}, \quad (16)$$

where $0 < D_q/(D_q - S_q) < 1$ since $S_q > 0$ and $D_q < 0$. Thus, the equilibrium price response attenuates but does not reverse the partial equilibrium response of capital sales.

Corollary 1 (Bonus depreciation and entry). *Around an interior stationary equilibrium, under the assumptions of Proposition 1 and Assumptions 4 and 5, a marginal increase in bonus depreciation raises investment in new capital. When the lock-in channel dominates in Proposition 1, bonus reduces capital sales, purchases per entrant, the entry cutoff, and the entry rate, while raising the used price. When the capital stock expansion channel dominates, these responses are reversed.*

Proof. See Appendix B.6.

If lock-in dominates, then bonus contracts incumbent supply. The resulting excess demand from entrants raises q , which moderates the initial contraction but is not enough to overcome it. Consequently, if lock-in dominates, then entry declines as well. Since the evidence from Compustat and used asset transactions in Texas correspond to the case in which lock-in dominates, the model predicts that entry should also decline, particularly where potential entrants rely more heavily on used capital. The next section tests that prediction.

8 Consequences for Firm Entry

The preceding evidence shows that (i) bonus episodes induce greater investment and reduce capital sales and (ii) bonus-vintage capital transacts less than nonbonus capital. By Proposition 2, that evidence suggests that entry should decline because reallocation and entry are linked. This section explores that with data from the Business Dynamics Statistics.

8.1 Data

I use the Business Dynamics Statistics (BDS), published annually by the Census Bureau from the Longitudinal Business Database, to study entry. The BDS provides industry-level counts of firms,

establishments, entry, and exit for the near-universe of U.S. employer businesses. I use it to test the effects of bonus depreciation on entry, focusing on the 4-digit NAICS level. Within the BDS there are several options for defining entry. For example, one could look at the startup rate for firms or establishments, or one could analyze a stock-flow measure. Guided by the model, in which entry is not necessarily from newly created legal entities, I focus on a stock-flow measure defined by the flow of firms into an industry’s active population rather than only the number of firms legally born that year. Letting $F_{N,t}$ denote the number of firms operating in industry N and $D_{N,t}$ the number of firm deaths, the entry rate is

$$\frac{F_{N,t} - F_{N,t-1} + D_{N,t}}{F_{N,t-1}}.$$

At the four-digit industry level, this includes both de novo startups and the net entry of continuing firms into an industry, which is consistent with the broader participation margin in the model. By contrast, the BDS count of age-zero firms captures only de novo startups and therefore imposes a narrower definition of entry than the model requires. Nevertheless, I report robustness to other definitions of the entry rate in the appendix. To study the effects of bonus on entry, I merge the BDS panel with industry-level bonus depreciation exposure from Zwick and Mahon (2017). This yields 4,984 industry-years across 227 industries from 1996–2017. Appendix E describes the construction in detail. I also link the data to a time-invariant measure of secondary asset market liquidity from Kermani and Ma (2022). Summary statistics appear in Table E.1.

8.2 Results

The empirical effect of bonus depreciation on entry nevertheless need not be negative. Entrants purchase only used equipment in the model, whereas in practice some potential entrants can purchase new equipment and benefit directly from bonus depreciation. The estimates therefore capture the net effect of these competing forces. The used capital mechanism also has a more specific implication: any decline in entry should be larger in industries where capital is more readily traded on secondary markets.

I test these predictions using panel local projections of the four-digit NAICS industry entry rate on the present value of depreciation deductions $\tilde{z}_{N,t}$ from Zwick and Mahon (2017). I estimate

$$y_{N,t+h} - y_{N,t-1} = \alpha_N + \gamma_t + \beta_h \tilde{z}_{N,t} + \text{Controls} + \varepsilon_{N,t}, \quad (17)$$

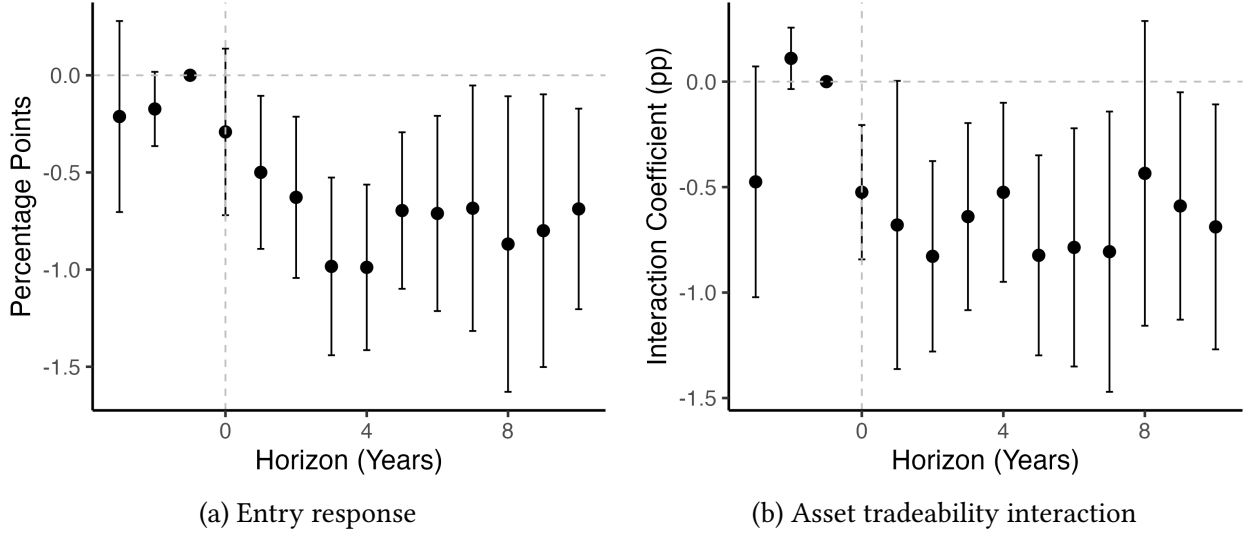
where $y_{N,t}$ is the stock-flow entry rate defined above, α_N and γ_t are industry and year fixed effects, and the controls include two-digit NAICS quadratic time trends. The identifying variation comes from national changes in bonus depreciation interacted with predetermined differences in the

assets used by each industry. The identifying assumption is therefore that industries with different exposure would otherwise have experienced similar changes in entry after accounting for the fixed effects and industry trends. Standard errors are clustered by four-digit NAICS industry. Because bonus policies are persistent, the coefficients trace the dynamic reduced-form response to greater depreciation generosity rather than the effect of an isolated one-year policy shock.

Figure 3 reports the results. Panel (a) shows that the two pre-period coefficients are not jointly significant, and neither is individually significant at the five-percent level. The entry response is negative on impact but initially imprecise. After one year, it becomes significantly negative, declines to -1 at horizons three and four, and remains significantly negative thereafter. Since identification comes from cross-sectional variation in exposure across industries, a coefficient of -1 means that an industry experiencing a one-percentage-point larger increase in $\tilde{z}$ experiences a one-percentage-point larger decline in its entry rate, relative to a less-exposed industry. The delayed response is also consistent with the timing of the secondary market mechanism, which predicts a gradual buildup in the response.

While the estimate is economically large, it is consistent with other evidence on the effects of tax policy on entry. In the context of our sample, a 50-percent bonus episode would cause the interquartile range of $\tilde{z}$ to increase by approximately 0.9 percentage points. Three years out from the shock, a coefficient of roughly -1 would imply a 0.9-percentage-point larger cumulative decline in entry in a more-exposed industry, relative to a mean entry rate of 7.9 percent. This corresponds to a relative decline of approximately 11 percent. By comparison, Sapollnik and Swonder (2025) find that state corporate tax cuts raise entry by approximately 10 percent after five years, while Fairlie et al. (2026) find that large corporate tax cuts raise employer startups by approximately 5 percent. Thus, my estimate is similar in absolute magnitude, although it has the opposite sign. Nevertheless, that may result from differences in both the policy experiments and level of variation.

Figure 3: Bonus Depreciation and Firm Entry



Notes: Panel (a) plots coefficients from panel local projections of the four-digit NAICS entry rate on the present value of depreciation deductions $\tilde{z}_{N,t}$. Panel (b) plots coefficients on the interaction of $\tilde{z}_{N,t}$ with the demeaned PPE recovery rate from Kermani and Ma (2022), expressed as a fraction. All specifications include industry and year fixed effects and two-digit NAICS quadratic time trends. Bars report 95% confidence intervals using standard errors clustered by four-digit NAICS industry.

The baseline response is consistent with the model, but it does not by itself establish that used capital is the operative channel. Panel (b) therefore asks whether the decline is larger in industries where secondary markets are more important. I measure asset tradeability using the PPE recovery rate from Kermani and Ma (2022), which reports the fraction of asset value recoverable upon liquidation. Industries with high recovery rates use standardized capital that retains substantial value in resale, while industries with low recovery rates use more specialized capital with limited secondary market value. The interaction is negative across the post-reform horizons and is statistically significant over much of the medium run. The two pre-period interaction coefficients have opposite signs and neither is statistically distinguishable from zero at the five-percent level. Thus, the decline in entry is larger in industries where capital is more readily traded, as the secondary market mechanism predicts.

Recovery rates may also be related to other features of an industry's capital and firms. I therefore estimate specifications that add interactions with average firm size, capital duration, entry cyclicity, and external finance dependence, both individually and jointly. Appendix Figure E.1 reports representative specifications, and the medium-run tradeability interaction remains negative. The establishment-level estimates are directionally similar, although establishment entry also captures expansion by incumbent firms, so I do not treat them as a separate test of firm entry. Finally, Appendix Figure E.3 shows that the decline is concentrated among firms aged zero

to five, while the stock of older firms changes little. Because the stock of young firms can also change through exit and aging, I interpret this decomposition as supporting evidence rather than a separate measure of firm entry.

These findings may initially appear to contrast with Basu, Kim, and Singh (2024), who find that bonus depreciation increases the number of establishments with five to nine employees. However, they capture the contemporaneous response of a stock measure of establishment entry across regions and industries rather than a dynamic, national, and industry-level analysis of entry flows, which is what I study.

Altogether, I find that the model's prediction about entry holds, namely that industries more exposed to bonus see larger declines in entry, and those declines seem to be mediated by secondary markets, although these patterns are more suggestive than conclusive. This pattern is consistent with the model's prediction that reduced circulation makes used capital more costly and discourages entry by marginal firms.

9 Aggregating the Effects of Bonus on Reallocation

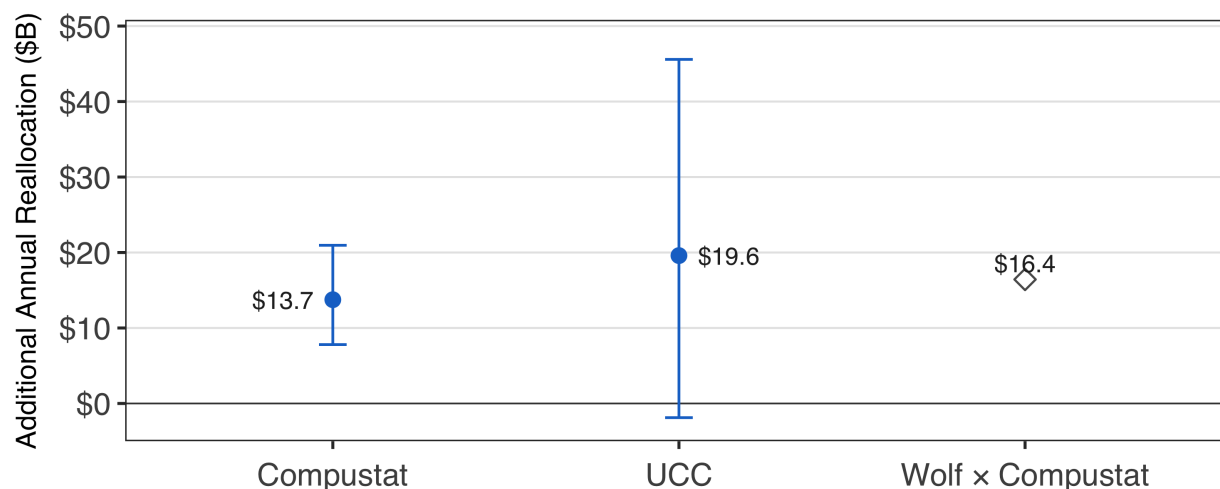
The empirical results largely conform to the theoretical model and suggest that bonus depreciation reduces capital reallocation. From 2012 through most of 2017, bonus depreciation was a constant 50 percent. Over the same time period, average annual reallocation, as measured by aggregate gross sale proceeds for Property A on IRS Form 4797, was roughly \$104 billion. I use that figure as a baseline to gauge how much extra reallocation my estimates imply there may have been under lower-bonus counterfactuals from 2012–17 via three different back-of-the-envelope calculations. All three should be interpreted as back-of-the-envelope accounting exercises rather than estimates of the full general equilibrium response.

The first exercise is a deliberately conservative calculation in the spirit of Zwick and Mahon (2017). I assign every industry the bonus increment received by the least-exposed quintile of industries while preserving its ordinary MACRS depreciation schedule. Applying the Compustat capital sales coefficient from Column 1 of Table 1 implies that this would increase reallocation by \$13.7 billion relative to observed policy, with a 95% confidence interval spanning \$7.8 billion to \$21 billion.

As a next estimate, I use the direct comparison in transactions between vintages receiving 50-percent bonus and zero-bonus vintages from Column 1 of Table 3. That specification implies that zero-bonus vintages transact about 19 percent more than vintages with 50 percent bonus. Applied to the baseline level of reallocation and holding the average value of used asset transactions fixed, this implies that reallocation would rise by roughly \$20 billion, though with a 95% confidence interval that includes zero.

Finally, I compute a third point of comparison using Koby and Wolf (2020) and Wolf (2023). The former supplies a partial equilibrium path of the investment response to bonus depreciation, while the latter uses demand equivalence to estimate the corresponding general equilibrium investment path in his macroeconomic environment.¹² I multiply the first year of that general equilibrium path by a lagged-sales/current-investment response ratio estimated on a common Compustat sample, following the timing in Appendix Table C.2, to estimate the response of reallocation to bonus. The resulting estimated \$16.4 billion in additional reallocation should be interpreted cautiously for four reasons. First, the sales-to-investment response ratio is treated as a partial equilibrium object and held fixed even though, in general equilibrium, both capital sales and investment respond to the used price. Second, a ratio of two coefficients blows up standard errors, so I do not propagate uncertainty. Third, the first year of the investment path is the largest in Wolf’s impulse response function, so it overstates the persistent change in bonus from 2012–17. Finally, the mapping from partial to general equilibrium relies on a particular macroeconomic closure and invariance conditions in Wolf (2023) which need not hold in this setting. Hence, it should largely be interpreted as a sensitivity exercise.

Figure 4: Economic Magnitude of the Reallocation Response



Notes: Dollar values use average annual Form 4797 Part III Property A gross sale proceeds from 2012 through 2017. Confidence intervals propagate coefficient uncertainty only. No interval is reported for the Wolf × Compustat calculation because the sales-to-investment ratio is imprecisely estimated.

Across the three approaches, the point estimates are on the same order of magnitude and imply roughly \$13–\$20 billion in additional reallocation per year under lower-bonus or no-bonus

12. The partial equilibrium investment and output responses determine the excess-demand path created by the investment subsidy. Wolf (2023) chooses a weighted combination of empirically identified demand shocks that generates the same path and adds their investment response to the partial equilibrium response to obtain the general equilibrium path.

counterfactuals relative to observed policy from 2012–17. While these estimates are not the general equilibrium effect of reducing bonus, they do point toward an economically meaningful effect of bonus depreciation on reallocation.

10 Conclusion

This paper studies how the tax system shapes capital reallocation. I show that tax policies which stimulate new capital formation may simultaneously impede its subsequent reallocation. That ambiguity arises from the actual implementation of investment subsidies through the tax code. Accelerated depreciation lowers the after-tax cost of purchasing new capital, generating a capital stock expansion channel, but it also leaves the subsidized asset with less tax basis when it later reaches the secondary market, generating a lock-in channel that strengthens the incentive to retain it. The general equilibrium model shows that equilibrium price adjustment attenuates but does not reverse the resulting response of capital sales and links reductions in reallocation to reductions in entry. Thus, a policy’s effect on investment alone does not reveal its effect on the allocation of capital.

I find consistent evidence for the lock-in channel dominating across three datasets. In Compustat, greater exposure to bonus depreciation raises investment while reducing capital sales. Looking at used asset transactions in Texas, I find that equipment from bonus vintages subsequently transacts approximately 13 percent less than equipment from nonbonus vintages. The transaction gap is concentrated at ages two through seven and attenuates among older assets, when differences in remaining tax basis are less relevant. Aggregated upward, my estimates imply approximately \$13–\$20 billion of additional reallocation per year under lower-bonus or no-bonus counterfactuals. This lock-in propagates through to industries more exposed to bonus depreciation, leading to persistent reductions in entry.

In sum, investment subsidies can create more capital while causing less of that capital to be subsequently reallocated. This matters because tax policy is a potentially important instrument to reduce the misallocation of capital and may ease the financial constraints of young entrants through a larger supply of used assets. Nevertheless, fully answering exactly how much the reallocation channel matters would require establishing the aggregate effects of the policy, characterizing optimal policy, and quantifying its consequences for welfare. This paper establishes that future studies of those areas would be important to undertake.

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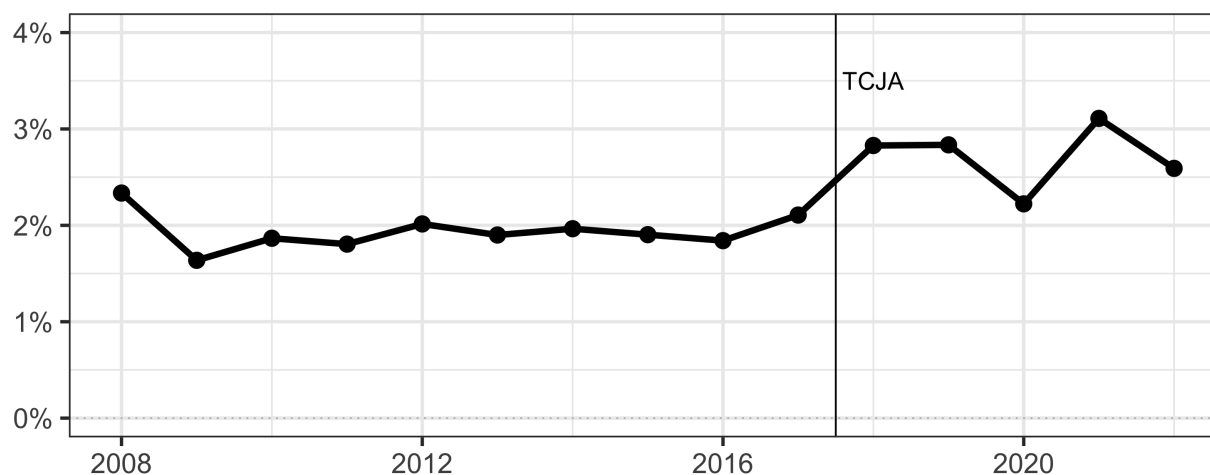
A Institutional Detail

Table A.1: Depreciation of Used and New Five-Year Assets under 50% Bonus for New Capital

Year:	0	1	2	3	4	5	Total	z_i
<i>Used assets</i>								
Deductions (000s)	200	320	192	115	115	58	1,000	
Tax benefit ($\tau = 35\%$)	70	112	67.2	40.3	40.3	20.2	350	0.890
Tax Benefit (Present Value)	70	104.7	58.7	32.9	30.7	14.4	311.4	
<i>New Assets (50 percent bonus)</i>								
Deductions (000s)	600	160	96	57.5	57.5	29	1,000	
Tax benefit ($\tau = 35\%$)	210	56	33.6	20.2	20.2	10	350	0.945
Tax Benefit (Present Value)	210	52.3	29.3	16.5	15.4	7.1	330.7	

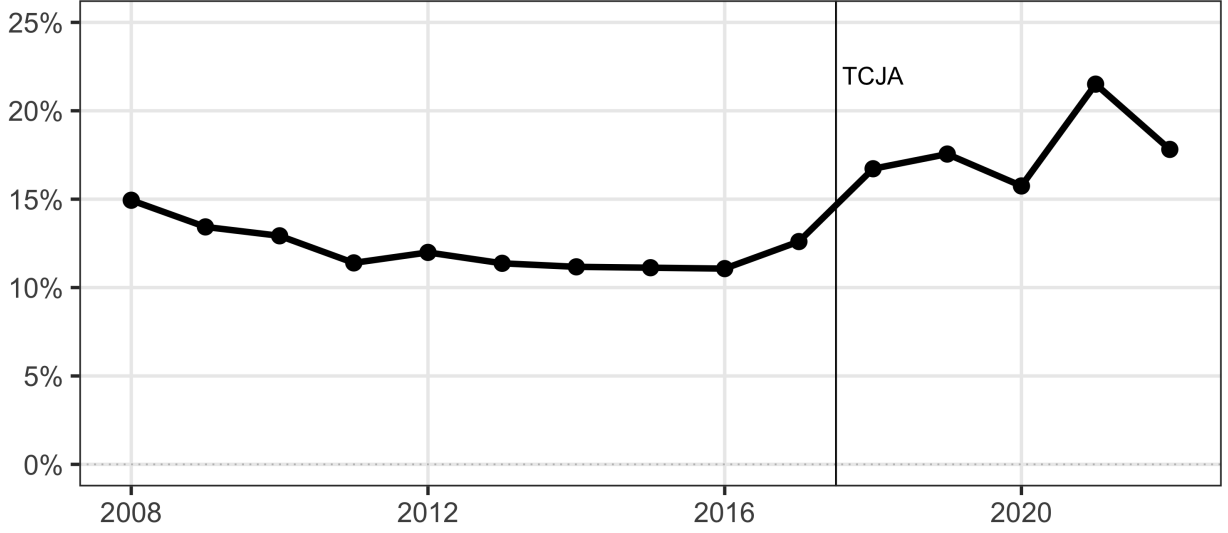
Notes: This table adapts Table 1 from Zwick and Mahon (2017). The used asset follows ordinary MACRS, while the new asset receives 50% bonus depreciation with the remainder deducted under MACRS. Tax savings use a 35% tax rate and a 7% discount rate. The z_i column gives the present value of depreciation deductions per dollar of investment.

Figure A.1: Used Equipment Sales Relative to Total Equipment Capital Stock (%)



Note: Used equipment sales are aggregate Section 1245 dispositions from Form 4797 as published in Statistics of Income Publication 5108. The equipment capital stock is from Fixed Asset Table 4.1 Line 18. The 2015 value of the Section 1245 series is linearly interpolated.

Figure A.2: Used Equipment Sales Relative to New Investment (%)



Note: Used equipment sales are aggregate Section 1245 dispositions from Form 4797 as published in Statistics of Income Publication 5108. New investment is from Fixed Asset Table 4.7 Line 18. The 2015 value of the Section 1245 series is linearly interpolated.

B Theoretical Derivations

This appendix derives the main theoretical results. Throughout, I evaluate all derivatives of the production function with respect to its two arguments at the stationary allocation. Define

$$\Delta \equiv F_{11}F_{22} - F_{12}^2.$$

Assumption 2 implies $F_{11} < 0$, $F_{22} < 0$, and $\Delta > 0$. It also implies that the Hessian with respect to the two physical inputs, K^n and K^o , is

$$H_\phi = \begin{pmatrix} F_{11} & \phi F_{12} \\ \phi F_{12} & \phi^2 F_{22} \end{pmatrix}, \quad \det H_\phi = \phi^2 \Delta > 0.$$

B.1 The No-Secondary-Market Benchmark

Derivation of (4). Write $x = K_{t-1}^n$ for inherited capital and abbreviate $d_j = d_j(\theta)$. The first-order condition for new investment in (3) is

$$0 = (1 - \tau)F_1(K_t^n, \phi x) - (1 - \tau d_0) + \beta V_x^N(K_t^n). \quad (\text{B.1})$$

The envelope condition is

$$V_x^N(x) = (1 - \tau)\phi F_2(K_t^n, \phi x) + \tau d_1 + \beta \tau d_2. \quad (\text{B.2})$$

At an interior stationary allocation, $K_t^n = x = K^n$. Substituting (B.2) into (B.1) and rearranging gives

$$F_1(K^n, \phi K^n) + \beta \phi F_2(K^n, \phi K^n) = \frac{1 - \tau [d_0 + \beta d_1 + \beta^2 d_2]}{1 - \tau},$$

which is (4).

To obtain the comparative static, let

$$L(K) \equiv F_1(K, \phi K) + \beta \phi F_2(K, \phi K)$$

and let $u(\theta)$ denote the right-hand side of (4). The bonus schedule in (2) implies

$$u'(\theta) = -\frac{\tau}{1 - \tau} [(1 - \beta)\bar{d}_1 + (1 - \beta^2)\bar{d}_2].$$

Thus $u'(\theta) < 0$ when $\tau > 0$, $\beta < 1$, and some ordinary depreciation is delayed. Moreover,

$$L'(K) = F_{11} + \phi(1 + \beta)F_{12} + \beta\phi^2 F_{22}, \quad (\text{B.3})$$

where the derivatives are evaluated at $(K, \phi K)$. Consequently, whenever the regularity condition $L'(K) < 0$ holds, the implicit function theorem gives

$$\frac{dK^n}{d\theta} = \frac{u'(\theta)}{L'(K^n)} > 0.$$

The sign restriction on $L'(K)$ is not implied by concavity alone because the current and future marginal products receive different discount weights. $\square$

B.2 Incumbent Optimality with a Secondary Market

Derivation of (6) and (7). Let $x = K_{t-1}^n$, let $\tilde{K}_t^o = x - S_t$, and again abbreviate $d_j = d_j(\theta)$. At an interior sale decision, differentiating (5) with respect to S_t gives

$$0 = -(1 - \tau)\phi F_2(K_t^n, \phi \tilde{K}_t^o) + (1 - \tau)q + \tau(d_1 + d_2) - \tau d_1 - \beta \tau d_2.$$

It follows that

$$\phi F_2(K_t^n, \phi \tilde{K}_t^o) = q + \frac{\tau(1 - \beta)}{1 - \tau} d_2.$$

The envelope condition for inherited capital is

$$V_x(x) = (1 - \tau)\phi F_2(K_t^n, \phi \tilde{K}_t^o) + \tau d_1 + \beta \tau d_2.$$

Using the sale condition to substitute for the marginal value of retention reduces this expression to

$$V_x(x) = (1 - \tau)q + \tau(d_1 + d_2). \quad (\text{B.4})$$

The first-order condition for K_t^n is

$$0 = (1 - \tau)F_1(K_t^n, \phi \tilde{K}_t^o) - (1 - \tau d_0) + \beta V_x(K_t^n).$$

At a stationary allocation, substitute (B.4), use $d_0 + d_1 + d_2 = 1$, and write $K^o = \tilde{K}_t^o$. The result is

$$F_1(K^n, \phi K^o) = 1 + \frac{\tau(1-\beta)}{1-\tau} (d_1 + d_2) - \beta q.$$

Together with the sale condition above, this proves (6) and (7). $\square$

B.3 Bonus Depreciation in Partial Equilibrium

Proof of Proposition 1. Let

$$a \equiv \frac{\tau(1-\beta)}{1-\tau} > 0, \quad m \equiv d_1(\theta) + d_2(\theta).$$

For $\theta < 1$, the bonus schedule implies

$$d'_j(\theta) = -\bar{d}_j = -\frac{d_j(\theta)}{1-\theta}, \quad j = 1, 2.$$

Differentiate (6) and (7) with respect to θ , holding q fixed. The resulting system is

$$\begin{pmatrix} F_{11} & \phi F_{12} \\ \phi F_{12} & \phi^2 F_{22} \end{pmatrix} \begin{pmatrix} K_\theta^n \\ K_\theta^o \end{pmatrix} = -\frac{a}{1-\theta} \begin{pmatrix} m \\ d_2(\theta) \end{pmatrix}. \quad (\text{B.5})$$

Inverting the physical-input Hessian gives

$$K_\theta^n = \frac{a}{\phi \Delta (1-\theta)} [d_2(\theta) F_{12} - \phi m F_{22}], \quad (\text{B.6})$$

$$K_\theta^o = \frac{a}{\phi^2 \Delta (1-\theta)} [\phi m F_{12} - d_2(\theta) F_{11}]. \quad (\text{B.7})$$

Assumption 3 implies

$$d_2 F_{12} - \phi m F_{22} = d_2 (F_{12} - \phi F_{22}) - \phi d_1 F_{22} \geq 0.$$

Hence bonus weakly raises new investment. The response is strictly positive if $d_1(\theta) > 0$, or if $d_2(\theta) > 0$ and $F_{12} > \phi F_{22}$. If $m = 0$, the policy is locally irrelevant; if $d_1 = 0$ and $F_{12} = \phi F_{22}$, the investment response is the knife-edge zero case.

Because $S = K^n - K^o$, subtracting (B.7) from (B.6) gives

$$S_\theta^{PE} = \frac{a}{\phi^2 \Delta (1-\theta)} \{d_2(\theta) F_{11} - \phi d_1(\theta) F_{12} - \phi^2 [d_1(\theta) + d_2(\theta)] F_{22}\}. \quad (\text{B.8})$$

The factor outside braces is strictly positive. The expression in braces therefore gives the necessary and sufficient sign condition in (8). $\square$

B.4 Stationary Supply, Entrant Demand, and Equilibrium

Proof of Lemma 1. Fix the stationary policy parameters and a used price q in the interval specified by Assumption 5. Under that assumption, incumbent and entrant choices are finite and interior.

Differentiating the two stationary incumbent conditions with respect to q gives

$$\begin{pmatrix} F_{11} & \phi F_{12} \\ \phi F_{12} & \phi^2 F_{22} \end{pmatrix} \begin{pmatrix} K_q^n \\ K_q^o \end{pmatrix} = \begin{pmatrix} -\beta \\ 1 \end{pmatrix}.$$

Consequently,

$$K_q^n = -\frac{F_{12} + \beta\phi F_{22}}{\phi\Delta}, \quad (\text{B.9})$$

$$K_q^o = \frac{F_{11} + \beta\phi F_{12}}{\phi^2\Delta}, \quad (\text{B.10})$$

$$S_q = -\frac{F_{11} + \phi(1 + \beta)F_{12} + \beta\phi^2 F_{22}}{\phi^2\Delta} > 0, \quad (\text{B.11})$$

where the final inequality follows from Assumption 5. Thus stationary incumbent supply is continuous and strictly increasing in q .

For entrants, define

$$\chi \equiv 1 + \frac{\tau(1 - \beta)}{1 - \tau}(1 - \theta^U) > 0.$$

The entrant first-order condition is

$$\phi h'(\phi U) = q\chi.$$

Strict concavity and the Inada conditions imply that the conditional choice $U(q) > 0$ is unique. Differentiation gives

$$U_q = \frac{\chi}{\phi^2 h''(\phi U)} < 0. \quad (\text{B.12})$$

The marginal entrant's cutoff is

$$\bar{c}(U) = h(\phi U) - \phi h'(\phi U)U,$$

so

$$\bar{c}_U = -\phi^2 U h''(\phi U) > 0. \quad (\text{B.13})$$

Strict concavity and $h(0) = 0$ also imply $h(\phi U) - \phi U h'(\phi U) > 0$ for every $U > 0$. Hence $\bar{c} > 0$, and Assumption 4 gives $0 < G(\bar{c}) < 1$. Let $g = G'$ and define aggregate entrant demand by

$$D(q) \equiv G(\bar{c}(U(q)))U(q).$$

Because

$$D_U = G(\bar{c}) + U g(\bar{c}) \bar{c}_U > 0,$$

equations (B.12) and (B.13) imply

$$D_q = D_U U_q < 0. \quad (\text{B.14})$$

Excess demand $Z(q) = D(q) - S(q)$ is therefore continuous and strictly decreasing on the admissible price interval. The crossing condition in Assumption 5 puts opposite signs on Z at two prices in this interval. The intermediate value theorem gives an interior clearing price, and strict monotonicity makes it unique. The unique conditional choices then give a unique stationary allocation.

At the clearing price, $U > 0$, $\bar{c} > 0$, and $0 < \lambda = G(\bar{c}) < 1$. Market clearing gives $S = \lambda U > 0$. Interiority of the incumbent choice gives $K^n > 0$, $K^o > 0$, and, because $S = K^n - K^o$, $0 < S < K^n$. Finally,

$$Z_q = D_q - S_q < 0.$$

The nonsingularity of the firm-level Hessians and the implicit function theorem imply that the clearing price and allocation are continuously differentiable in sufficiently small admissible policy changes. Interiority is preserved locally. $\square$

B.5 Reallocation and Entry

Proof of Proposition 2. For the policy changes considered in the model, h and G are primitives. Equation (B.13) implies that the entry cutoff is strictly increasing in purchases per entrant. Therefore

$$\lambda_U = g(\bar{c})\bar{c}_U > 0.$$

Using market clearing,

$$S = G(\bar{c}(U))U,$$

and hence

$$\frac{dS}{dU} = G(\bar{c}) + U g(\bar{c})\bar{c}_U > 0.$$

It follows that, along any sufficiently small equilibrium response to the policy parameters of the model, S , U , $\bar{c}$, and λ move in the same direction. $\square$

B.6 Bonus Depreciation in General Equilibrium

Proof of Corollary 1. Let $S(q, \theta)$ denote stationary incumbent supply and let $D(q)$ denote aggregate entrant demand. Entrant demand has no direct dependence on θ because θ^U is held fixed. Define the fixed-price supply response

$$S_\theta^{PE} \equiv \left. \frac{\partial S(q, \theta)}{\partial \theta} \right|_{q \text{ fixed}},$$

which is given by (B.8). Market clearing requires

$$S(q, \theta) = D(q).$$

Differentiating at the initial stationary equilibrium gives

$$S_\theta^{PE} + S_q q_\theta = D_q q_\theta.$$

Thus

$$q_\theta = \frac{S_\theta^{PE}}{D_q - S_q}, \quad (\text{B.15})$$

$$S_\theta^{GE} = D_q q_\theta = \frac{D_q}{D_q - S_q} S_\theta^{PE}. \quad (\text{B.16})$$

Since $S_q > 0$ and $D_q < 0$,

$$0 < \frac{D_q}{D_q - S_q} < 1.$$

The endogenous price therefore preserves the sign of the partial equilibrium sales response and strictly attenuates its magnitude whenever that response is nonzero. It moves in the opposite direction from incumbent supply at the original price.

It remains to establish the investment response because attenuation of sales alone does not sign investment. Let $m = d_1(\theta) + d_2(\theta)$ and $a = \tau(1 - \beta)/(1 - \tau)$. Combining (B.6), (B.9), and (B.15), and simplifying, gives

$$\frac{dK^n}{d\theta} = \frac{a \{m + \beta d_2 - \phi D_q [d_2 F_{12} - \phi m F_{22}]\}}{(1 - \theta) \phi^2 \Delta (S_q - D_q)}. \quad (\text{B.17})$$

Assumption 3 implies

$$d_2 F_{12} - \phi m F_{22} = d_2 (F_{12} - \phi F_{22}) - \phi d_1 F_{22} \geq 0.$$

Because $D_q < 0$, $S_q - D_q > 0$, and $\Delta > 0$, the derivative in (B.17) is strictly positive whenever $m > 0$. If $m = 0$, the ordinary schedule already expenses the entire purchase and changing θ has no effect.

When the lock-in channel dominates, $S_\theta^{PE} < 0$. Equation (B.15) then gives $q_\theta > 0$, while (B.16) gives $S_\theta^{GE} < 0$. Entrant purchases fall along their unchanged demand curve, and Proposition 2 implies that $\bar{c}$ and λ fall as well. When the capital stock expansion channel dominates, all of these signs reverse. $\square$

B.7 Comparative Statics for Used Asset Subsidies and Corporate Taxes

The main analysis varies bonus depreciation for new capital, θ , while holding the tax treatment of used purchases, θ^U , fixed. The government also has access to bonus depreciation for used purchases and changing the corporate tax rate. This section gives local comparative statics for those two additional policy components. The exercises hold the other policy parameters fixed. They therefore do not, without further assumptions along the entire policy path, sign the effect of the finite simultaneous TCJA package.

Accelerated Depreciation for Used Capital

Proposition 3 (Accelerated depreciation for used capital). *Suppose $\tau > 0$, $\beta < 1$, and $\theta^U < 1$. Around an interior stationary equilibrium satisfying Assumptions 2, 4, and 5, a marginal increase*

in θ^U raises the used price, capital sales, purchases per entrant, the entry cutoff, and the entry rate. It raises investment in new capital if and only if

$$F_{12} + \beta\phi F_{22} < 0.$$

If this inequality is reversed, new investment falls; at equality it has no first-order response.

Proof. Let

$$a = \frac{\tau(1-\beta)}{1-\tau} > 0, \quad \chi = 1 + a(1-\theta^U) > 0.$$

Holding q fixed and differentiating $\phi h'(\phi U) = q\chi$ with respect to θ^U gives

$$U_{\theta^U} = -\frac{aq}{\phi^2 h''(\phi U)} > 0. \quad (\text{B.18})$$

Since aggregate demand is strictly increasing in U ,

$$D_{\theta^U} = D_U U_{\theta^U} > 0.$$

The reform does not enter incumbent supply directly. Differentiating market clearing therefore gives

$$q_{\theta^U} = \frac{D_{\theta^U}}{S_q - D_q} > 0. \quad (\text{B.19})$$

It follows that $S_{\theta^U} = S_q q_{\theta^U} > 0$. Proposition 2 then gives the positive responses of U , $\bar{c}$, and λ .

Finally, used-purchase expensing affects incumbent investment only through the equilibrium price. Equation (B.9) gives

$$K_{\theta^U}^n = -\frac{F_{12} + \beta\phi F_{22}}{\phi\Delta} q_{\theta^U}.$$

Since $q_{\theta^U} > 0$, the stated condition is necessary and sufficient for investment to rise. $\square$

Corporate Tax Cuts

For this experiment, hold θ , and therefore each $d_j(\theta)$, fixed, and also hold θ^U fixed. Define

$$E_\phi \equiv d_2(\theta)F_{11} - \phi d_1(\theta)F_{12} - \phi^2 [d_1(\theta) + d_2(\theta)] F_{22}. \quad (\text{B.20})$$

This is the same expression that signs the fixed-price supply response to new-capital bonus in (8).

Proposition 4 (Corporate tax cuts). *Suppose $\tau \in (0, 1)$ and $0 \leq \theta^U < 1$. Around an interior stationary equilibrium satisfying Assumptions 2, 4, and 5, a marginal permanent reduction in τ has the following effects:*

(i) *New investment rises if*

$$\phi F_{22} \leq F_{12} < -\beta\phi F_{22}.$$

(ii) *If $E_\phi > 0$, capital sales and entry rise, while the response of q is ambiguous. If $E_\phi = 0$, q , capital sales, and entry all rise. If $E_\phi < 0$, q rises, while the responses of capital sales and entry are ambiguous.*

Proof. Let

$$a_\tau \equiv \frac{1 - \beta}{(1 - \tau)^2} > 0, \quad m \equiv d_1(\theta) + d_2(\theta),$$

and retain $\chi = 1 + \tau(1 - \beta)(1 - \theta^U)/(1 - \tau)$. Define

$$\rho \equiv \frac{q(1 - \theta^U)}{\chi} > 0.$$

All derivatives in the proof are with respect to τ , so a tax cut reverses their signs.

At a fixed used price, differentiation of the entrant first-order condition gives

$$U_\tau = \frac{qa_\tau(1 - \theta^U)}{\phi^2 h''(\phi U)} < 0.$$

Comparison with (B.12) yields

$$U_\tau = a_\tau \rho U_q.$$

Because D depends on τ only through U at a fixed q ,

$$D_\tau = a_\tau \rho D_q < 0. \tag{B.21}$$

Thus a tax cut shifts entrant demand outward when used capital is not fully expensed.

Differentiating the incumbent conditions and market clearing gives

$$F_{11}K_\tau^n + \phi F_{12}K_\tau^o = a_\tau m - \beta q_\tau, \tag{B.22}$$

$$\phi F_{12}K_\tau^n + \phi^2 F_{22}K_\tau^o = a_\tau d_2 + q_\tau, \tag{B.23}$$

$$K_\tau^n - K_\tau^o = D_q(q_\tau + a_\tau \rho). \tag{B.24}$$

Solving this system gives

$$q_\tau = \frac{a_\tau [E_\phi + \phi^2 \Delta \rho D_q]}{\phi^2 \Delta (S_q - D_q)}, \tag{B.25}$$

$$S_\tau = \frac{a_\tau D_q [E_\phi + \phi^2 \Delta \rho S_q]}{\phi^2 \Delta (S_q - D_q)}, \tag{B.26}$$

$$K_\tau^n = \frac{a_\tau}{\phi^2 \Delta (S_q - D_q)} \left[-\phi D_q \left\{ \phi m F_{22} - d_2 F_{12} + \rho (F_{12} + \beta \phi F_{22}) \right\} - (m + \beta d_2) \right]. \tag{B.27}$$

Suppose $\phi F_{22} \leq F_{12} < -\beta \phi F_{22}$. The lower bound implies

$$d_2 F_{12} - \phi m F_{22} = d_2 (F_{12} - \phi F_{22}) - \phi d_1 F_{22} \geq 0,$$

while the upper bound implies $F_{12} + \beta \phi F_{22} < 0$. Therefore the expression in braces in (B.27) is strictly negative. Since $D_q < 0$, the entire numerator in (B.27) is strictly negative, so $K_\tau^n < 0$. A tax cut consequently raises investment.

For the used market, first suppose $E_\phi > 0$. Since $S_q > 0$, $D_q < 0$, and $\rho > 0$, equation (B.26)

gives $S_\tau < 0$, while (B.25) leaves the sign of q_τ ambiguous. A tax cut therefore raises capital sales. If $E_\phi = 0$, equations (B.25) and (B.26) give $q_\tau < 0$ and $S_\tau < 0$, so a tax cut raises both. Finally, if $E_\phi < 0$, then $q_\tau < 0$, while the sign of S_τ is ambiguous. A tax cut then raises the used price, with an ambiguous sales response. Proposition 2 supplies the corresponding entry responses. $\square$

Full expensing of used capital. When $\theta^U = 1$, entrant demand is independent of τ , so $\rho = 0$ and $D_\tau = 0$. Equations (B.25) and (B.26) reduce to

$$q_\tau = \frac{a_\tau E_\phi}{\phi^2 \Delta(S_q - D_q)}, \quad S_\tau = \frac{a_\tau D_q E_\phi}{\phi^2 \Delta(S_q - D_q)}.$$

A tax cut reverses these signs. Consequently, if used purchases are fully expensed, then the response is governed entirely by the incumbent supply shift.

The investment derivative at $\theta^U = 1$ is

$$K_\tau^n = \frac{a_\tau}{\phi^2 \Delta(S_q - D_q)} \left[-\phi D_q (\phi m F_{22} - d_2 F_{12}) - (m + \beta d_2) \right].$$

Under $\phi F_{22} \leq F_{12}$, this derivative is negative whenever $m > 0$, so a tax cut raises investment. At joint full expensing, however, $\theta = 1$ implies $d_1(\theta) = d_2(\theta) = m = 0$, while $\theta^U = 1$ makes entrant demand independent of τ . The stationary incumbent and entrant conditions then reduce to

$$F_1 = 1 - \beta q, \quad \phi F_2 = q, \quad \phi h'(\phi U) = q,$$

which contain no τ . The entire stationary equilibrium is therefore locally invariant to the corporate tax rate at the joint-full-expensing corner.

C Evidence from Compustat

I use S&P Global's Compustat North America Annual files to construct separate samples for the bonus-depreciation and TCJA analyses. The first contains U.S. firms; the second adds foreign firms as a comparison group. Both use the same outcomes but different sources of policy variation.

C.1 Outcome Measurement and Sample Construction

I measure new investment with capital expenditures (`capx`) and the value of capital sales with sales of property, plant, and equipment (`sppe`). Both variables are price times quantity, so they do not separately identify either component. The sales variable also understates total dispositions because it excludes noncash transactions like asset-for-equity swaps, involuntary conversions, and retirements (Bai et al. 2022). I set missing or negative capital sales to zero.

The sample is U.S.-domiciled firms outside NAICS 52 with positive total assets, net property, plant, and equipment (`ppent`), and sales, and nonnegative reported capital expenditures. To get variation in exposure to bonus, I match each firm to the four-digit industry-level present value of depreciation deductions from Zwick and Mahon (2017). The exposure, $\tilde{z}_{N,t}$, combines predetermined differences in industry capital composition with the bonus rate in year t . The

1995–2016 sample contains 102,466 firm-years from 11,674 firms. I winsorize `sppe` at the 3rd and 97th percentiles but do not winsorize capital expenditures in the baseline. Table C.1 reports summary statistics.

Table C.1: Summary Statistics: Compustat Sample

	Mean	Median
<i>Panel A: Firm Characteristics</i>		
Total Assets (\$M)	2143.9	152.9
Net PPE (\$M)	803	23.4
Sales (\$M)	1683.4	137
Capital Expenditure (\$M)	120.8	5.4
Asset Sales (\$M)	1.2	0
% Firm-Years with Positive Asset Sales	30.2%	
% Firm-Years with Positive CapEx	98%	
<i>Panel B: Scaled Ratios (%)</i>		
Asset Sales / Total Assets	0.46	0
CapEx / Total Assets	5.83	3.45
Asset Sales / PPE	6.39	0
CapEx / PPE	37.15	21.57
<i>Panel C: Tax and Heterogeneity Variables</i>		
PV of Depreciation Deductions ($\tilde{z}$), Mean (SD)	0.908 (0.041)	
% Profitable (OIBDP > 0)	71.3%	
% with Net Operating Losses	42.7%	
PPE Recovery Rate (% , Kermani–Ma)	34.551	34.87

Notes: Sample consists of U.S. non-financial Compustat firms with positive sales, assets, and PPE, 1995–2016. Firm-years: 102,466. Unique firms: 11,674. $\tilde{z}$ is the bonus-adjusted present value of depreciation deductions per dollar of investment, computed from industry-level MACRS recovery schedules and the bonus depreciation rate. Asset sales (`sppe`) are set to zero when missing or negative and winsorized at the 3rd and 97th percentiles. PPE recovery rate is the industry-level liquidation value from Kermani and Ma (2022).

The model places the investment response at purchase and the sales response after capital becomes old. Table C.2 accordingly uses current bonus exposure for investment and the previous year’s schedule for sales, both separately and jointly. Tables C.3 and C.4 vary the treatment of missing `sppe`, reporting histories, upper tails, scaling, and the exposure denominator.

Table C.2: Bonus Depreciation: Timing of Investment and Capital Sales

	Capital Sale Proceeds (sppe)			Investment (capx)
	Unconditional (1)	Any Proceeds (2)	Positive Proceeds (3)	Unconditional (4)
<i>Panel A: Inherited-cohort timing</i>				
$\tilde{z}_{t-1}$	-10.134*** (1.628)	-0.577** (0.269)	-1.466* (0.862)	
$\tilde{z}_t$				3.242** (1.632)
<i>Panel B: Joint current and lagged exposure</i>				
$\tilde{z}_t$	-8.600*** (2.277)	-0.371** (0.148)	-2.483 (1.708)	2.779** (1.186)
$\tilde{z}_{t-1}$	-4.344*** (0.732)	-0.343* (0.192)	-0.048 (0.770)	0.743 (0.761)
Sum	-12.944*** (2.262)	-0.714** (0.319)	-2.531* (1.450)	3.522* (1.904)
Estimator	PPML	LPM	PPML	PPML
Firm FE	X	X	X	X
Year FE	X	X	X	X
Panel A observations	64,497	96,323	28,926	96,070
Panel B observations	64,497	96,323	28,926	96,070

Notes: Panel A isolates the model's inherited-cohort timing: capital sale outcomes use the previous year's bonus generosity, $\tilde{z}_{t-1}$, while investment uses current generosity, $\tilde{z}_t$, over the same lag-available years, 1996–2016. Panel B includes current and lagged generosity jointly, allowing for the contemporaneous equilibrium response; Sum reports their linear combination. All specifications include firm and year fixed effects. Parenthesized values are standard errors clustered by four-digit NAICS industry. Columns (1), (3), and (4) are estimated by Poisson pseudo-maximum likelihood, and column (2) is a linear probability model. Column (3) conditions on positive proceeds. Capital sale proceeds are winsorized at the 3rd and 97th percentiles. ***, **, and * denote significance at the 1, 5, and 10 percent levels.

Table C.3: Bonus Depreciation and the Reporting of Capital Sale Proceeds

	Sale Proceeds		Any Proceeds		Observed Sample
	Baseline (1)	Observed SPPE (2)	Baseline (3)	Observed SPPE (4)	Indicator (5)
<i>Panel A: Alternative treatment of missing SPPE</i>					
$\tilde{z}$	-11.858*** (2.286)	-9.187*** (2.584)	-0.641*** (0.242)	-0.413 (0.435)	-1.828*** (0.151)
Estimator	PPML	PPML	LPM	LPM	LPM
Firm FE	X	X	X	X	X
Year FE	X	X	X	X	X
Observations	69,445	59,696	102,466	83,092	102,466
<i>Panel B: Firms with SPPE observed in every sample year</i>					
$\tilde{z}$	-7.382*** (2.469)		-0.205 (0.657)		
Observations	36,627		52,068		

Notes: In the baseline sample, missing and negative values of Compustat capital sale proceeds (sppe) are coded as zero. Observed-SPPE specifications instead exclude observations for which the raw item is missing or negative; sale proceeds are then winsorized within that sample at the 3rd and 97th percentiles. The outcome in column (5) is an indicator that raw SPPE is nonmissing and nonnegative, and thus that the observation enters columns (2) and (4). Panel B restricts the sample to firms for which raw SPPE is nonmissing and nonnegative in every observed firm-year. Because reporting is endogenous, these restrictions are sensitivity checks rather than corrections for selection. The sample spans 1995–2016. All specifications include firm and year fixed effects; parenthesized values are standard errors clustered by four-digit NAICS industry. ***, **, and * denote significance at the 1, 5, and 10 percent levels.

Table C.4: Bonus Depreciation: Outcome Construction and Scale

	Capital Sale Proceeds (sppe)			Investment (capx)
	Unconditional (1)	Any Proceeds (2)	Positive Proceeds (3)	Unconditional (4)
<i>Panel A: Alternative treatment of the SPPE upper tail</i>				
No winsorization	-28.375*** (5.722)	-0.641*** (0.242)	-11.787* (6.882)	3.402** (1.684)
1st/99th percentiles	-14.585*** (3.026)	-0.641*** (0.242)	-2.452 (2.123)	3.402** (1.684)
3rd/97th percentiles (baseline)	-11.858*** (2.286)	-0.641*** (0.242)	-2.470 (1.580)	3.402** (1.684)
5th/95th percentiles	-10.071*** (2.310)	-0.641*** (0.242)	-1.541 (1.455)	3.402** (1.684)
<i>Observations</i>	69,445	102,466	30,917	102,210
<i>Panel B: PPML exposure offsets</i>				
Lagged total assets offset	-9.991*** (1.221)	-0.601** (0.233)	-1.646 (1.059)	5.179*** (0.687)
<i>Observations</i>	65,750	96,926	29,675	96,666
Lagged net PPE offset	-11.584*** (1.201)	-0.604** (0.234)	-2.573** (1.057)	3.491*** (0.496)
<i>Observations</i>	65,501	96,411	29,639	96,150
<i>Panel C: Alternative outcome transformations</i>				
Inverse hyperbolic sine	-4.476*** (1.221)	-0.641*** (0.242)	-4.004 (3.686)	5.047*** (0.398)
$\log(1 + y)$	-3.914*** (1.087)	-0.641*** (0.242)	-3.570 (3.256)	5.028*** (0.369)
<i>Observations</i>	102,466	102,466	30,917	102,466

Notes: Each row reports a separate set of regressions. Panel A varies the upper-tail treatment of SPPE; capital expenditures remain unwinsorized. Panel B estimates PPML specifications with the logarithm of the indicated lagged firm size measure as an exposure offset. The denominators are taken from the preceding calendar year, and the extensive-margin regressions use the corresponding lag-available sample without an offset. Panel C estimates OLS regressions after applying the indicated transformation to each continuous outcome; its coefficients are not directly comparable in scale to Panels A and B. Panels A and C span 1995–2016; Panel B uses lag-available years 1996–2016. All specifications include firm and year fixed effects; parenthesized values are standard errors clustered by four-digit NAICS industry. Column (2) is a linear probability model, and column (3) conditions on positive proceeds. ***, **, and * denote significance at the 1, 5, and 10 percent levels.

D UCC Filings

D.1 Parser Details

This appendix describes how I transform raw Texas UCC financing statements into the analysis sample. The raw data are fixed-width text files originally from the Texas Secretary of State’s office, but are provided by Jensen (2024). Each row in a text file begins with a one-digit label indicating the type of information it contains. There are seven substantive labels distinguishing filing metadata, debtors, secured parties, amendments, and collateral descriptions. There is also a final row which summarizes the data and allows me to verify that the entire file has been parsed correctly. I proceed in several steps to construct the data.

Reconstructing Collateral Descriptions and Filing Dates

The first step is to recover a complete collateral description and the correct observation date for each filing event, which I treat as a transaction. This is challenging because a description may be divided across multiple rows, and it is possible that the equipment either appears on an original financing statement or in a later amendment.

As a first step, I verify that every row has the expected format and that the number of rows of each kind agrees with the totals supplied by the Texas Secretary of State, which is a summary available at the bottom of each file.

Each file divides collateral descriptions into consecutive 80-character pieces which may span multiple rows. Each row contains the original filing number, an event number, a sequence number, and the 80-character piece. I group rows with the same filing and event numbers, which ensures that they are part of the same transaction, and join their text fields end to end without inserting spaces. After reconstructing the full description, I remove padding and standardize white space. Doing it this way (rather than padding *ex ante*) preserves model and serial numbers that cross an 80-character boundary. The sequence numbering indicates the order in which the descriptions should be concatenated; I require these within a group to be consecutive.

Finally, I label each description with the filing event in which it appears. If the filing is indicated to be an original financing statement, I use the initial filing date. Sometimes financing statements are amended. In that case, I link the description using the original filing number and the state’s UCC3 event number and use the date of the event. I retain the original filing year separately, but define the observation year as the year in which the equipment appears. Asset age is therefore the observation year minus the model year. In the paper, I refer to the observation year as the filing year.

Excluding Blanket Liens

The analysis requires collateral descriptions that identify particular pieces of equipment. Some filing events instead contain blanket liens, which give the lender a security interest in all or broad categories of the borrower’s property without identifying any specific machine. References to “equipment” or “inventory” in these descriptions therefore do not establish that a particular asset was involved in the transaction.

I identify blanket liens using the complete collateral description associated with each filing event. I classify an event as a blanket lien when the description contains broad pledging language such as “all personal property,” “all inventory,” “all assets,” “general intangibles,” “accounts receivable,” or “now owned or hereafter acquired.” I then exclude every potential equipment item

parsed from that event. Splitting all collateral descriptions produces 7,022,114 potential items; 2,981,088 remain after removing those associated with blanket liens.

Separating Multiple Pieces of Equipment

The unit of observation is an individual piece of equipment, but a single collateral description may list several machines. I therefore divide each description at points that indicate the beginning of a new machine. The rules are deliberately conservative: punctuation alone is not treated as a boundary unless it is accompanied by evidence that the preceding item has ended or that a new item has begun.

I apply the following rules sequentially, beginning with the clearest boundaries:

1. Split at numbered-item prefixes, such as “1-NEW” and “2-USED”;
2. Split at semicolons;
3. Split at a comma when the text following it begins with a recognized manufacturer name;
4. Split at an ampersand when the preceding text ends with a serial-number-like identifier and the following text begins with a quantity or manufacturer name;
5. Split when a quantity followed by a manufacturer name appears immediately after a serial-number-like identifier; and
6. Split when a recognized manufacturer name follows a serial-number-like identifier after two or more spaces.

The manufacturer names used in these rules come from a dictionary of approximately 100 equipment manufacturers. The conditions involving serial numbers, quantities, and manufacturers prevent ordinary commas, ampersands, and spaces within an equipment description from being mistaken for boundaries between machines.

Splitting the descriptions also produces fragments that do not describe equipment. I exclude fragments containing only legal language, such as “together with all accessions,” “products and proceeds,” or “purchase money security interest”; financing terms, such as payment schedules or dollar amounts; and text containing fewer than 10 characters. After excluding blanket liens and these non-equipment fragments, 2,109,751 potential equipment descriptions remain. The sequence-number check described above excludes a further 96, leaving 2,109,655.

Extracting Equipment Characteristics

For each potential equipment description that survives the preceding filters, I extract the characteristics needed to date the equipment, assign its tax treatment, compare similar machines, and track repeat appearances. The five principal fields are equipment type, manufacturer, model year, serial number or VIN, and any explicit indication that the equipment is new or used.

Equipment type. Equipment type is needed to assign the appropriate MACRS recovery class and to compare similar assets. I match each description against an ordered dictionary of approximately 80 equipment types covering construction and agricultural machinery, trucks and trailers, material-handling and forestry equipment, and office and information-technology equipment. More specific descriptions are matched before broader ones, so that, for example, “wheel loader” is not classified simply as “loader” and “gooseneck trailer” is not classified simply as “trailer.” I also use manufacturer-specific model prefixes to identify equipment whose description contains a model number but no equipment-type term. A named model series is used only when the corresponding manufacturer has also been identified, which prevents generic model names from generating false matches. This procedure identifies an equipment type for 971,632 potential items.

Manufacturer. Manufacturer identity allows me to compare equipment produced by the same company and helps identify equipment type from manufacturer-specific model numbers. I match descriptions against a dictionary of approximately 100 manufacturers using patterns that accommodate common abbreviations and misspellings. The dictionary includes major construction and agricultural equipment manufacturers, truck manufacturers, and trailer and specialty-equipment producers. For example, the patterns recognize variants such as “MASSEY FERGERTON,” “ARTIC CAT,” and “NEWHOL.” In the preferred specifications, I compare equipment within groups defined by manufacturer, equipment type, recovery class, and basis schedule.

Model year. Model year proxies for the original purchase vintage and, together with the filing year, determines asset age. I identify plausible four-digit years from 1940 through 2029 in each description. I retain a year when the surrounding text identifies it as equipment information or when it is the only plausible equipment year in the description. I exclude years appearing in legal dates or other non-equipment contexts and leave descriptions with conflicting equipment years unclassified.

A valid 17-character VIN provides an additional source of model-year information. I first verify its check digit and then decode the character identifying the model year. Because that character repeats every 30 years, I use the filing-event year to select the feasible cycle. The enrichment procedure described below supplies additional model-year information. Before enrichment, 242,766 potential items have both an identified equipment type and model year; enrichment raises this count to 288,137.

Serial number and VIN. Serial numbers allow me to identify repeated appearances of the same machine. I extract an identifier only when it follows an explicit label such as S/N, SER, SERIAL, VIN, PIN, or S#. A bare number sign is not sufficient because it may introduce a model number or other numbered text. I separately retain 17-character VINs that pass the NHTSA check-digit test.

Tracking repeat appearances. Before constructing the analysis data, I replace filing, equipment, and debtor identifiers with keyed hashes. This permits the same entity or machine to be matched across records without retaining the original identifier. I define a machine using either

a validated VIN or an explicitly labeled serial number combined with its manufacturer or equipment type. To reduce false matches, the tracking sample retains only identifiers associated with a single model year, equipment type, recovery class, and depreciation method. Borrower-pair analyses also exclude filings with multiple debtors because a particular machine cannot be assigned unambiguously to one borrower. These identifiers allow me to count a machine only once within a filing event and to identify the first observed appearance of a machine–borrower pair. Such an appearance need not represent a verified sale because a UCC filing may also arise from refinancing or a later collateral event.

New/used label. The main analysis determines whether equipment is used from its model year and age. Explicit NEW and USED labels provide a more restrictive classification used in robustness tests. I exclude phrases in which these words do not describe equipment condition, such as “whether new or used,” “now owned or hereafter acquired,” “used in the operation of,” and “as used herein.” I also remove variants of the manufacturer name “New Holland” before searching for NEW. When both labels appear, USED takes priority, so that a description such as “1 USED NEW HOLLAND TRACTOR” is correctly classified as used.

Quantity. I record a quantity when the description states one explicitly. Because a number preceding an equipment name may instead be part of its model designation, I do not use the quantity field as an analytical weight. The regressions count parsed equipment mentions rather than summing reported quantities.

Completing Missing Equipment Information

The initial text parser does not recover every item’s manufacturer, equipment type, or model year. I supplement these fields using information tied to the identifiers already present in the collateral description. The purpose of this step is to increase coverage without replacing classifications that can be obtained directly from the filing text.

First, I use validated 17-character VINs to obtain manufacturer, vehicle type, body class, model year, and gross vehicle weight from the National Highway Traffic Safety Administration. To make the results reproducible, I save the NHTSA response for each VIN and reuse that fixed set of responses rather than querying the service each time the data are rebuilt. The resulting cache covers all 148,252 unique valid VINs in the parsed data. Because the same VIN may appear in more than one filing event, these results supply information to 154,042 equipment observations.

Second, Caterpillar serial-number prefixes identify particular product lines. For Caterpillar equipment with a serial number but no equipment type recovered from the text, I use these prefixes to identify the corresponding model and equipment type.

Third, when a manufacturer and model number are observed but the equipment type is missing, I use manufacturer-specific model conventions to identify the type. These rules are limited to model patterns associated with a known manufacturer so that a generic model number cannot generate a classification by itself.

Fourth, the identity of the secured party can provide information about how otherwise ambiguous equipment is used. I identify Farm Credit System lenders and use that information only as a fallback indicator of agricultural use when assigning the MACRS recovery class. Lender iden-

tity never overrides an equipment description that identifies the asset as a truck or as construction equipment.

Information recovered directly from the collateral description generally takes priority over these additional sources. The relevant exception is a generic “tractor” or “truck” classification that a validated VIN identifies unambiguously as an over-the-road tractor unit. I correct those classifications because road tractor units receive a different MACRS recovery period and fall outside the analysis sample.

Assigning MACRS Schedules

The purpose of the MACRS classification is to reconstruct the ordinary depreciation schedule of the seller’s original purchase cohort and therefore the tax basis remaining when the equipment appears in a UCC transaction. A complete schedule requires both a recovery period and a depreciation method. I assign these only after combining the available information on equipment type, manufacturer, model year, and secured party. When the available information does not determine a schedule, I leave the item unclassified rather than impose one.

Recovery period. I assign equipment directly when its observed type maps to a specific MACRS category. Trailers, for example, are five-year property. Over-the-road tractor units are three-year property and therefore fall outside the paper’s five- and seven-year analysis sample. I identify road tractor units from explicit collateral descriptions, manufacturers associated with road tractors, or a validated NHTSA truck-tractor body classification.

Some descriptions do not identify the relevant asset category precisely enough. A generic “semi” or “tractor-trailer” may refer either to the three-year tractor unit or to the five-year trailer, so I leave it unclassified. I also leave terminal tractors unclassified when their use is unknown. Generic aircraft and boats are excluded because their recovery periods depend on uses that are not reported in the collateral description.

Farm machinery. The recovery period for farm machinery depends on when it was originally placed in service. I use model year as a proxy for that purchase vintage. Farm machinery from vintages through 2017 is assigned to the seven-year class, except for qualifying machinery from 2009, when a temporary rule assigned it to the five-year class. Farm machinery from vintages beginning in 2018 is assigned to the five-year class under the maintained assumption that its original use began with the owner associated with its model-year cohort.

I identify farm use first from the equipment description, including agricultural equipment and agricultural trailer categories. When the equipment itself does not determine its use, an agricultural secured party provides a fallback classification. Secured-party identity never overrides an explicit truck or construction-equipment classification.

Depreciation method. Recovery period and depreciation method are assigned separately. Pre-2018 farm-use property follows the 150-percent declining-balance schedule for its assigned five- or seven-year recovery period under the half-year convention. Nonfarm property and farm machinery from vintages beginning in 2018 follow the corresponding 200-percent declining-balance schedule. Each observation retains its recovery period, depreciation method, and the rule used to assign them.

Explicit NEW or USED labels in a UCC description refer to the equipment’s condition when it appears in that filing, not necessarily when it was originally purchased. I therefore do not use those labels to determine the seller’s original recovery period or depreciation method.

Constructing Purchase Vintage and Tax Exposure

The purpose of this step is to assign each machine the tax treatment that would have applied when it was originally purchased and to calculate the resulting difference in remaining basis when it later appears in the UCC data. I treat model year as the original purchase year and match it to the federal bonus-depreciation rate in effect for that year. This produces a measure of statutory eligibility: it records whether the machine would have qualified for bonus depreciation, not whether its owner actually claimed the deduction. I do not observe bonus elections, Section 179 deductions, prior owners, or taxable transfers that may have reset the asset’s basis.

Let $B_s(k)$ denote the fraction of ordinary basis remaining after k annual deductions under depreciation schedule s . If an asset’s purchase vintage received bonus rate θ_v , bonus immediately deducted that share of its basis and applied the ordinary schedule to the remainder. The predicted difference between the basis of a nonbonus asset and an otherwise identical bonus asset is therefore

$$\theta_v B_s(k).$$

I refer to this difference as the bonus-induced basis gap.

For an asset of age a , the baseline measure assumes that the filing-year deduction has already been taken and therefore sets $k = a + 1$. An alternative measure uses the basis at the beginning of the filing year and sets $k = a$. Under the baseline convention, the basis gap is exhausted at age five for five-year property and age seven for seven-year property. For bonus analysis, I multiply the basis gap by a 35-percent corporate tax rate to obtain a statutory measure of the associated tax wedge. Because the UCC data also contain pass-through businesses and do not report each seller’s tax status, the 35-percent rate is a common benchmark rather than an estimate of each seller’s actual tax liability.

The UCC data report model year but not the exact date on which a machine was placed in service. I therefore assign one bonus rate to each calendar year using the rate that applied during most of that year. Under this convention, 2001 is coded as having no bonus, while 2003, 2010, and 2017 are coded at 50 percent despite changes in the statutory rate during those years.

The main pre-TCJA specifications use an indicator equal to one for model-year vintages eligible for any positive bonus depreciation. Appendix specifications isolate vintages receiving exactly 50-percent bonus and compare them with zero-bonus vintages. Other appendix specifications use the continuous basis-gap measures defined above.

Constructing the Analysis Sample

The purpose of the final sample restrictions is to retain equipment mentions for which I can identify the purchase vintage, calculate asset age, and assign the ordinary depreciation schedule needed to measure remaining basis. I retain filing events from 2001 through 2019 whose collateral descriptions pass the sequence-number check described above. Each retained equipment mention must have an identified equipment type and model year, an age between zero and 50, and an assigned five- or seven-year MACRS recovery period and depreciation method.

The unit of observation is an equipment mention in a filing event. A filing that lists several machines contributes one observation for each machine, and the same machine may contribute multiple observations if it appears in multiple filing events. The alternative counting specifications described below use the hashed identifiers to remove repeat appearances.

The resulting sample contains 233,845 equipment mentions. Table D.1 reports how each parsing and sample restriction changes the number of potential items. Tables D.2 and D.3 describe the final sample and its largest equipment categories.

Table D.1: UCC Sample Construction

Step	Items
Total parsed equipment items	7,022,114
After dropping blanket liens	2,981,088
After dropping item-level boilerplate	2,109,751
After validating collateral sequences	2,109,655
With identified equipment type	971,632
With parser type and model year	242,766
With enriched type and model year	288,137
In the filing-year and age window	273,648
Restricted to MACRS 5- or 7-year property	233,845

Notes: Filing years 2001–2019. Items are individual pieces of equipment parsed from collateral descriptions in Texas UCC financing statements. Blanket liens pledge all assets without naming specific equipment. Boilerplate items are parsed legal or financing text rather than identifiable machines. Collateral sequence validation excludes items whose continuation records cannot be linked unambiguously. The last step retains five- and seven-year property. It excludes three-year road tractor units, ambiguous tractor-trailer composites, and generic aircraft or boats whose use is not observed. Trailers are five-year property. For farm machinery, model year proxies the seller’s original purchase cohort: pre-2018 cohorts are seven-year property except for the statutory 2009 five-year exception, and cohorts from 2018 are five-year property under the maintained first-owner interpretation. Pre-2018 farm-use five- and seven-year property uses 150-percent declining balance; nonfarm property and post-2017 farm-use cohorts use 200-percent declining balance. Current NEW/USED collateral labels do not determine seller basis.

Table D.2: Summary Statistics for the UCC Analysis Sample

Equipment	Filing years	Items	Mean age	Median age	Positive future basis (%)	Future years
New equipment	2001–2017	50,323	0.0	0	100.0	5.6
New equipment	2018–2019	8,514	0.0	0	100.0	5.1
Used equipment	2001–2017	147,172	9.8	7	37.3	1.2
Used equipment	2018–2019	27,836	8.3	5	48.0	1.5

Notes: The analysis sample contains equipment items with an identified equipment type, model year, and five- or seven-year MACRS class and basis method. Positive Future Basis indicates that the ordinary MACRS schedule leaves basis after the filing-year deduction. Future Years counts deductions after that year, censored at zero.

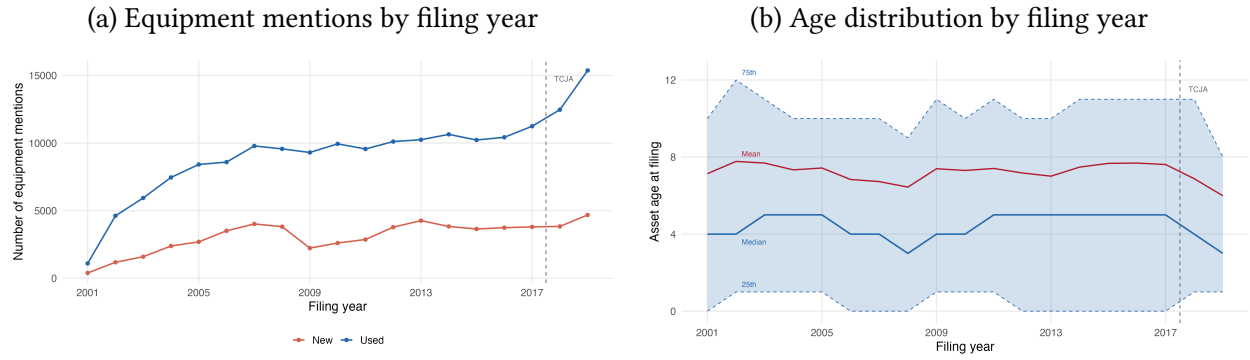
Table D.3: Largest Equipment Categories in the UCC Analysis Sample

Equipment type	MACRS	Method	Items	New (%)	Mean used age	Positive basis (%)
Trailer	5	200-percent DB	74,609	32.2	8.0	58.0
Tractor	7	150-percent DB	39,391	11.2	14.2	40.7
Truck	5	200-percent DB	13,791	28.1	8.6	50.7
Mower / Cutter	7	150-percent DB	7,438	38.7	4.9	86.9
Dozer	5	200-percent DB	6,332	3.5	14.5	16.5
Excavator	5	200-percent DB	6,138	15.1	7.6	46.7
Skid Steer	5	200-percent DB	5,467	16.6	6.6	49.7
Loader	7	150-percent DB	5,426	18.5	9.7	55.4
Gooseneck Trailer	5	200-percent DB	5,135	22.8	9.5	47.8
Forklift	5	200-percent DB	4,215	21.6	8.9	39.1
Backhoe	5	200-percent DB	3,723	6.1	11.9	23.6
Trailer	5	150-percent DB	3,038	55.4	6.4	81.3
Baler	7	150-percent DB	2,766	18.8	8.7	56.6
Laptop	5	200-percent DB	2,564	11.9	3.0	85.5
Center Pivot	7	150-percent DB	2,493	49.8	12.3	66.0

Notes: Categories are equipment-type $\times$ MACRS-class $\times$ basis-method cells ranked by the number of items in the analysis sample. Mean Used Age excludes age-zero items.

Figure D.1 documents the sample’s coverage over time and the ages of the equipment observed in each filing year. Panel (a) shows a sharp increase in equipment mentions early in the sample. The bulk files do not allow me to distinguish an increase in underlying filing activity from an expansion in data coverage, so I do not interpret this change as an economic response. The regressions include filing-year fixed effects and therefore do not use aggregate changes in the number of records for identification. Panel (b) shows that the age distribution is relatively stable through 2017 before shifting toward younger equipment in 2018 and 2019.

Figure D.1: UCC Sample Coverage and Asset Ages



Notes: Panel (a) reports equipment mentions by filing year. “New” denotes age zero and “used” denotes positive age; both are model-year proxies rather than observed purchase status. Panel (b) reports mean and median age together with the interquartile range. Asset age is filing year minus model year. The vertical line marks 2018, the first filing year following enactment of the TCJA.

D.2 Additional Results

Table D.4 reports robustness of the preferred specification to alternative samples, classification rules, and counting units.

Table D.4: Preferred UCC Bonus-Vintage Specification: Sample and Counting Robustness

Specification	Coef.	Standard error	Percent effect	Cells	Counted items	Mfrs.	Vintage clusters
<i>Panel A: Sample and classification restrictions</i>							
Baseline preferred	-0.141**	0.071	-13.2	46,465	110,453	2,945	66
Initial filings only	-0.131*	0.070	-12.3	43,663	99,162	2,829	66
Exclude age one	-0.144*	0.077	-13.4	42,233	96,773	2,703	65
Explicit USED text only	-0.042	0.056	-4.1	4,043	5,260	621	56
Exclude farm-use proxy	-0.198***	0.066	-18.0	32,028	65,817	2,079	65
Exclude all trailer types	-0.159*	0.081	-14.7	30,276	77,561	1,597	66
Direct high-confidence MACRS rules	-0.123**	0.055	-11.6	17,427	35,154	1,410	58
Omit model-year 2001	-0.158**	0.066	-14.6	44,338	105,859	2,906	65
<i>Panel B: Alternative counting units</i>							
Baseline mention count	-0.141**	0.071	-13.2	46,465	110,453	2,945	66
Trackable single-debtor mentions	-0.141**	0.067	-13.2	32,594	64,371	2,414	66
Unique asset-event counts	-0.141**	0.067	-13.2	32,594	64,344	2,414	66
First observed asset-borrower pair	-0.136**	0.067	-12.7	32,111	61,568	2,409	66
Validated VIN first-borrower pairs	-0.159**	0.066	-14.7	15,202	28,194	1,729	49

Notes: Each row reports a separate PPML regression of UCC equipment counts in observed manufacturer by filing-year by asset-age by model-year-vintage cells. Manufacturer combines the manufacturer identifier, equipment type, MACRS class, and basis schedule. All regressions include manufacturer by asset-age and filing-year fixed effects and use positive-age filings from 2001–2017. Bonus vintage equals one when the model-year purchase vintage received any federal bonus depreciation. Percent effect is $100[\exp(\hat{\beta}) - 1]$.

Panel A changes the analytic sample. Explicit USED labels are selective text matches. The nonfarm restriction selects the same analytic records as retaining ordinary 200DB schedules. Direct high-confidence MACRS rules are deterministic assignments retained in the frozen cache; the original confidence flag itself is not retained. Panel B changes the counted unit. An asset-borrower pair is an observed filing pair, not a verified sale. The VIN sample is concentrated in transport equipment.

Standard errors clustered by model-year vintage are shown in their own column. Stars use two-sided $t(G - 1)$ p-values, where G is the number of vintage clusters: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table D.5 restricts the analysis to exact 50-percent and zero-bonus purchase vintages and considers alternative samples and treatments of zero-count cells.

Table D.5: Exact 50-Percent Bonus Versus Zero: Cohort-Supported Robustness

	Baseline counts		Initial filings		First asset-borrower	
	Observed	Completed	Observed	Completed	Observed	Completed
50-percent bonus vintage	-0.111** (0.041)	-0.172*** (0.047)	-0.104** (0.043)	-0.165*** (0.049)	-0.109* (0.058)	-0.196** (0.066)
Log age-zero cohort size	0.210*** (0.019)	0.266*** (0.027)	0.219*** (0.022)	0.277*** (0.029)	0.209*** (0.028)	0.277*** (0.033)
Manufacturer by age FE	X	X	X	X	X	X
Filing-year FE	X	X	X	X	X	X
Supported zeros completed		X		X		X
Estimation cells	13,523	23,861	12,754	22,909	9,288	17,175
Counted items	47,593	47,593	42,522	42,522	25,792	25,792
Manufacturers	908	908	872	872	713	713
Vintage clusters	14	14	14	14	14	14
Zero-bonus vintages	4	4	4	4	4	4

Notes: Each column reports a PPML regression comparing model-year vintages receiving exactly 50 percent bonus depreciation with zero-bonus vintages. The comparison excludes the 30-percent 2002 vintage and the 100-percent 2011 vintage. The sample covers filing years 2001–2017. A manufacturer by vintage enters only when it has an age-zero filing count, which is included flexibly in logs. Observed columns retain positive used-filing cells. Completed columns fill feasible filing-year by ages 1-25 with zeros; estimation cells report the cells retained by PPML after separation checks.

The first-pair columns count only the first observed appearance of an asset-borrower pair; this is not necessarily a verified sale. Standard errors clustered by model-year vintage are in parentheses. Stars use two-sided $t(G - 1)$ p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The design has 14 vintage clusters but only four zero-bonus vintages, so inference should be read cautiously.

Table D.6 examines smooth purchase-vintage curvature and the policy episodes underlying the bonus vintage comparison. The results show that the 2005–07 lapse provides the principal identifying variation.

Table D.6: Purchase-Vintage Timing and the Bonus-Filing Gap

Specification	Coefficient (SE)	<i>t-p</i>	Cells	Treated / zero vintages
<i>Baseline and smooth purchase-vintage curvature</i>				
Baseline	-0.141** (0.071)	0.050	46,465	12 / 54
Add quadratic vintage curvature	-0.166** (0.071)	0.023	46,465	12 / 54
Add quadratic and cubic vintage curvature	-0.166** (0.071)	0.022	46,465	12 / 54
<i>Which control vintages carry the comparison?</i>				
Retain post-2001 vintages (lapse controls only)	-0.212*** (0.064)	0.005	22,603	12 / 3
Omit 2005–07 lapse	0.031 (0.080)	0.703	39,188	12 / 51
<i>One treated episode omitted at a time</i>				
Omit 2002 transition (30 percent)	-0.127 (0.081)	0.121	44,481	11 / 54
Omit 2003–04 bonus episode (50 percent)	-0.162* (0.093)	0.087	42,209	10 / 54
Omit 2008–10 bonus episode (50 percent)	-0.153** (0.065)	0.022	42,063	9 / 54
Omit 2011 full-expensing vintage	-0.139* (0.072)	0.058	45,326	11 / 54
Omit observed 2012–16 bonus episode (50 percent)	-0.140* (0.072)	0.057	42,920	7 / 54

Notes: Each row reports the bonus-vintage coefficient from a separate PPML regression with manufacturer $\times$ raw asset-age fixed effects and filing-year fixed effects. Standard errors clustered by purchase vintage are shown in parentheses; the displayed *p*-values and stars use t_{G-1} . A linear purchase-vintage trend is exactly collinear with filing-year and age fixed effects because vintage equals filing year minus age. The curvature rows therefore add the estimable squared term, then squared and cubed terms, in purchase vintage centered at 2000 and scaled by ten years. The post-2001 row leaves the three 2005–07 lapse vintages as the only zero-bonus controls. Conversely, the lapse-omission row removes those three controls, leaving only vintages through 2001 as zero-bonus observations. Episode omissions are leverage diagnostics, not separate causal designs.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.7 reports the age-specific coefficients and formal tests underlying the lifecycle figure in the main text.

Table D.7: Lifecycle Pattern in the Bonus-Vintage Filing Gap

Age	Coefficient (SE)	Age	Coefficient (SE)
1	-0.098 (0.090)	8	-0.099 (0.110)
2	-0.210*** (0.077)	9	-0.042 (0.114)
3	-0.223*** (0.081)	10	-0.069 (0.089)
4	-0.199** (0.080)	11	-0.032 (0.122)
5	-0.205*** (0.069)	12	0.019 (0.104)
6	-0.151** (0.073)	13+	0.017 (0.054)
7	-0.199* (0.105)		
<i>Joint and direct lifecycle tests</i>			
Ages 2–7 jointly zero	$F_{6,65} = 2.251$		$p = 0.049$
Ages 8+ jointly zero	$F_{6,65} = 0.914$		$p = 0.491$
Coefficients at ages 2–7 equal	$F_{5,65} = 0.544$		$p = 0.743$
Equal-weight mean, ages 2–7	-0.198		
Equal-weight mean, ages 8+	-0.034		
Early minus late mean	-0.164*** (0.060)		$p = 0.008$
Cells		46,465	
Equipment mentions		110,453	
Vintage clusters		66	

Notes: Each coefficient is the PPML log count gap between bonus and zero-bonus purchase vintages for the indicated age group. The sample covers filing years 2001–2017 and asset ages 1–50. The regression absorbs manufacturer $\times$ raw asset-age fixed effects and filing-year fixed effects. Age 13+ pools ages 13–50 only for the policy interaction; fixed effects remain defined at each raw age. Standard errors clustered by purchase vintage are in parentheses. Coefficient stars and confidence intervals use a t_{G-1} reference distribution. The early and late means give equal weight to six reported age coefficients: ages 2–7 and age groups 8–12 plus 13+.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Business Dynamics Statistics

E.1 Data Construction

The BDS reports annual counts of firms, establishments, and firm deaths by four-digit NAICS industry. Within each industry, I calculate the entry rate as $(F_{N,t} - F_{N,t-1} + D_{N,t})/F_{N,t-1}$ and the exit rate as $D_{N,t}/F_{N,t-1}$, where $F_{N,t}$ is the number of active firms and $D_{N,t}$ is the number of firm deaths. Because the bonus-exposure series begins in 1995 and constructing these rates requires the lagged firm stock, 1996 is the first usable year.

To construct bonus depreciation exposure, I merge the BDS panel by four-digit NAICS with the ordinary MACRS present value z_N from Zwick and Mahon (2017) and the annual statutory bonus rate θ_t . I then calculate $\tilde{z}_{N,t} = \theta_t + (1 - \theta_t)z_N$. After requiring a valid entry rate and bonus exposure and restricting the sample to years before 2018, the resulting unbalanced panel contains 4,984 industry-years across 227 industries from 1996 through 2017.

For the tradeability analysis, I map each four-digit NAICS industry to a BEA industry using the four-digit match where available and otherwise the corresponding three- or two-digit match. I then merge the time-invariant PPE recovery rate from Kermani and Ma (2022). This produces recovery-rate matches for 186 industries and 4,091 industry-years in the bonus sample. The appendix robustness exercises also use the BDS establishment entry rate and firm counts by age. Section G.1 describes the separate exposure measure used for the TCJA analysis.

Table E.1 reports summary statistics for the bonus-depreciation and TCJA samples described in the main text.

Table E.1: Summary Statistics: Business Dynamics Statistics

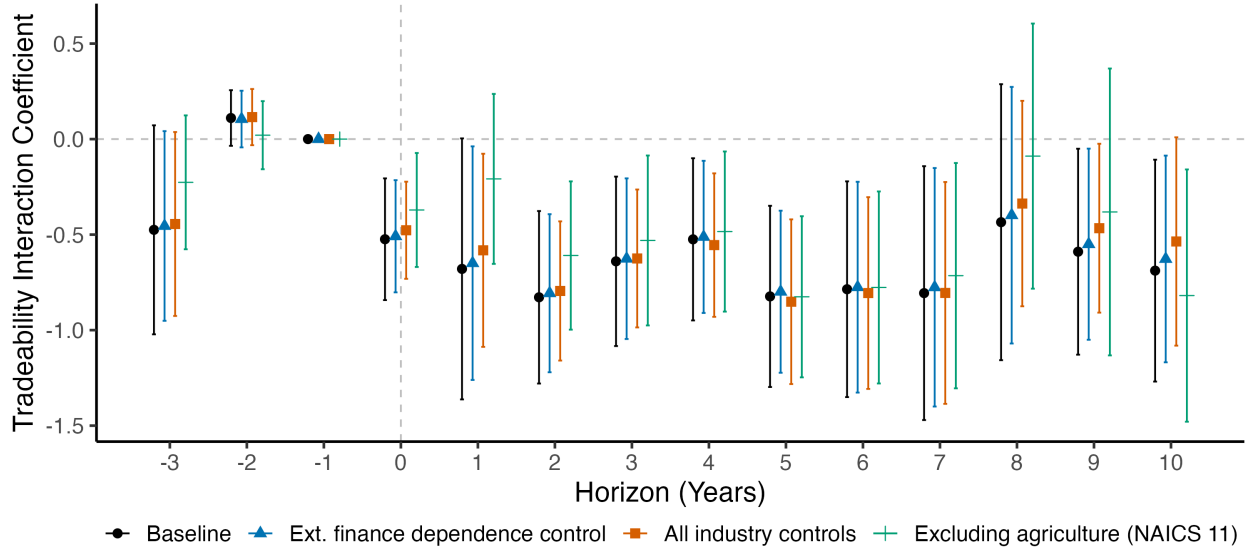
	Mean	SD
<i>Panel A: Bonus Depreciation Sample (1996–2017)</i>		
Entry Rate (%)	7.9	6.06
Exit Rate (%)	7.94	2.94
$\tilde{z}_{N,t}$	0.918	0.035
PPE Recovery Rate (%)	32.18	17.61
Industries	227	
Industry-Years	4,984	
<i>Panel B: TCJA Sample (2012–2019)</i>		
Entry Rate (%)	7.19	5.05
Exit Rate (%)	7.02	2.37
Δg_N (Demand Wedge Change)	0.032	0.004
Δg_N (Standardized)	0	1
Industries	201	
Industry-Years	1,608	
Recovery-Matched Industries	168	
Recovery-Matched Industry-Years	1,344	

Notes: Data are from the Census Bureau’s Business Dynamics Statistics at the 4-digit NAICS level. Entry rate is $(\text{firms}_t - \text{firms}_{t-1} + \text{deaths}_t) / \text{firms}_{t-1}$. Exit rate is firm deaths divided by lagged firm count. In Panel A, $\tilde{z}_{N,t} = \theta_t + (1 - \theta_t)z_N$ is the present value of depreciation deductions from Zwick and Mahon (2017). In Panel B, Δg_N is the change in the demand-side tax wedge between new and used capital induced by the TCJA, computed using 2014–2016 BEA fixed-investment weights. The standardized measure is demeaned and divided by its standard deviation across 4-digit industries. Recovery-matched counts describe the common sample used for the tradeability interaction. PPE recovery rate is the industry-level liquidation value from Kermani and Ma (2022).

E.2 Bonus Depreciation

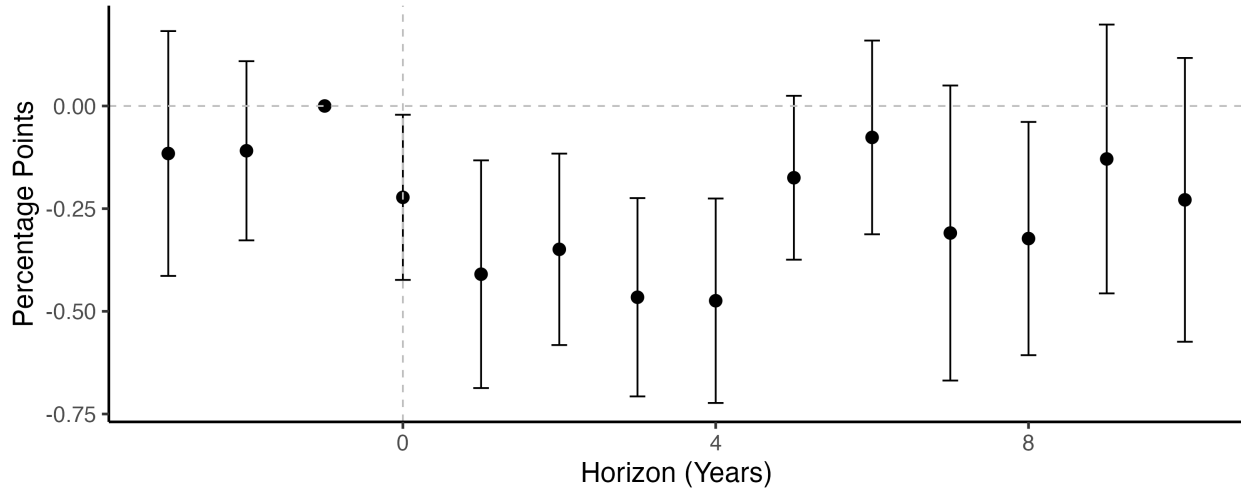
The main text shows that the entry response is larger in industries with more tradeable capital. Figure E.1 tests whether this interaction is explained by other industry characteristics or by agriculture.

Figure E.1: The Entry–Tradeability Interaction: Robustness



Note: This figure reports the coefficient on $\tilde{z}$ interacted with the PPE recovery rate from the panel local projection in Figure 3b. The specifications move from the baseline to a control for external finance dependence, controls for all four competing characteristics, and a sample excluding agriculture (NAICS 11). The competing characteristics are external finance dependence, average firm size, capital duration, and entry cyclicality; external finance dependence is the Rajan–Zingales measure constructed from U.S. Compustat firms over 1980–2001. All specifications include $\tilde{z}$, industry and year fixed effects, and two-digit NAICS quadratic trends. Bars report 95% confidence intervals from standard errors clustered by four-digit industry; horizon -1 is the omitted period.

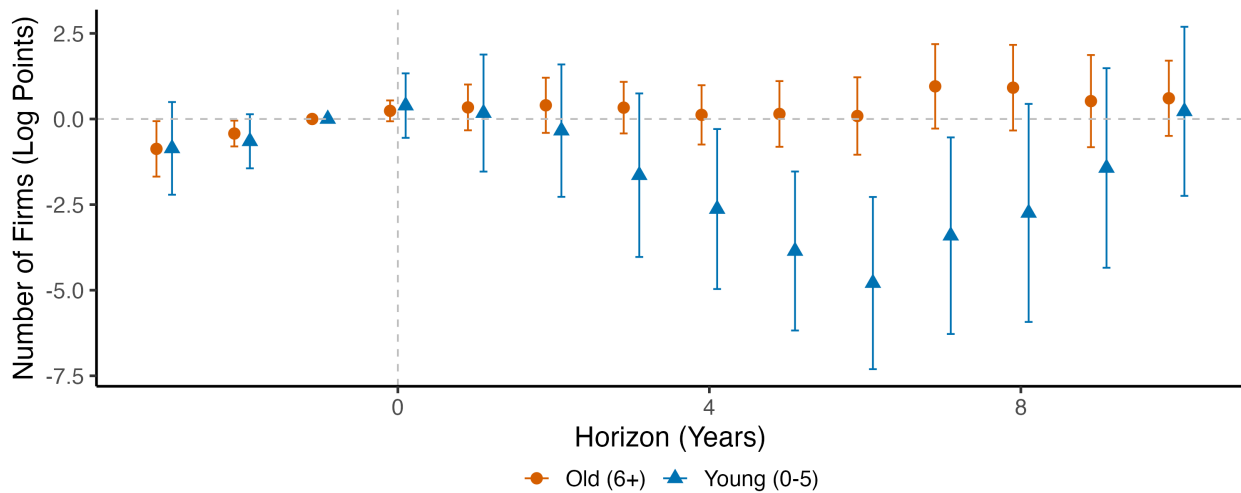
Figure E.2: Dynamic Effect of Bonus Depreciation on Establishment Entry



Note: This panel local projection uses the establishment entry rate as the outcome and the industry-level present value of depreciation allowances, $\tilde{z}_{N,t}$, as the treatment. It includes four-digit NAICS and year fixed effects and two-digit NAICS linear and quadratic trends. Bars report 95% confidence intervals from standard errors clustered by four-digit NAICS industry.

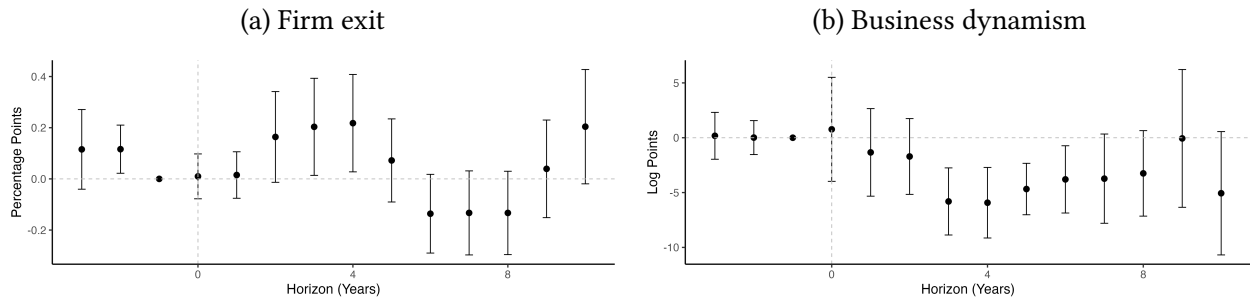
Figures E.3 and E.4 show that the response is concentrated among young firms and distinguish it from broader changes in firm turnover.

Figure E.3: Age Composition of Firms



Note: These panel local projections use the log number of firms in each age group as the outcome and the industry-level present value of depreciation allowances, $\tilde{z}_{N,t}$, as the treatment. They include four-digit NAICS and year fixed effects and two-digit NAICS linear and quadratic trends. Bars report 95% confidence intervals from standard errors clustered by four-digit NAICS industry.

Figure E.4: Bonus Depreciation, Firm Exit, and Business Dynamism



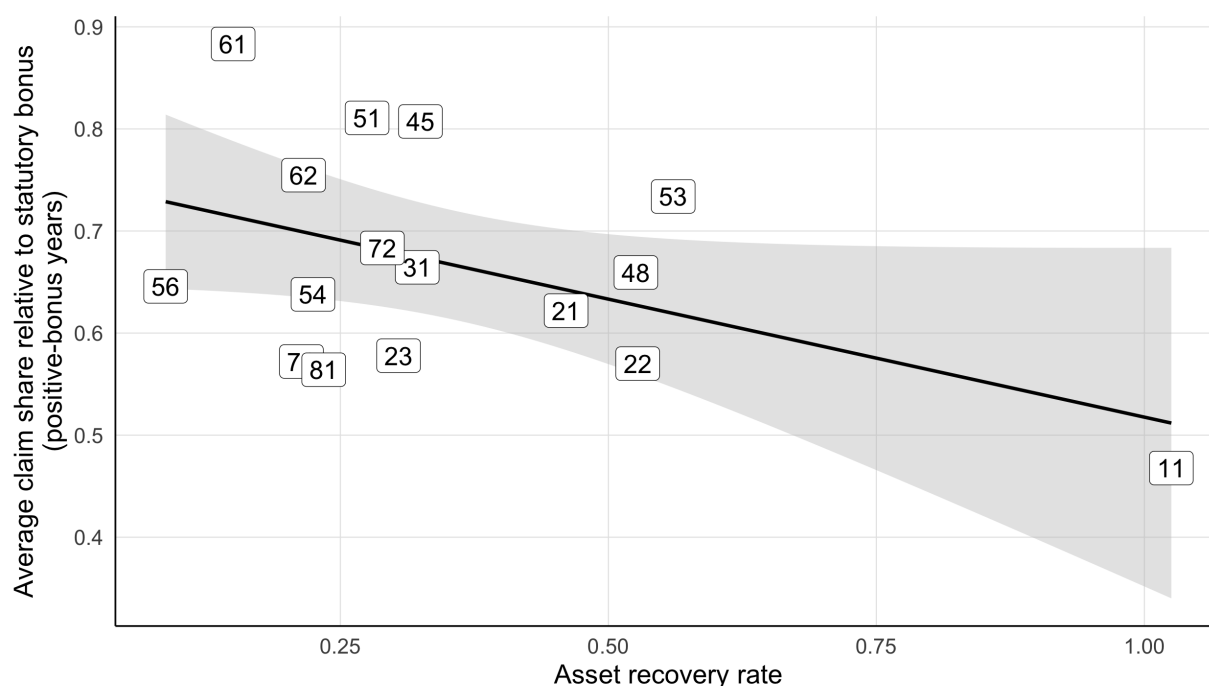
Notes: These panel local projections use the specification from the main entry analysis. The outcomes are the firm exit rate in panel (a) and the log sum of the firm entry and exit rates in panel (b). Both specifications include four-digit NAICS and year fixed effects and two-digit NAICS quadratic time trends. Bars report 95% confidence intervals from standard errors clustered by four-digit NAICS industry.

F Statistics of Income Evidence on Bonus Claims

The public Statistics of Income tabulations report Form 4562 special depreciation allowances by two-digit industry. I divide Line 14 by Line 14 plus the reported basis for three- through twenty-year GDS property. The tabulations neither isolate basis eligible for nationwide bonus depreciation nor exclude other special allowances from Line 14, so I interpret the variable as an industry-level special-allowance claim share rather than a literal firm-level take-up rate.

Figure F.1 plots the average claim share relative to the statutory bonus rate across industries during positive-bonus years. This comparison of claiming intensity and asset tradeability is descriptive; it does not plot the fixed-effect estimates.

Figure F.1: Bonus Depreciation Claims and Asset Tradeability



Notes: Each point is a two-digit NAICS industry. The vertical axis is the industry's average special-allowance claim share relative to the statutory bonus rate during years with positive bonus depreciation. The horizontal axis is the PPE recovery rate from Kermani and Ma (2022). The fitted line is unweighted.

Table F.1 reports panel regressions of the claim share on statutory bonus rates and asset tradeability. Given the small number of industry clusters, I report exact null-imposed wild-cluster p -values. I treat these estimates as supplementary evidence on claiming behavior, not as identification of the reallocation effects in the main text.

Table F.1: Bonus Depreciation Claims and Asset Tradeability

	Full Sample		Bonus Years	Excl. Agriculture
	(1)	(2)	(3)	(4)
$\theta_i \times \ell_i$	-0.269*	-0.280*	-0.329**	-0.285
	(0.074)	(0.082)	(0.172)	(0.209)
Exact wild-cluster p -value	0.073	0.067	0.044	0.216
SOI claiming-capacity controls		X		
Industry linear trends	X	X	X	X
Year FE	X	X	X	X
Industry FE	X	X	X	X
Observations	316	298	236	295
Industry clusters	16	16	16	15

Notes: The dependent variable is the special depreciation allowance claimed on Form 4562, Line 14, divided by Line 14 plus the reported basis for three-, five-, seven-, ten-, fifteen-, and twenty-year GDS property. Line 21 is excluded because it reports depreciation claimed on listed property rather than depreciable basis; ADS property is also excluded because it is not part of the broad bonus-eligible base. Because Line 14 includes special allowances beyond the nationwide bonus policy and the public SOI tabulations do not separately identify bonus-eligible basis—including new versus ineligible used property before 2018—the outcome is an industry-level claim share rather than a literal firm-level take-up rate. The variable θ_i is the statutory bonus rate and ℓ_i is the PPE recovery rate from Kermani and Ma (2022). Column (2) controls for the NOL-to-sales ratio, the share of returns reporting net income, sales relative to capital, and sales growth. Column (3) retains only years with a positive nationwide bonus rate, and column (4) excludes agriculture. All specifications include industry and year fixed effects and industry-specific linear trends. The source window is 2000–2021; 2008 is excluded because the available tabulation does not provide usable Line 14 amounts, and industry-years with a disclosure-suppressed claim-share input are dropped. Dashes in the source tables are treated as reported zeros. Standard errors clustered by two-digit NAICS are in parentheses. Reported p -values and significance markers use an exact null-imposed Rademacher wild-cluster bootstrap over all 2^G sign assignments.

G The Tax Cuts and Jobs Act

The Tax Cuts and Jobs Act bundled several reforms simultaneously. First, it increased bonus depreciation for new assets from 50% to 100%. Second, it cut the corporate tax rate from 35% to 21%, and modestly reduced the top tax rate on passthrough income. Third, it extended accelerated depreciation to used asset purchases. That bundle makes the overall package less clean to evaluate than bonus depreciation. This appendix considers evidence related to reallocation around TCJA.

G.1 Evidence from the Tax Cuts and Jobs Act

Evidence from Compustat

In the model, policy affects investment and used asset sales through the tax treatment of new and used capital and through the corporate tax rate. Among other provisions, the 2017 Tax Cuts and Jobs Act (TCJA) cut the corporate tax rate and extended full expensing to used asset purchases. The model unambiguously predicts that used asset expensing increases reallocation, whereas the effect of a corporate tax cut is generally ambiguous. The preceding bonus results are consistent with the region in which retention dominates capital creation. In that region, both additional expensing for new capital and the incumbent-side response to the tax cut contract used supply, while the tax cut and expensing of used purchases expand demand. The TCJA therefore raises the used price but has an ambiguous effect on physical reallocation. Because Compustat observes qS , this section identifies only the bundled reform’s net effect on capital sale proceeds.

Throughout the TCJA exercise, I compare U.S. firms with foreign firms in Compustat. I require at least four usable firm-years and nonzero pre-reform variation in $sppe$. I also require each four-digit NAICS industry to contain at least two foreign firm-years. Within the U.S. and foreign groups, I winsorize capital sales and investment at the 97th percentile, preserving zeros, and sales, assets, and net PPE at the 3rd and 97th percentiles. The sample contains 17,480 firm-years from 1,620 U.S. and 660 foreign firms. Table G.1 reports summary statistics.

Table G.1: Summary Statistics: TCJA Difference-in-Differences Sample

	US Firms	Foreign Firms
Firms	1,620	660
Firm-Years	12,496	4,984
Total Assets (\$M)	2,938.2	13,026.5
Net PPE (\$M)	908.5	4,820.1
Sales (\$M)	2,243.3	7,431.3
Capital Expenditure (\$M)	151.6	727.5
Capital Sale Proceeds (\$M)	6.70	24.44
% with Positive Capital Sale Proceeds	55.3%	68.3%
Sample Period	2011–2019	

Notes: The U.S. sample contains nonfinancial Compustat firms with positive sales, assets, and PPE, at least four usable firm-years, and nonzero pre-period variation in capital sale proceeds. Foreign firms are foreign-domiciled firms in Compustat North America satisfying the same restrictions. All values are reported in U.S. dollars. Both groups are restricted to four-digit NAICS industries containing at least two foreign firm-years. Capital sale proceeds and investment are winsorized at the within-group 97th percentile, preserving zeros; other dollar amounts are winsorized at the within-group 3rd and 97th percentiles.

Because the TCJA applied to all U.S. firms at the same time, we cannot use the same identification strategy as before. Instead, following Crawford and Markarian (2024) and Chodorow-Reich

et al. (2025), I construct a variety of tests comparing US-domiciled firms, which are exposed to the tax cut, to a control group of foreign firms. For the first test, I estimate

$$\mathbb{E}[Y_{it} | \text{US}_i, \text{Post}_t, \alpha_i, \gamma_t] = \exp[\alpha_i + \gamma_t + \beta(\text{US}_i \times \text{Post}_t)], \quad (\text{G.1})$$

where Post_t indicates fiscal years 2018 and 2019, α_i are firm fixed effects, and γ_t are year fixed effects. I estimate (G.1) by PPML for capital sale proceeds and investment and use a linear probability model for whether the firm reports positive proceeds. Standard errors are clustered by firm. The identifying assumption is that, absent the TCJA, outcomes for U.S. and foreign firms represented in the same industries would have changed similarly. Because some foreign firms have U.S. operations, any exposure of the control group to the reform would tend to attenuate the estimated difference. The results in Table G.2 show that investment and capital sale proceeds both increased following the TCJA, drawing a sharp contrast to the effects of bonus. The investment results are similar to those obtained by a variety of studies also showing higher investment following TCJA (Chodorow-Reich et al. 2025; Kennedy et al. 2023; Hartley, Hassett, and Rauh 2025; Crawford and Markarian 2024). However, unlike the bonus results, the capital sales response is concentrated along the intensive margin rather than the extensive margin.

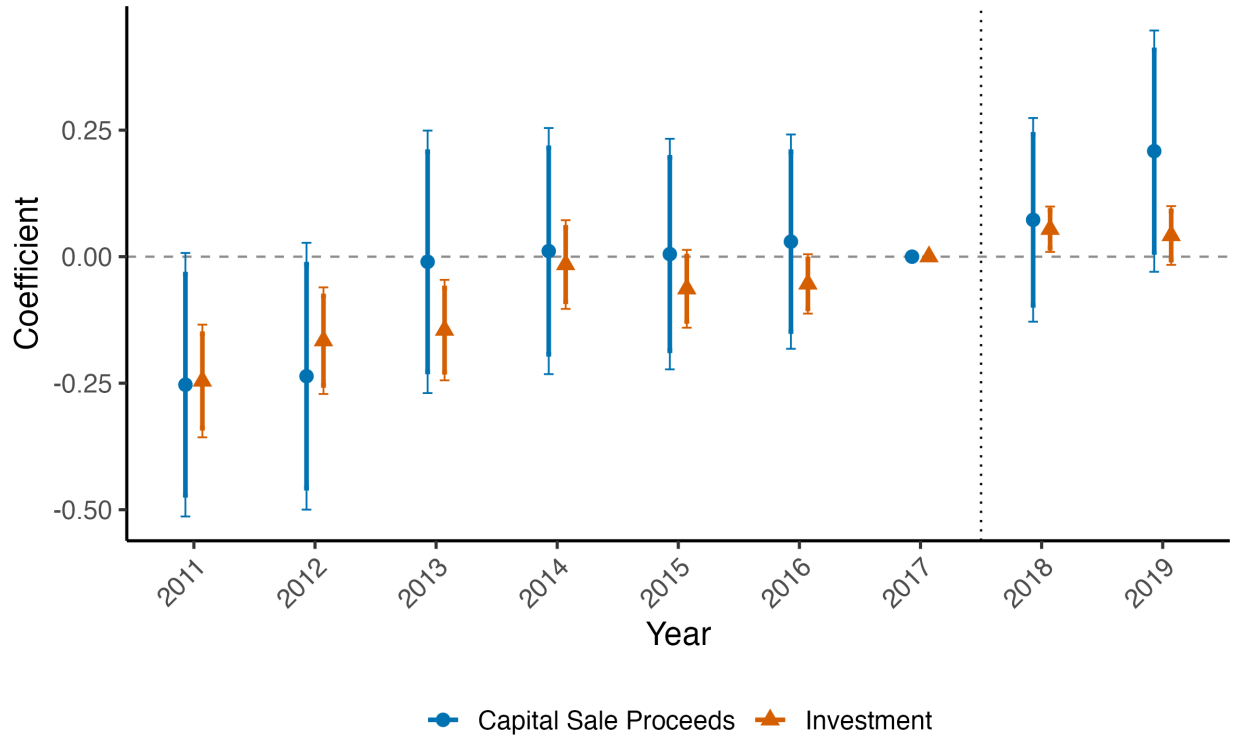
Table G.2: TCJA, Capital Sale Proceeds, and Investment

	Capital Sale Proceeds (sppe)			Investment (capx)
	Unconditional	Any Proceeds	Positive Proceeds	Unconditional
	(1)	(2)	(3)	(4)
Post-TCJA $\times$ US	0.196** (0.084)	-0.015 (0.019)	0.123* (0.069)	0.144*** (0.033)
Estimator	PPML	LPM	PPML	PPML
Firm FE	X	X	X	X
Year FE	X	X	X	X
Observations	17,480	17,480	10,310	17,471

Notes: Capital sale proceeds are cash proceeds from sales of property, plant, and equipment (sppe); investment is capital expenditures (capx). Post-TCJA indicates fiscal years 2018 and later, and US identifies U.S.-domiciled firms; foreign-domiciled Compustat firms form the control group. The sample spans 2011–2019. Parenthesized values are standard errors clustered by firm. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The difference-in-differences analysis in Table G.2 relies on parallel trends for the comparison to be valid. Appendix Figure G.1 gives an event study version of the table. The capital sale proceeds coefficients are close to zero from 2013 through 2016 and are jointly insignificant, supporting parallel trends for the paper’s central outcome. The investment coefficients instead rise toward zero before enactment and are jointly different from zero, so the investment event study does not support the same interpretation. Both outcomes turn positive after enactment, although the year-specific sales estimates are imprecise.

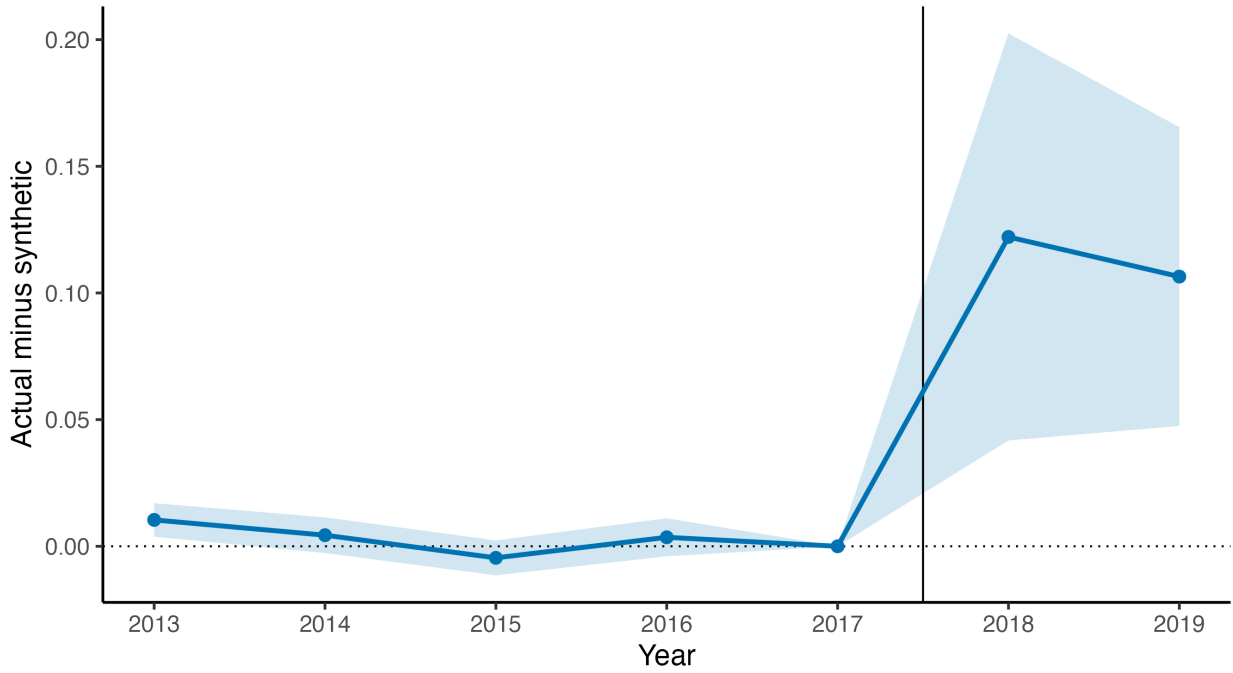
Figure G.1: Event Study of Capital Sales and Investment Around TCJA



Notes: This figure plots event-study coefficients for U.S. firms relative to non-U.S. firms around the TCJA. Outcomes are capital sales (sppe) and capital expenditures (capx), estimated by Poisson pseudo-maximum likelihood with firm and year fixed effects. The reference year is 2017. Thick bars report 90% confidence intervals and thin bars report 95% confidence intervals from standard errors clustered by firm. The vertical dotted line marks enactment of the TCJA.

In a third test, I construct a firm-specific synthetic control using foreign firms from the same three-digit industry. Each synthetic control matches the U.S. firm on pre-TCJA capital sales, investment, revenue, the stock of property, plant, and equipment, and total assets. Figure G.2 shows that among the firms with close matches, the actual-minus-synthetic gap is approximately zero throughout 2013–2017 and increases considerably in both post-reform years. Compared to their synthetic counterparts, U.S. firms increase capital sales by roughly 0.1 inverse-hyperbolic sine units.

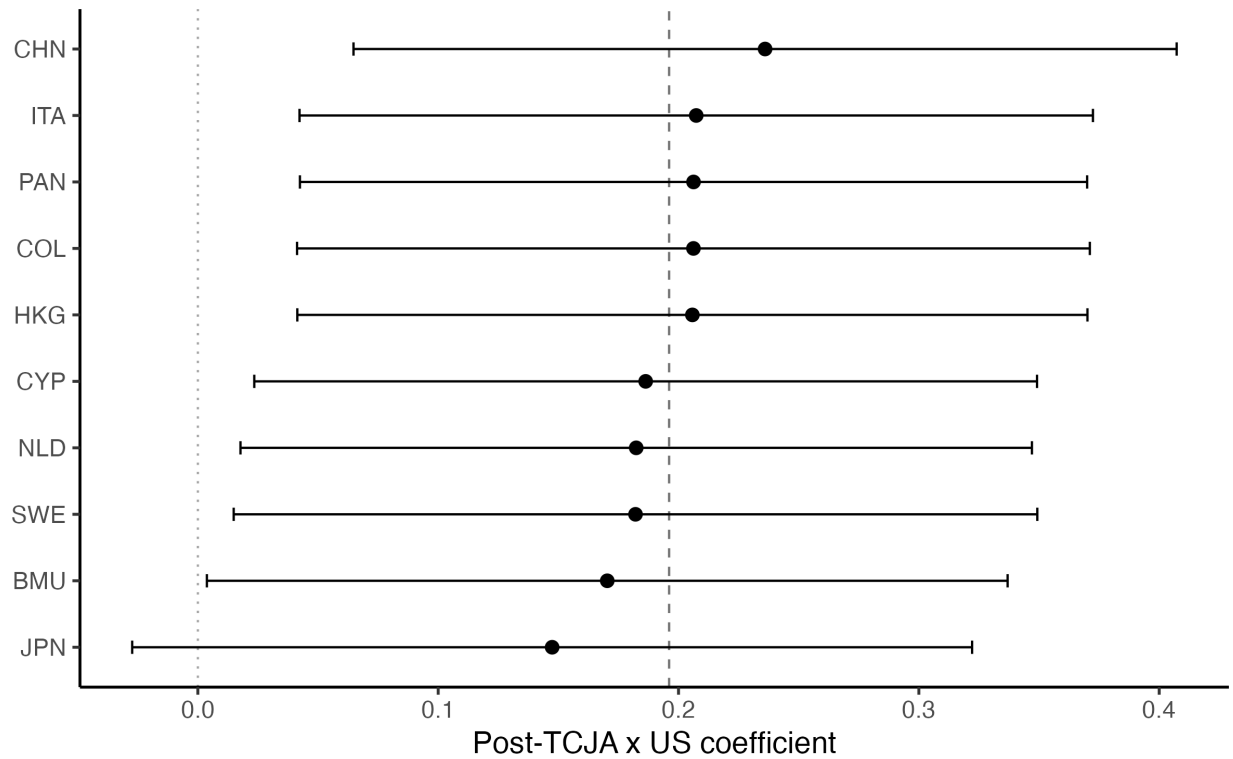
Figure G.2: U.S. Capital Sales Relative to Synthetic Control around the 2017 Tax Cuts and Jobs Act



Note: This figure plots coefficients from a regression of the firm-level gap between actual and synthetic capital sales, $\text{asinh}(\text{SPPE}) - \text{asinh}(\text{SPPE}^{\text{synth}})$, on year indicators with firm fixed effects, with 2017 as the reference year. Each U.S. firm's synthetic control is built from foreign firms listed on U.S. exchanges in the same 3-digit NAICS industry, and the sample keeps firms whose mean absolute pre-period gap is below 0.10. The shaded region is the 95% confidence interval from standard errors clustered by firm. The vertical line marks the enactment of the TCJA.

In short, the TCJA results diverge from those under bonus, though they are consistent with each other in theory. However, there are several important shortcomings. First, in contrast to bonus, which solely changed expensing for new investment, TCJA bundled several reforms simultaneously, so it is difficult to distinguish the effects of any individual provision on capital sales and investment. Second, Compustat solely measures the volume of capital sales qS , so the increase in proceeds could reflect a higher used price, greater physical reallocation, or both. The UCC evidence in the following section isolates S .

Figure G.3: TCJA Difference-in-Differences: Leave One Donor Country Out



Note: Each point re-estimates the U.S. $\times$ Post-TCJA capital sales coefficient after excluding one of the 56 donor countries. The figure shows the five smallest and five largest estimates; the dashed line is the full-sample coefficient. Bars report 95% confidence intervals from standard errors clustered by firm.

Table G.3: Summary Statistics: Synthetic-Control Sample

	Actual Capital Sale Proceeds	Synthetic Capital Sale Proceeds	Actual Investment	Synthetic Investment
Pre-TCJA	1.44	1.47	48.15	59.93
Post-TCJA	1.84	1.19	58.31	61.14

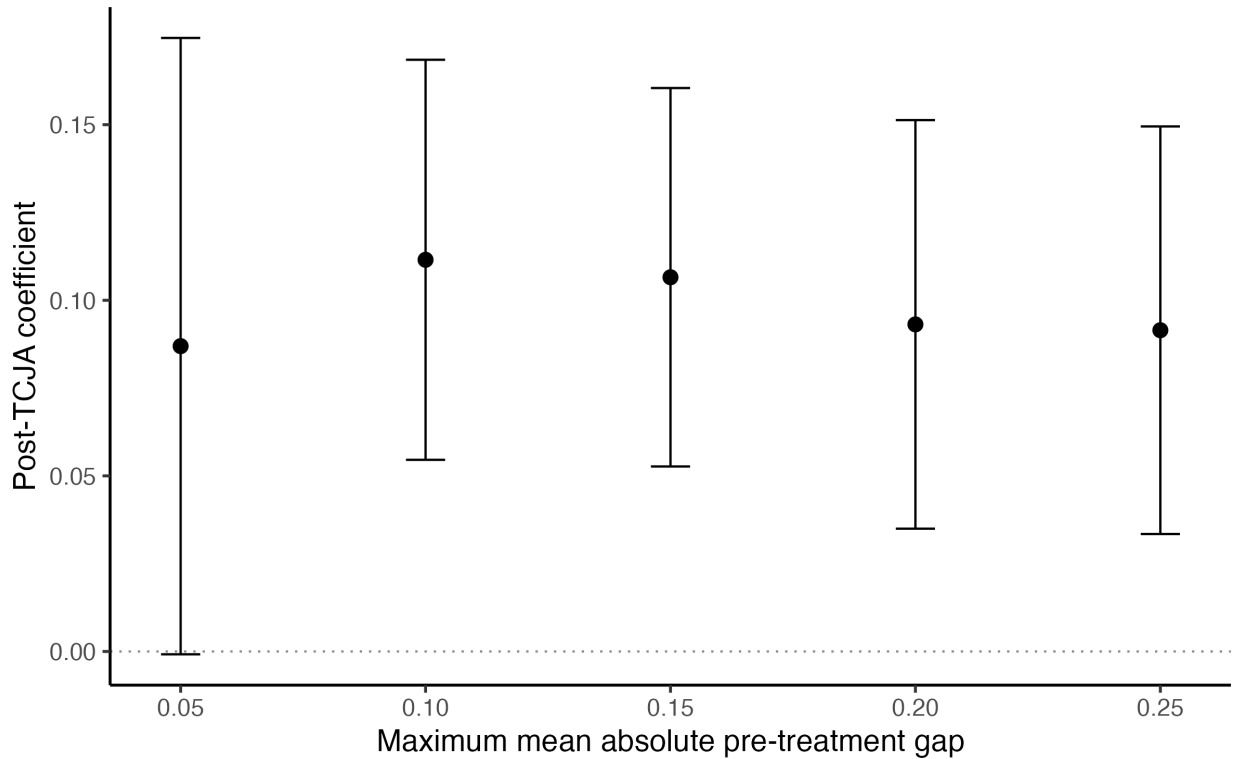
Notes: The sample contains the 383 U.S. firms satisfying the pre-treatment fit screen used in column (1) of Table G.4. Entries are unweighted firm-year means in millions of U.S. dollars.

Table G.4: Synthetic Control: Post-TCJA Capital Sale Proceeds Gap

	Good Pre-Treatment Fit (1)	All Matched Firms (2)
Post-TCJA	0.112*** (0.029)	0.039 (0.040)
Firm FE	X	X
Pre-treatment fit screen	X	
Firms	383	816
Observations	2,681	5,712

Notes: The dependent variable is $\text{asinh}(\text{sppe}) - \text{asinh}(\text{sppe}^{\text{synth}})$ over 2013–2019. Each U.S. firm is matched to a weighted combination of foreign firms in the same three-digit NAICS industry using 2013–2017 capital sale proceeds, investment, revenue, PPE, and assets. Column (1) keeps firms whose mean absolute pre-treatment outcome gap does not exceed 0.10; column (2) keeps every successfully matched firm. Both specifications contain firm fixed effects. Parenthesized values are standard errors clustered by firm. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure G.4: Synthetic Control: Sensitivity to Pre-Treatment Fit Threshold



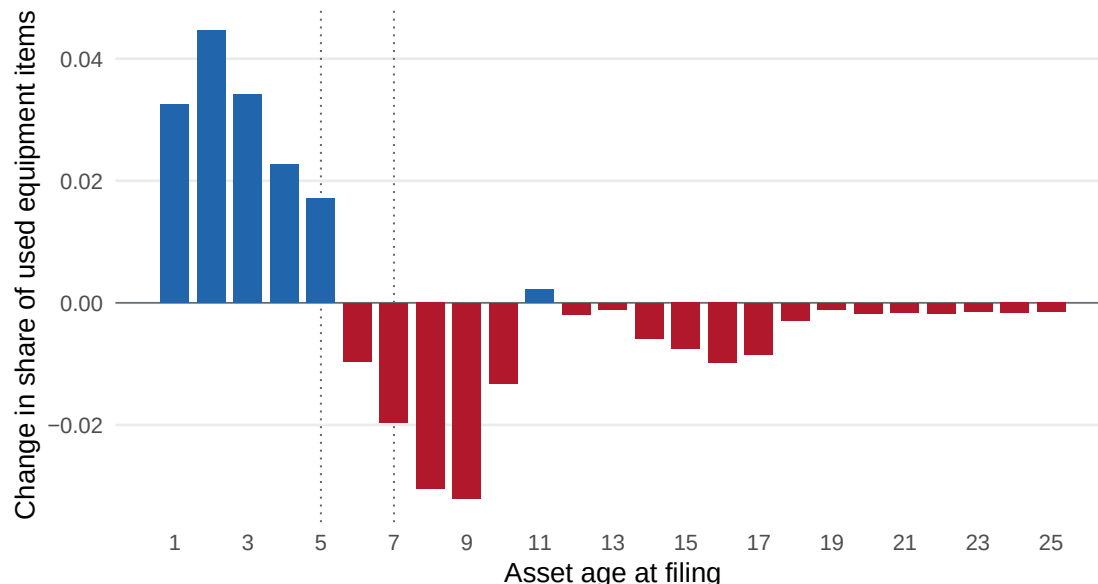
Note: This figure plots the post-TCJA coefficient from the firm fixed-effects regression of the gap between actual and synthetic capital sales on a post indicator, at varying thresholds for the pre-treatment fit screen. The x-axis denotes the maximum allowed mean absolute inverse-hyperbolic-sine gap between actual and synthetic capital sales over the pre-period (2013–2017). Bars represent 95% confidence intervals based on standard errors clustered by firm.

Evidence from Texas UCC Filings

The Compustat results show that U.S. firms increased both investment and capital sale proceeds following the TCJA relative to foreign firms. That finding is important because TCJA bundled several reforms with ambiguous effects on reallocation, but Compustat cannot distinguish between the price and quantity responses. Since the preceding subsection and the Compustat evidence find that bonus depreciation for new capital reduced subsequent used equipment sales and transactions, we can take that result through the model to sign the effect of the tax cut theoretically since they are governed by the same condition. In particular, a corporate tax cut pushes used capital supply inward, since the same condition governs the two responses. On the buyer side, however, the rate cut and the extension of expensing to used purchases shift demand outward. Thus, there are two countervailing effects on the market in equilibrium, and it is ex ante unclear which will dominate the reallocative force.

Figure G.5 provides initial descriptive evidence that the composition of used asset sales substantially changed post-TCJA. It plots the change in each age's share of used equipment transactions between 2012–2017, when policy was stable, and 2018–2019.

Figure G.5: The Age Distribution of Used Equipment Transactions Following the TCJA



Notes: The figure plots the change in each asset age’s share of used equipment transactions between 2012–2017 and 2018–2019. The sample contains used equipment items between ages one and 25. Positive values indicate that an age accounted for a larger share of transactions following the TCJA. The dotted lines mark ordinary basis exhaustion for five- and seven-year MACRS property.

The distribution shifts sharply toward transactions of younger equipment post-TCJA. Ages one through five, which would typically have some basis remaining in the pre-TCJA era, gain approximately 15 percentage points of transaction shares, while assets aged six through ten lose 11 percentage points. This shift is potentially significant because the rate cut associated with TCJA narrowed the tax disadvantage of selling bonus vintages compared to nonbonus vintages and therefore may provide evidence of reduced lock-in. Providing further evidence of that, Appendix Table G.7 estimates a corresponding 9.8-percentage-point increase in the probability that used equipment falls within the ordinary recovery period, while Appendix Figure G.6b shows that similar movements appear in some earlier placebo windows.

Nevertheless, the figure does not establish that total transactions increased because the figure plots each age as a share of total transactions. Importantly, it also does not account for the capital stock expansion channel, which may have enlarged the unobserved stock of young equipment available for subsequent sale. I address this concern by restricting the sample to model-year vintages no later than 2016 and following transactions involving these preexisting cohorts over time using the same PPML framework as in the previous subsection. A cohort is defined by equipment type, MACRS class, and model-year vintage, while filing year records when its transactions occur. Column (1) of Table G.5 asks whether bonus vintages experienced a larger post-TCJA change in transactions than nonbonus vintages.

The remaining columns ask whether the post-TCJA response was larger for bonus vintages that still had a bonus-induced basis gap. All nonbonus vintages were fully depreciated by 2018 and therefore provide a common comparison group. After netting that out, the reported “positive minus zero” estimate compares younger bonus vintages with remaining basis to older bonus vintages with no remaining basis gap. The older vintages provide a placebo because there is no

longer a differential incentive to retain the capital. Cohort fixed effects absorb permanent differences in cohort size and composition, equipment-type $\times$ filing-year fixed effects absorb common market changes, and age fixed effects absorb the normal life-cycle transaction pattern. The identifying assumption is that, absent TCJA, the two types of bonus vintages would have experienced similar changes relative to nonbonus vintages and that those changes would not otherwise have differed with the remaining basis gap.

Column (1) provides a pooled benchmark by comparing the post-TCJA change in transactions involving bonus vintages with the contemporaneous change for the much older nonbonus vintages that remain in the UCC data. The coefficient of 0.131 implies a 14 percent larger age-adjusted change for bonus vintages, although the nonoverlapping vintage support makes this comparison less informative than the within-bonus comparisons that follow. This reflects a relative catch-up by preexisting cohorts rather than an estimate of the aggregate change in transactions. Column (2) shows that the response is 0.148 log points larger among bonus vintages with a positive future basis gap at the filing date than among bonus vintages whose gap has been exhausted. Because vintages can change groups as their basis is exhausted, column (3) fixes each cohort's status using its basis gap in 2018. The difference remains positive at 0.120 log points, or approximately 13 percent, but is imprecisely estimated and cannot be distinguished from zero. Finally, column (4) shows that every additional ten percentage points of the 2018 basis gap is associated with a 0.155-log-point larger response, while the relative response at a zero gap is small and statistically insignificant.

The central empirical result is that the post-TCJA reallocation response is larger for bonus vintages whose prior tax treatment still implies a statutory basis gap. The firm's first-order condition for resale motivates this comparison because remaining basis enters the sale decision, but it does not imply that the rate cut alone generated the observed ordering; it could have also come from the demand-side expensing provision. The result is therefore suggestive rather than dispositive, but it is compelling alongside the Compustat evidence. Appendix Table G.6 shows that the current-gap and continuous-gap estimates are similar when restricting the sample to initial filings, adding supported zero-count cells, omitting 2017, and adding saturated fixed effects. The fixed-2018 indicator remains positive across specifications but is generally imprecise.

Table G.5: TCJA and Used Equipment Transactions Within Fixed Cohorts

	Average response (1)	Future gap at filing (2)	Future gap in 2018 (3)	2018 gap amount (4)
<i>Post-TCJA response of bonus vintages</i>				
All bonus vintages	0.131** (0.060)			
No future basis gap		0.088 (0.062)	0.095 (0.069)	
Positive future basis gap		0.237*** (0.066)	0.215*** (0.071)	
Difference: positive minus zero		0.148** (0.066)	0.120 (0.091)	
Response at zero 2018 basis gap				0.077 (0.059)
Additional response per 10 pp of basis gap				0.155*** (0.047)
Filing-year future-basis-gap control		X		
Type-class-method-vintage FE	X	X	X	X
Type × filing year FE	X	X	X	X
Asset age FE	X	X	X	X
Observations	9,780	9,780	9,780	9,780

Notes: The dependent variable is the number of used equipment transactions in equipment-type × MACRS-class × basis-method × model-year-vintage × filing-year cells from 2012–2019; all cohorts predate the TCJA. The future basis gap is the reduction in ordinary MACRS basis that would remain after the filing-year deduction if the asset were retained. Column (2) uses the filing-year gap; columns (3) and (4) fix the gap after the 2018 deduction, with column (4) scaled per ten percentage points. PPML standard errors clustered by model-year vintage are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table G.6: Robustness of the Exact-Cohort TCJA Comparisons

	Observed all filings (1)	Initial filings (2)	Supported with zeros (3)	Drop 2017 (4)	Saturated fixed effects (5)
Current gap: positive minus exhausted	0.148** (0.066)	0.184*** (0.057)	0.144** (0.062)	0.178** (0.079)	0.133* (0.071)
Fixed 2018 gap: positive minus exhausted	0.120 (0.091)	0.178** (0.081)	0.119 (0.087)	0.138 (0.111)	0.103 (0.085)
Fixed 2018 gap: response per 10 pp	0.155*** (0.047)	0.169*** (0.040)	0.145*** (0.044)	0.245*** (0.067)	0.097** (0.047)
Exact type-class-method-vintage FE	X	X	X	X	X
Equipment type $\times$ year FE	X	X	X	X	
Product $\times$ year and product $\times$ age FE					X
Observations	9,780	9,196	17,775	8,503	9,780

Notes: Each entry is a relative post-TCJA comparison within exact equipment-type $\times$ MACRS-class $\times$ basis-method $\times$ vintage cohorts. Column (1) reproduces the main observed-cell sample. The sample covers filing years 2012–2019; all cohorts predate the TCJA. Column (2) retains initial filings. Column (3) defines the cohort risk set using only cohorts observed by 2017 and fills feasible absent cells with zeros. Column (4) omits 2017, which is potentially contaminated by the September 27 effective date and anticipation of Section 1031 repeal. Column (5) replaces equipment-type by year and common age effects with product-by-year and product-by-age effects. The current-gap regressions include the time-varying gap level. Standard errors clustered by vintage are in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table G.7 provides a formal counterpart to the descriptive figure showing the age shift in the composition of used asset transactions.

Table G.7: Age Composition of Equipment Filings Around the TCJA

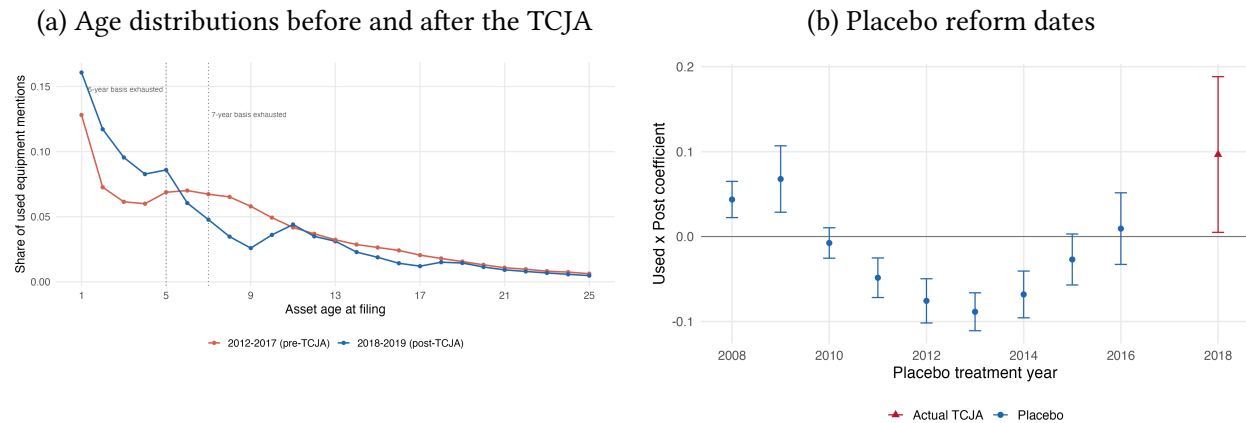
	Positive future MACRS basis			Future deduction years
	Raw	Linear trend	Quadratic trend	Linear trend
	(1)	(2)	(3)	(4)
Used $\times$ Post-TCJA	0.098** (0.047)	0.097** (0.047)	0.097** (0.047)	0.612* (0.334)
Macro controls		X	X	X
Type-class-method FE	X	X	X	X
Observations	122,298	122,298	122,298	122,298

Notes: Item-level linear regressions over 2012–2019. Positive Future MACRS Basis indicates that the ordinary five- or seven-year schedule leaves basis after the filing-year deduction if the asset is retained. The final column counts future deduction years after the filing year, censored at zero. Macro controls include GDP growth, oil and agricultural price changes, agricultural income growth, the ten-year Treasury rate, and Texas unemployment. Standard errors clustered by equipment-type by MACRS-class by basis-method cells are in parentheses. This is descriptive evidence on the composition of filings, not a separate causal design.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure G.6 shows that the post-TCJA distribution shifts toward ages with future deductions remaining, although similar movements appear in some earlier placebo windows. Because this comparison does not hold model-year composition fixed, I use it as descriptive evidence about timing rather than as a separate causal design.

Figure G.6: Age Composition of Used Equipment Around the TCJA



Notes: Panel (a) plots the share of positive-age equipment mentions at each asset age in 2012–2017 and 2018–2019. Panel (b) assigns the reform to each placebo year from 2008 through 2016. Bars are 95% confidence intervals from standard errors clustered by equipment type, recovery class, and basis method.

TCJA and Firm Entry

This section examines the effects of the Tax Cuts and Jobs Act on firm entry, which provides a second, weaker test of the entry mechanism. In addition to cutting the corporate tax rate and increasing bonus depreciation for new capital, the Act extended full expensing to purchases of used equipment. Used capital expensing shifts demand outward in the model and therefore increases reallocation and entry, although the other provisions of the Act can generate offsetting effects. Appendix Figure G.7a shows that entry rises after 2018 in industries where the reform most reduced the preexisting tax disadvantage of used equipment. With only two post-reform years before the pandemic, the estimates are imprecise, so I treat this evidence as directional rather than conclusive. Moreover, quantitative evidence from Sedlacek and Sterk (2019) and Zeida (2022) points toward higher entry through other channels like higher returns to entrepreneurship post-TCJA, which I cannot distinguish from here.

By extending full expensing to used purchases, the TCJA narrowed the tax disadvantage of buying used rather than new capital. I measure this change using the tax-adjusted acquisition cost of each type of capital. Let z_N^0 denote the present value of ordinary MACRS deductions for industry N , and let

$$z_N(\theta) = \theta + (1 - \theta)z_N^0, \quad p_N(\theta, \tau) = 1 - \tau z_N(\theta).$$

The pre-TCJA wedge is

$$g_N^{\text{pre}} = 1 - \frac{p_N(0.5, 0.35)}{p_N(0, 0.35)}.$$

Full expensing of both new and used purchases makes the corresponding post-TCJA wedge zero. I therefore define $\Delta g_N = g_N^{\text{pre}} - g_N^{\text{post}} = g_N^{\text{pre}}$, which is larger in industries where the reform did more to lower the relative cost of used capital. I construct z_N^0 following the approach of Zwick and Mahon (2017) by weighting each BEA industry's fixed investment from 2014 through 2016, and mapping the resulting exposure to four-digit NAICS industries.

Let $\widehat{\Delta g}_N$ denote this exposure after demeaning and dividing by its cross-industry standard deviation. I estimate

$$y_{N,t} = \alpha_N + \gamma_t + \sum_{s \neq 2017} \beta_s \widehat{\Delta g}_N \mathbf{1}\{t = s\} + \eta_{n(N)}t + \varepsilon_{N,t}, \quad (\text{G.2})$$

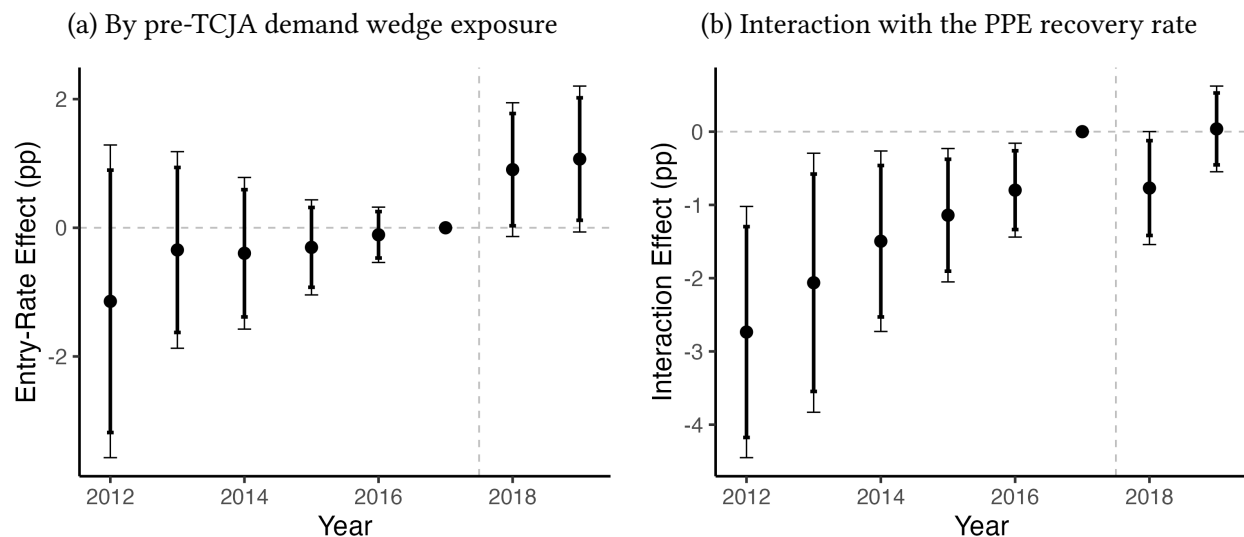
where $\eta_{n(N)}t$ denotes two-digit NAICS linear trends. Standard errors are clustered by four-digit NAICS industry. The sample contains 201 industries from 2012 through 2019.

Results. Panel (a) of Figure G.7 plots β_s . The coefficients from 2013 through 2016 are close to zero; the 2012 coefficient is more negative but imprecise. They rise to 0.9 percentage points in 2018 and 1.1 percentage points in 2019 for a one-standard-deviation change in the wedge. Both post-TCJA estimates are statistically significant at the 10% level but not the 5% level. With only two post-reform years before COVID, I interpret this evidence as suggestive.

Panel (b) asks whether the response is larger in industries with more tradeable capital. I add year interactions with the demeaned recovery rate and with its interaction with $\widehat{\Delta g}_N$. I scale the interaction to a one-standard-deviation change in the wedge and a ten-percentage-point difference in the recovery rate. The pretrends are negative and converge toward 2017 as TCJA

approaches. Because this differential pretrend undermines a parallel-trends interpretation, the post-TCJA interaction does not provide a credible separate test of the tradeability mechanism.

Figure G.7: Firm Entry Around the TCJA



Notes: Panel (a) reports β_s from (G.2). Panel (b) reports year-specific interactions between $\widehat{\Delta g}_N$ and the demeaned PPE recovery rate from Kermani and Ma (2022); the regression also includes the corresponding year interactions with each variable separately. Both specifications include four-digit NAICS and year fixed effects and two-digit NAICS linear trends. The reference year is 2017. Thick bars report 90% confidence intervals and thin bars report 95% confidence intervals from standard errors clustered by four-digit NAICS industry.